Exploratory Data Analysis for SPROJ

Leonardo Andrade

PACKAGES

In order to do this Exploratory Data Analysis, I will use the following packages:

```
library(ggplot2) # for plotting data
library(readxl) # for reading excel files if necessary
library(dplyr) # for data wrangling
# library(arrow)
library(janitor) # for cleaning messy names of datasets
library(readr) # for reading files
library(gt) # for making 'publication ready' tables
library(usethis) #enables git pushing to put on github
library(gitcreds)
library(purrr) # this package is to loop and automate this entire process, instead of going
library(fixest) #fixed fx
library(broom)
library(haven) ## allows to read .dta files meant for stata
library(tidyverse)
library(tidysynth)
library(gsynth)
```

This senior project document will serve as the backbone of methodology and analysis part of the Senior Project"

Motivation

According to the UN Trade Hub and Development:

(https://unctadstat.unctad.org/datacentre/reportInfo/US.RCA)

“Revealed comparative advantage (RCA) is based on Ricardian trade theory, which posits that patterns of trade among countries are governed by their relative differences in productivity. Although such productivity differences are difficult to observe, an RCA metric can be readily calculated using trade data to”reveal” such differences. While the metric can be used to provide a general indication and first approximation of a country’s competitive export strengths, it should be noted that applied national measures which affect competitiveness such as tariffs, non-tariff measures, subsidies and others are not taken into account in the RCA metric.”

“When a country has a revealed comparative advantage for a given product (RCA >1), it is inferred to be a competitive producer and exporter of that product relative to a country producing and exporting that good at or below the world average. A country with a revealed comparative advantage in product i is considered to have an export strength in that product. The higher the value of a country’s RCA for product i, the higher its export strength in product i.”

Part of this senior project aims to explain shifts in Revealed Comparative Advantage through a political economic lens, specifically in Mexico’s Agriculture industry As the UN mentions, this RCA is merely an inference variable, and is used as a ‘first approximation’ in comparative advantage. Many of the shifts in RCA over time are not captured. I will use classical economic theory, neoclassical theory, economic literature, and papers to describe shifts in the RCA through a political economic lens. My sector analysis in these documents will be done in order of sector code and number, starting from 01; Animals Live, throughout the rest of the agricultural sector.

The agricultural sector is one that is extremely broad- ranging from plants and livestock, to coffee beans and even honey. In order to further analyze the RCA index, we must discern what codes constitute the agriculture industry. Each sector has its own general product code at the 2 digit, 4 digit, and 6 digit level, with each digits being more specific in each product. Using the International Trade Administration (ITA) Harmonized System (HS) codes, we will use these definitions to discern what products are part of the agriculture industry. There is opinions in the literature that showws variations in the product codes that is part of the ‘agricultural sector.’ I will use the definitions from this paper (https://www.researchgate.net/publication/332551060_Changes_in_Foreign_Trade_in_Agricultural_Produc to classify the codes that belongs in the agricultural sector. Furthermore, I will analyze each sector at a general level, starting at the 2 digit, and then up to 4 digits. Due to constraints and limitations of the publicly available data, the maximum scope of analysis is up to the 4 digit HS code level. The data will also be split up into multiple time periods. Period A will be from 1992-2000, period B will be from 2000-2008, and period C will be from 2008-2016, period D will be from 2016-2024.

Functions

The section below describes the functions which are necessary in computing this analysis, I have made comments in the code chunks in order to further expand on each line of code and what it does.

Necessary Variables Function

```
## This is a function that takes in only the necessary variables for an RCA analysis, includ

necessary_variables <- function(df) { # inputs a dataframe
  df |>
    dplyr::mutate(cmdCode = as.character(cmdCode)) |>
    arrange(period) |> #arranges the df by year
    select(period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue) #sele
}
```

RCA Function

```
## RCA calculation function

#inputs country exports of product i , country's total exports, world exports of product i, a
calculate_rca <- function(country_product, country_total, world_product, world_total) {

  #join by period to align years
  rca_df <- country_product |>
    inner_join(country_total, by = "period", suffix = c("_prod", "_total")) |>
    inner_join(world_product, by = "period") |>
    inner_join(world_total, by = "period", suffix = c("_world_prod", "_world_total")) |>
    mutate(
      RCA = (fobvalue_prod / fobvalue_total) /
            (fobvalue_world_prod / fobvalue_world_total)
    ) |>
    select(period, RCA)

  return(rca_df)
}
```

generalized RCA Function

```
calculate_rca_generalized <- function(country_sector_df,
                                      world_sector_df,
                                      country_all_df,
                                      world_all_df) {

  #first, I aggregate by year and product, with variables period, and cmdCode, removing any r
  country_prod <- country_sector_df |>
    group_by(period, cmdCode) |>
    summarise(country_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  world_prod <- world_sector_df |>
    group_by(period, cmdCode) |>
    summarise(world_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  #next, I aggregate total exports using fobvalue  (denominator)
  country_all <- country_all_df |>
    group_by(period) |>
    summarise(total_country_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  world_all <- world_all_df |>
    group_by(period) |>
    summarise(total_world_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  # 3. next, I merge everytjing by year and commodity code
  rca_df <- country_prod |>
    left_join(world_prod, by = c("period", "cmdCode")) |>
    left_join(country_all, by = "period") |>
    left_join(world_all, by = "period") |>
    mutate(
      RCA = (country_product_value / total_country_value) /
            (world_product_value / total_world_value),
      has_CA = RCA >= 1
    ) |>
    arrange(period, cmdCode)

  return(rca_df) #outputs a dataframe
}
```

Recocded loading dataframes

```
#setup

base_path <- "~/Desktop/SPROJ/workplace/input"

time_periods <- c("a", "b", "c")

sector_codes <- c("01", "02", "04", "05", "07", "08", "09", "10", "11",
                  "12", "13", "14", "15", "16", "17", "18", "19", "20",
                  "21", "22", "23", "24", "50", "51", "52", "53")


# functions (unchanged from your original)

necessary_variables <- function(df) {
  df |>
    dplyr::mutate(cmdCode = as.character(cmdCode)) |>
    arrange(period) |>
    select(period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue)
}

calculate_rca_generalized <- function(country_sector_df,
                                      world_sector_df,
                                      country_all_df,
                                      world_all_df) {
  country_prod <- country_sector_df |>
    group_by(period, cmdCode) |>
    summarise(country_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  world_prod <- world_sector_df |>
    group_by(period, cmdCode) |>
    summarise(world_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  country_all <- country_all_df |>
    group_by(period) |>
    summarise(total_country_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

  world_all <- world_all_df |>
    group_by(period) |>
    summarise(total_world_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")
```

```r
  rca_df <- country_prod |>
    left_join(world_prod, by = c("period", "cmdCode")) |>
    left_join(country_all, by = "period") |>
    left_join(world_all, by = "period") |>
    mutate(
      RCA = (country_product_value / total_country_value) /
            (world_product_value / total_world_value),
      has_CA = RCA >= 1
    ) |>
    arrange(period, cmdCode)

  return(rca_df)
}


#LOAD AND COMBINE TOTAL EXPORT FILES ACROSS TIME PERIODS


load_combined_totals <- function(folder_prefix, file_prefix, time_periods, base_path) {
  # Loops over time periods a, b, c and stacks into one dataframe
  map_dfr(time_periods, function(t) {
    file_path <- file.path(base_path,
                           paste0(folder_prefix, t), # i.e mx_total_xports_world_time_a
                           paste0(file_prefix, t, ".csv")) # e.x mx_xports_all_time_a.csv
    if (!file.exists(file_path)) {
      message("File not found, skipping: ", file_path)
      return(NULL)
    }
    read_csv(file_path, show_col_types = FALSE) |> necessary_variables()
  })
}

# Combined Mexico total exports: 1990-2012
mx_xports_all <- load_combined_totals(
  folder_prefix = "mx_total_xports_world_time_",
  file_prefix   = "mx_xports_all_time_",
  time_periods  = time_periods,
  base_path     = base_path
)

# Combined world total exports: 1990-2012
world_xports_all <- load_combined_totals(
```

```r
  folder_prefix = "world_total_xports_time_",
  file_prefix   = "world_xports_all_time_",
  time_periods  = time_periods,
  base_path     = base_path
)


# PROCESS SECTOR ACROSS ALL THREE TIME PERIODS

process_sector <- function(sector) {
  sector_data <- map_dfr(time_periods, function(t) {

    # mx files: try short name first (time_a and time_c pattern)
    mx_file <- file.path(base_path,
                         paste0("mx_xports_prod_i_time_", t),
                         paste0("mx_sector_", sector, "_time_", t, ".csv"))

    # if not found, try long name (time_b pattern and sector 01)
    if (!file.exists(mx_file)) {
      mx_file <- file.path(base_path,
                           paste0("mx_xports_prod_i_time_", t),
                           paste0("mx_xports_sector_", sector, "_time_", t, ".csv"))
    }

    # world files: always world_xports_sector_XX_time_Y.csv
    world_file <- file.path(base_path,
                            paste0("world_xports_prod_i_time_", t),
                            paste0("world_xports_sector_", sector, "_time_", t, ".csv"))

    if (!file.exists(mx_file) | !file.exists(world_file)) {
      message("Skipping sector ", sector, " time_", t, " (file not found)")
      return(NULL)
    }

    mx_sector <- read_csv(mx_file, show_col_types = FALSE) |>
      necessary_variables()

    world_sector <- read_csv(world_file, show_col_types = FALSE) |> necessary_variables()

    rca_slice <- calculate_rca_generalized(
      mx_sector,
      world_sector,
```

```
      mx_xports_all,
      world_xports_all
    )

    return(rca_slice)
  })

  if (is.null(sector_data) | nrow(sector_data) == 0) return(NULL)
  sector_data$sector <- sector
  return(sector_data)
}


# BUILD THE BIG RCA DATAFRAME


dir.create("output/clean/rca_sectors", recursive = TRUE, showWarnings = FALSE)

rca_big_df <- map_dfr(sector_codes, function(s) {
  result <- process_sector(s)
  if (!is.null(result)) {
    write_csv(result, paste0("output/clean/rca_sectors/rca_sector_", s, ".csv"))
    message("Saved sector ", s)
  }
  return(result)
})

# Add sector column and NAFTA dummies
rca_big_df <- rca_big_df |>
  mutate(
    cmdCode = as.character(cmdCode),
    sector = substr(cmdCode, 1, 2),
    post_nafta = if_else(period >= 1994, 1, 0),
    nafta_phase = case_when(
      period < 1994                ~ "pre",
      period >= 1994 & period <= 2000 ~ "post"
    )
  )
```

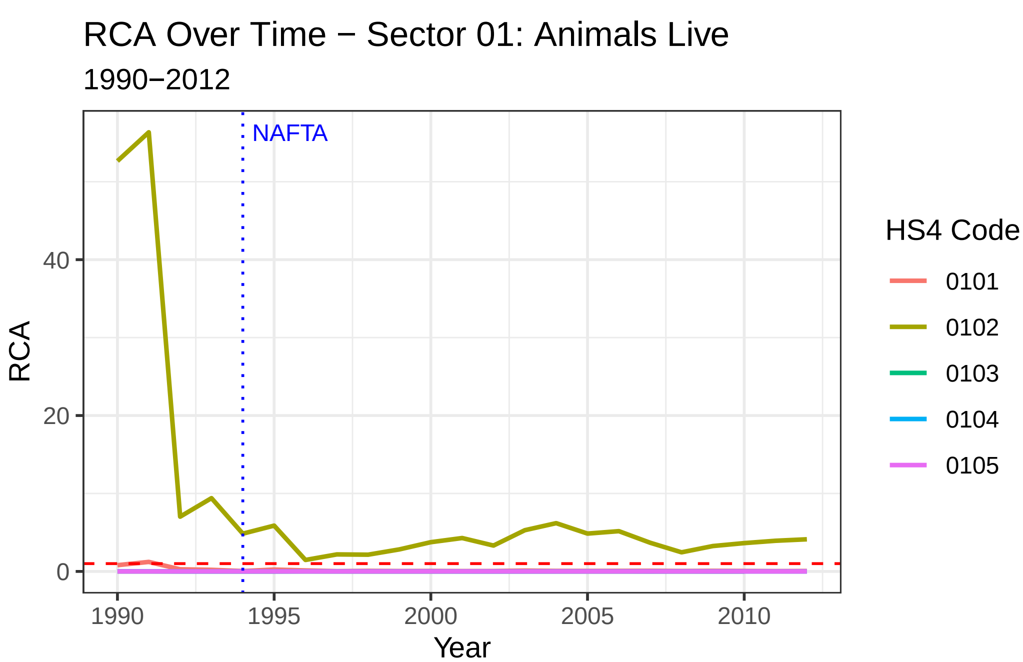
SECTORS

01 Animals; Live

This section will calculate RCA over time before NAFTA of the animal section, containing HS product code of 0101 to 0106.

```
## load the datasets
rca_sector_01 <- rca_big_df |>
  filter(sector == "01", nchar(cmdCode) == 4)
```

```
ggplot(rca_sector_01, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") + # NAFTA line
  annotate("text", x = 1994.3, y = max(rca_sector_01$RCA, na.rm = TRUE),
           label = "NAFTA", hjust = 0, size = 3, color = "blue") +
  labs(title = "RCA Over Time - Sector 01: Animals Live",
       subtitle = "1990-2012",
       x = "Year", y = "RCA",
       color = "HS4 Code") +
  theme_bw()
```

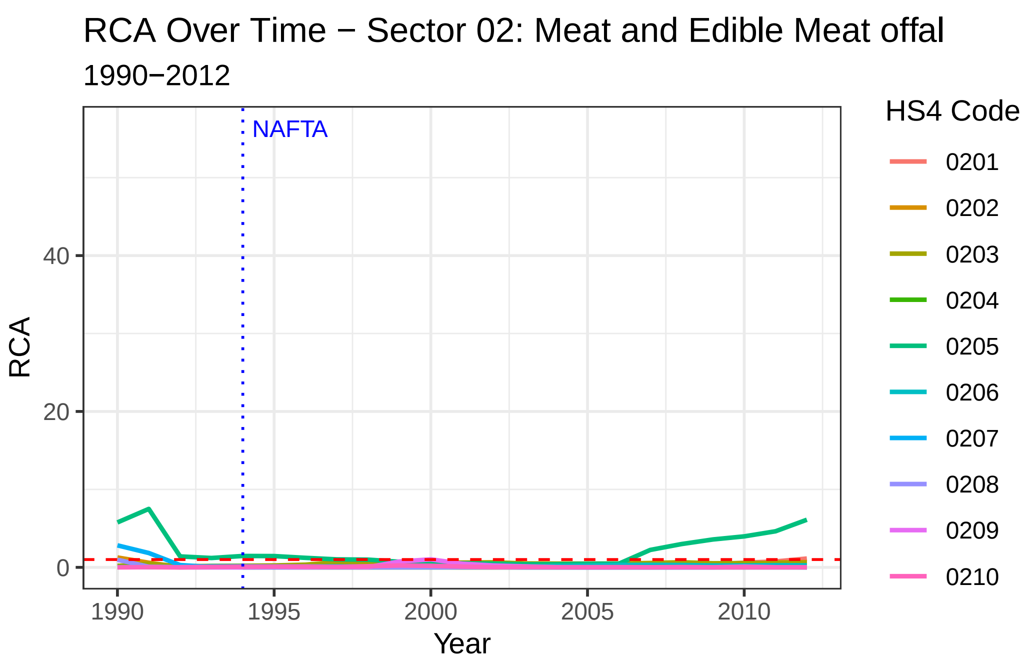


Interesting that Mexico seems to have a general RCA of Live Animals, but there was a sharop decline starting in 1993. NAFTA was enacted January 1, 1994- was it a driving factor in RCA shift? What about data from future years?

02 Meat and Edible Meat offal

```
## load the datasets
rca_sector_02 <- rca_big_df |>
  filter(sector == "02", nchar(cmdCode) == 4)
```

```
## graph RCA
ggplot(rca_sector_02, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") + # NAFTA line
  annotate("text", x = 1994.3, y = max(rca_sector_01$RCA, na.rm = TRUE),
           label = "NAFTA", hjust = 0, size = 3, color = "blue") +
  labs(title = "RCA Over Time - Sector 02: Meat and Edible Meat offal",
       subtitle = "1990-2012",
       x = "Year", y = "RCA",
       color = "HS4 Code") +
  theme_bw()
```



Sector of 02 seems to not have much of a comparative advantage in Mexico.

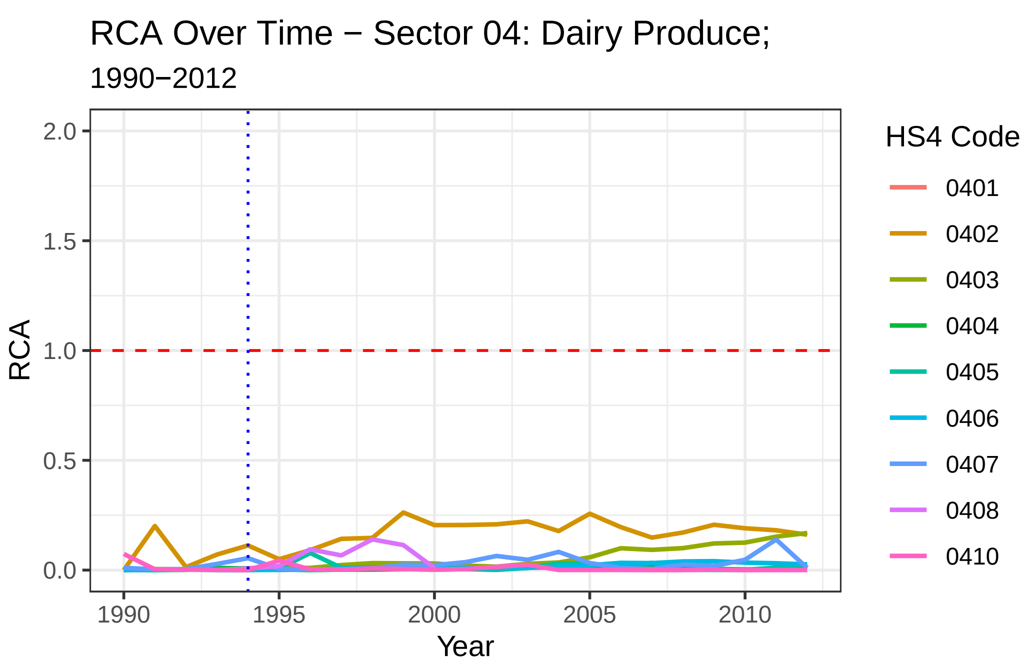
04 Dairy Produce; Birds' eggs; Natural Honey; Edible products of animal origin

- Omit 0409: Honey.

```r
rca_sector_04 <- rca_big_df |>
  filter(sector == "04", nchar(cmdCode) == 4)
```

Graph Sector 04

```r
## graph RCA
ggplot(rca_sector_04, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 2)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 04: Dairy Produce;",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



05 Oher Animal Products

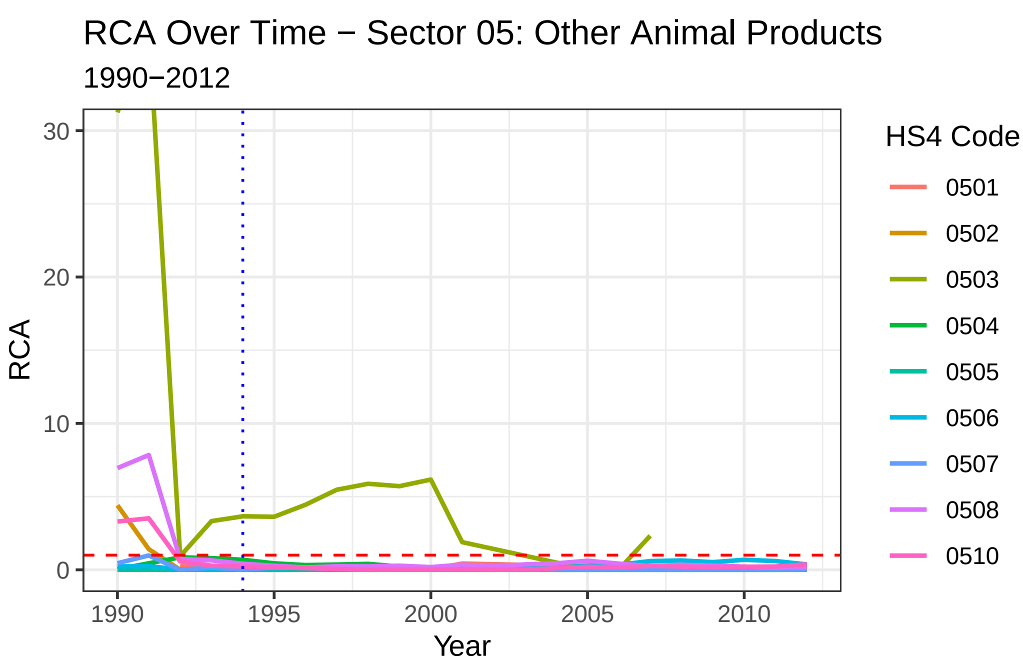
ANIMAL ORIGINATED PRODUCTS; NOT ELSEWHERE SPECIFIED OR INCLUDED

We omit the following

- 0501 Human hair; unworked, whether or not washed or scoured; waste of human hair
- 0509 Sponges
- 0511 Animal products not elsewhere specified or included; dead animals of chapter 1 or 3, unfit for human consumption

```r
rca_sector_05 <- rca_big_df |>
  filter(sector == "05", nchar(cmdCode) == 4)
```

```r
ggplot(rca_sector_05, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 30)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 05: Other Animal Products",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

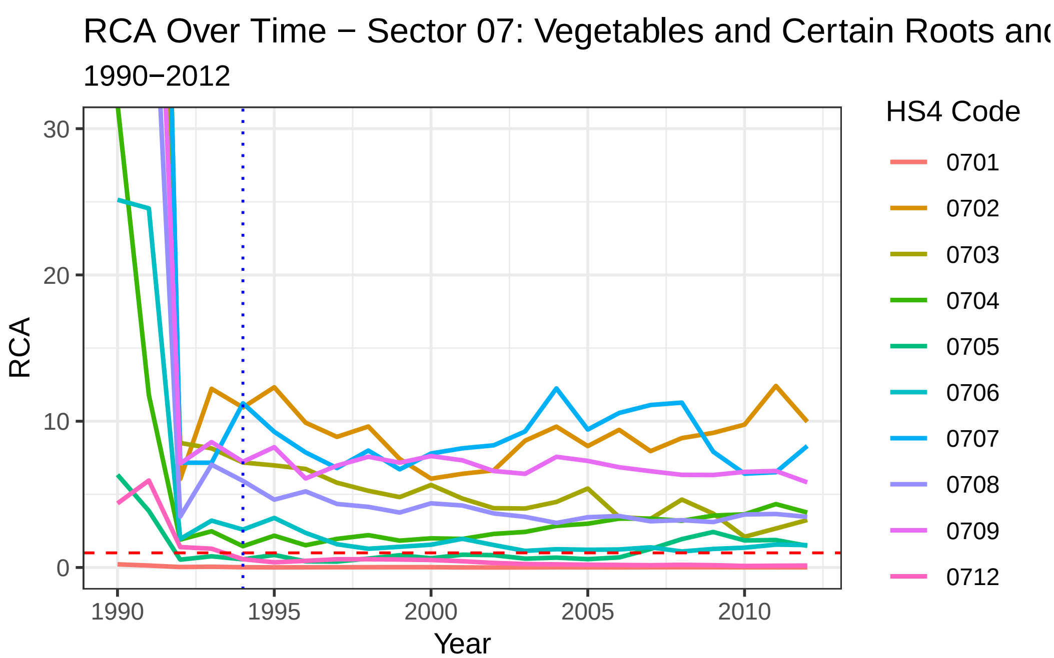


Interestingly, 0503 "Horsehair, waste" has a comparative advantage in that product, but the sector does not have a comparative advantage.

07- Vegetables and Certain Roots and Tubes; Edible

```
rca_sector_07 <- rca_big_df |>
  filter(sector == "07", nchar(cmdCode) == 4)
```

```
ggplot(rca_sector_07, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 30)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 07: Vegetables and Certain Roots and Tubes; Edible",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



Sector Analysis : 07

Now, I are tasked to clean the dataset and calculate an RCA given the dataset with the entire sector. Using the code above, find a way to generalize the RCA calculation to continue the calculation, with an ultimate goal of showing an RCA graph for each product in the sector.

Now, I must use the necessary variables for the RCA. necessary variables are the following: period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. Recall that the RCA variable is calculated with the following:

RCA = [(Country Exports of Product i / Country A Exports of ALL products) / (World Exports of product i / World Exports of all products)]

$$\text{RCA}_i = \frac{\frac{X_{A,i}}{X_{A,\text{all}}}}{\frac{X_{World,i}}{X_{World,\text{all}}}}$$

The RCA variable calculates a share of country exports a product over the world exports of said product. If the RCA variable is greater than, or equal to 1, then the country has a revealed comparative advantage in producing said good. According to neoclassical theory, gains from trade are still probable in comparative advantage. Theory says to export goods in which the country has a comparative advantage in producing.

Now, I have the necessary variables, Country MX Exports of product 0701 -> 0710. I have the data of Mexico's exports of all products, titled `mx_xports_all`. Now, I need World's exports of products 0701 -> 0710. I can do this by using UNCOMTRADE data as Ill. Repeating the same steps, I will grab the necessary variables.

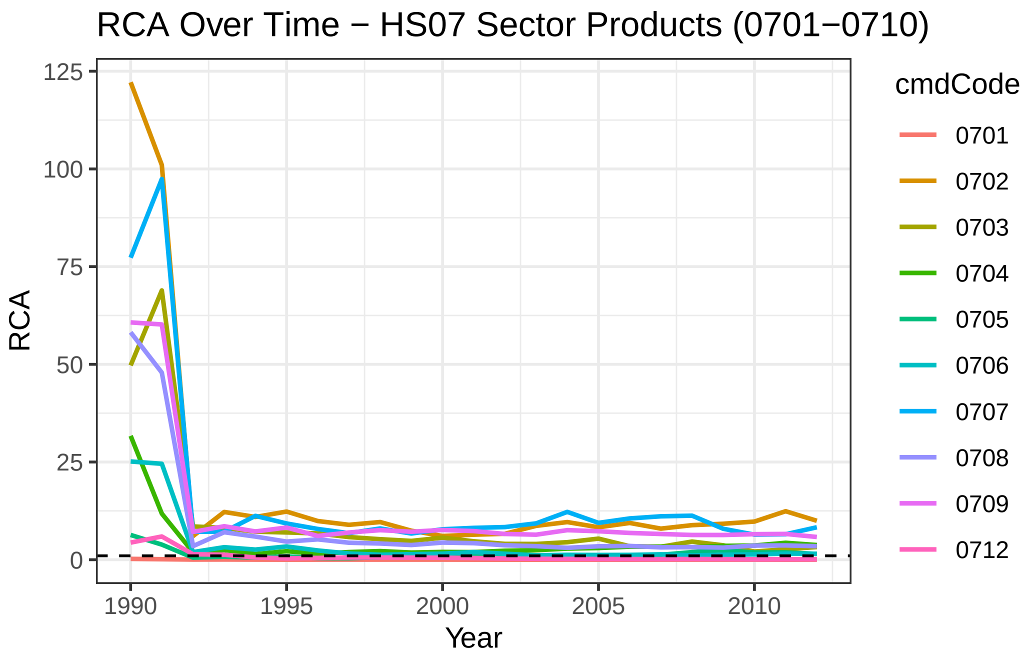
I now have World exports of product 0701 -> 0710 with the same time period of 1992 to 2000. Recall I have datasets of Mexico's exports of all products titled `mx_xports_all` with variables period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. I also have `world_xports_all` which contains world exports of all products with variables period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. HoIver, notice that `world_xports_sector_07` and `mx_xports_sector_07` have the cmdCode, hoIver, they are not aggregated, and their cmdCode varies from 0701 -> 0710. Below is an attempted geenralized calclation of RCA, with a clear difference of grouping by `group_by(period, cmdCode)` due to the nature of the dataframe with varying cmdCode

Let's test this in action to see if it works!

RCA of sector 0701 to 0710.

Visualize over time:

```r
## ggplot to bisualize over time
ggplot(rca_sector_07, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (0701-0710)",
       x = "Year", y = "RCA") +
  theme_bw()
```



I can now develop code to show which products have an RCA > 1 and RCA < 1.

I can do this by running a simple filter:

```
## show which years had an RCA > 1
table_RCA_gt1 <- rca_sector_07 |>
  filter(RCA > 1) |>
  arrange(period, cmdCode)
table_RCA_gt1
```

```
# A tibble: 173 x 11
   period cmdCode country_product_value world_product_value total_country_value
    <dbl> <chr>                   <dbl>               <dbl>               <dbl>
 1   1990 0702                428402080           732965989         26344665088
 2   1990 0703                 80259696           337482057         26344665088
 3   1990 0704                 13531930            89220241         26344665088
 4   1990 0705                  3586518           118230254         26344665088
 5   1990 0706                  6576971            54681540         26344665088
 6   1990 0707                 79643672           215427053         26344665088
 7   1990 0708                 18968764            68132928         26344665088
 8   1990 0709                245167568           843898327         26344665088
 9   1990 0712                  5677731           270895411         26344665088
10   1991 0702                261739168           736978466         26956705792
# i 163 more rows
```

[

Mexico's RCA, Sector 07 (RCA > 1)] Mexico's RCA, Sector 07 (RCA > 1)	
cmdCode	
0701	
0702	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 20
0703	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 20
0704	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 20
0705	19
0706	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 20
0707	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 20
0708	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 20
0709	1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 20
0712	

```
# i 6 more variables: total_world_value <dbl>, RCA <dbl>, has_CA <lgl>,
#   sector <chr>, post_nafta <dbl>, nafta_phase <chr>
```

or by showing which years had an RCA > 1

```
#summarize the comparative advantage by product by grouping by product code
summary_CA_by_product <- rca_sector_07 |>
  group_by(cmdCode) |>
  summarise(
    years_with_CA = paste(period[RCA > 1], collapse = ", "),
    n_years_CA = sum(RCA > 1),
    total_years = n(),
    share_years_CA = n_years_CA / total_years
  )
```

and a publication ready table:

```
summary_CA_by_product |>
  gt() |>
  tab_header(
    title = "Mexico's RCA, Sector 07 (RCA > 1)",
  )
```

I can fix this dataframe by changing the name of the previous one

Sector definitions

07 - VEGETABLES AND CERTAIN ROOTS AND TUBERS; EDIBLE

- 0701 - Potatoes; fresh or chilled
- 0702 - Tomatoes; fresh or chilled
- 0703 - Onions, shallots, garlic, leeks and other alliaceous vegetables; fresh or chilled
- 0704 - Cabbages, caulifloIrs, kohlrabi, kale and similar edible brassicas; fresh or chilled
- 0705 - Lettuce (lactuca sativa) and chicory (cichorium spp.) fresh or chilled
- 0706 - Carrots, turnips, salad beetroot, salsify, celeriac, radishes and similar edible roots; fresh or chilled
- 0707 - Cucumbers and gherkins; fresh or chilled
- 0708 - Leguminous vegetables; shelled or unshelled, fresh or chilled
- 0709 - Vegetables; n.e.c. in chapter 07, fresh or chilled
- 0710 - Vegetables (uncooked or cooked by steaming or boiling in water); frozen
- 0711 - Vegetables provisionally preserved; (e.g. by sulphur dioxide gas, in brine, in sulphur water or in other preservative solutions), but unsuitable in that state for immediate consumption
- 0712 - Vegetables, dried; whole, cut, sliced, broken or in powder, but not further prepared
- 0713 - Vegetables, leguminous; shelled, whether or not skinned or split, dried

08- FRUIT AND NUTS, EDIBLE; PEEL OF CITRUS FRUIT OR MELONS

We omit the following:

- 0812 Fruit and nuts provisionally preserved; e.g. by sulphur dioxide gas, brine, in sulphur water or in other preservative solutions, but unsuitable in that state for immediate consumption
- 0813 Fruit, dried, other than that of heading no. 0801 to 0806; mixtures of nuts or dried fruits of this chapter
- 0814 Peel of citrus fruit or melons (including watermelons); fresh, frozen dried or provisionally preserved in brine, in sulphur water or in other preservative solutions

```
# label: debug-check
exists("rca_big_df")
```

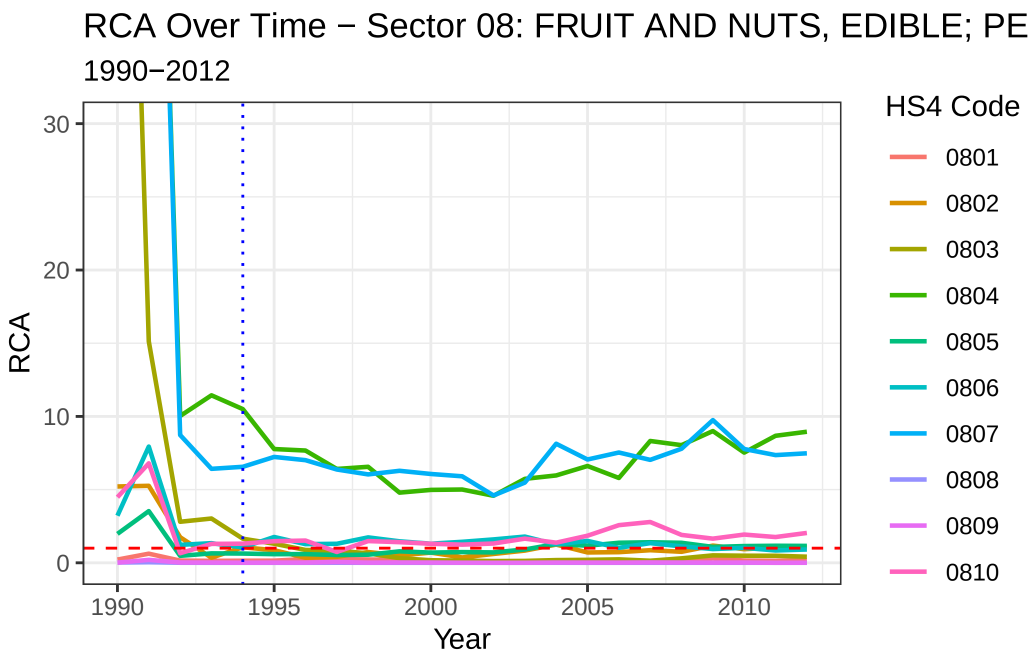
```
[1] TRUE
```

```r
nrow(rca_big_df)
```

```
[1] 4305
```

```r
rca_sector_08 <- rca_big_df |>
  filter(sector == "08", nchar(cmdCode) == 4)
```

```r
ggplot(rca_sector_08, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 30)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 08: FRUIT AND NUTS, EDIBLE; PEEL OF CITRUS FRUIT OR MI
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



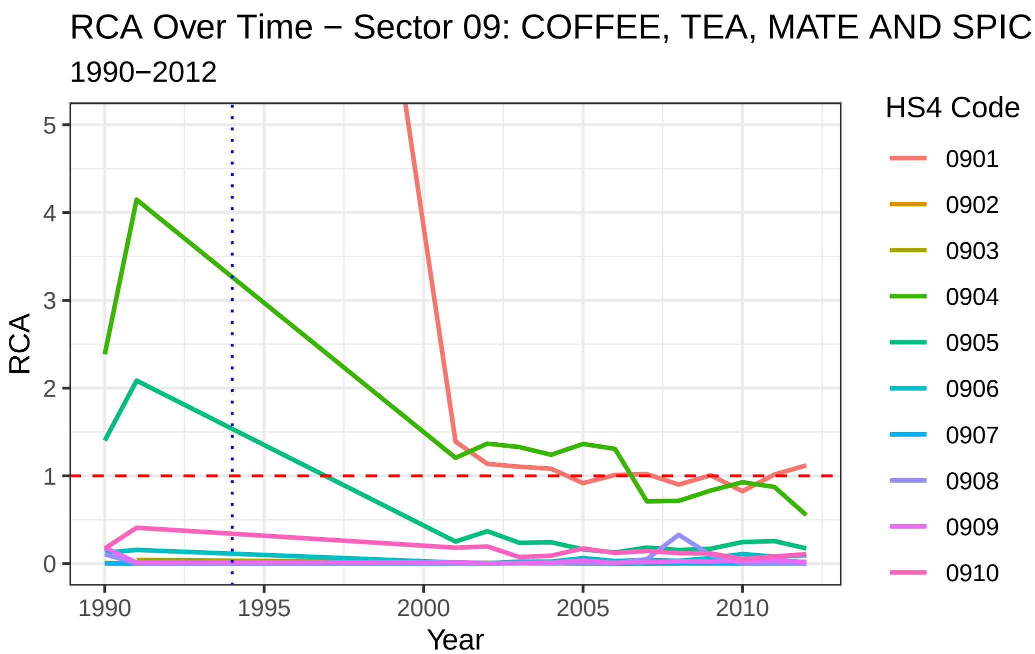
09- COFFEE, TEA, MATE AND SPICES

No 4 digit HS classification code is omitted.

```r
rca_sector_09 <- rca_big_df |>
  filter(sector == "09", nchar(cmdCode) == 4)
```

Plot sector 09

```r
ggplot(rca_sector_09, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 09: COFFEE, TEA, MATE AND SPICES",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



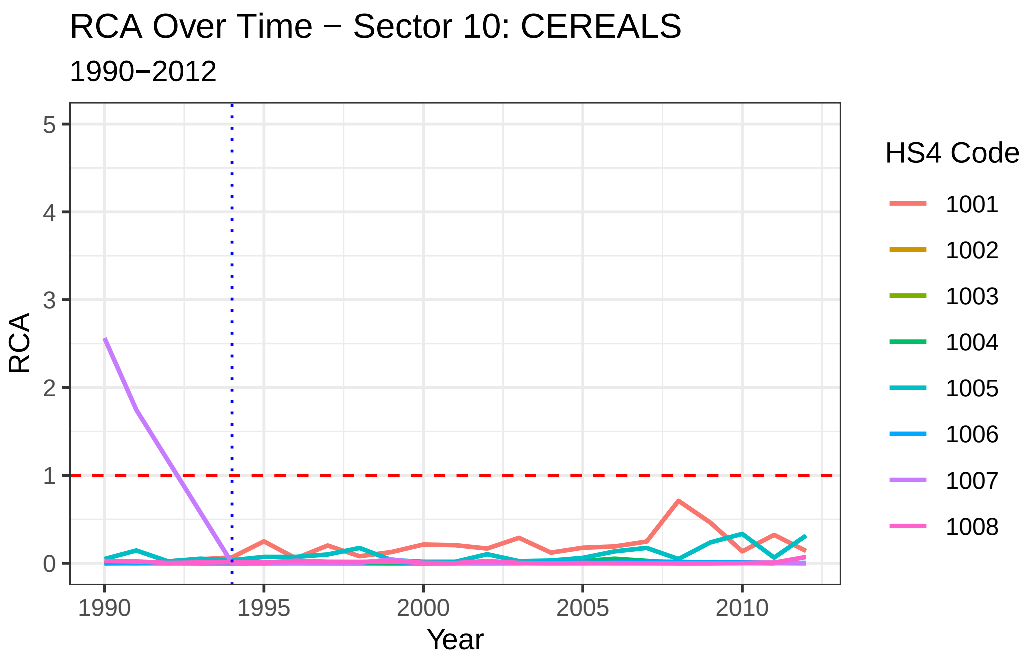
10 CEREALS

No 4 digit HS classification code is omitted.

```r
rca_sector_10 <- rca_big_df |>
  filter(sector == "10", nchar(cmdCode) == 4)
```

Plot sector 10

```r
ggplot(rca_sector_10, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 10: CEREALS",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



11 Milling products; malt; starch, etc.; gluten

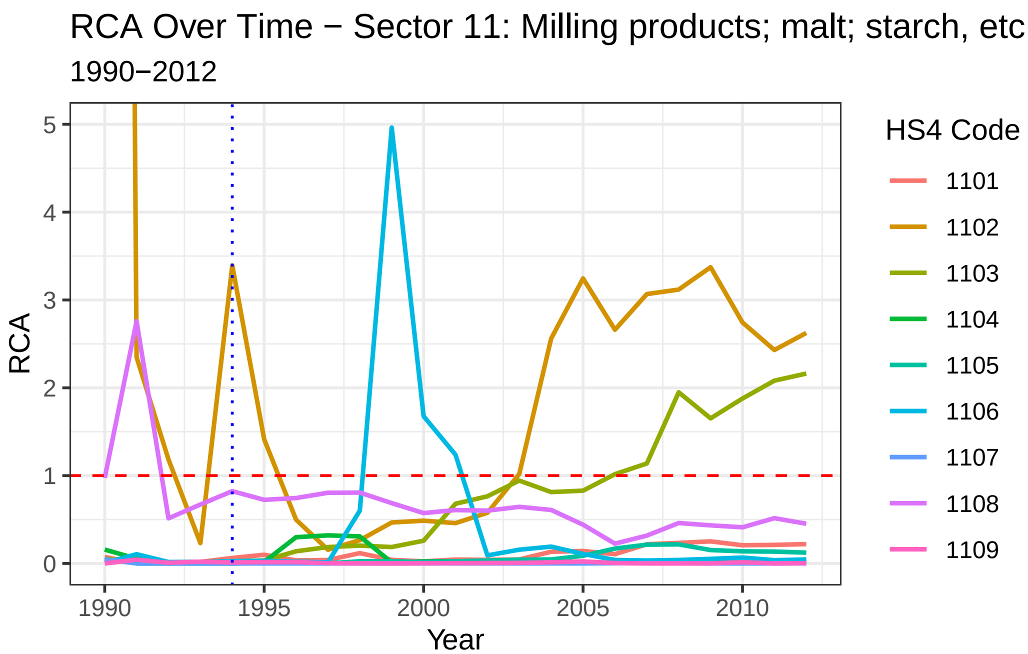
No 4 digit HS classification code is omitted.

No 4 digit HS classification code is omitted.

```r
rca_sector_11 <- rca_big_df |>
  filter(sector == "11", nchar(cmdCode) == 4)
```

Plot sector 11

```r
ggplot(rca_sector_11, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 11: Milling products; malt; starch, etc.; gluten",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



12 Oilseeds; kernels; industrial medicinal plants; feed

We omit the following HS Codes:

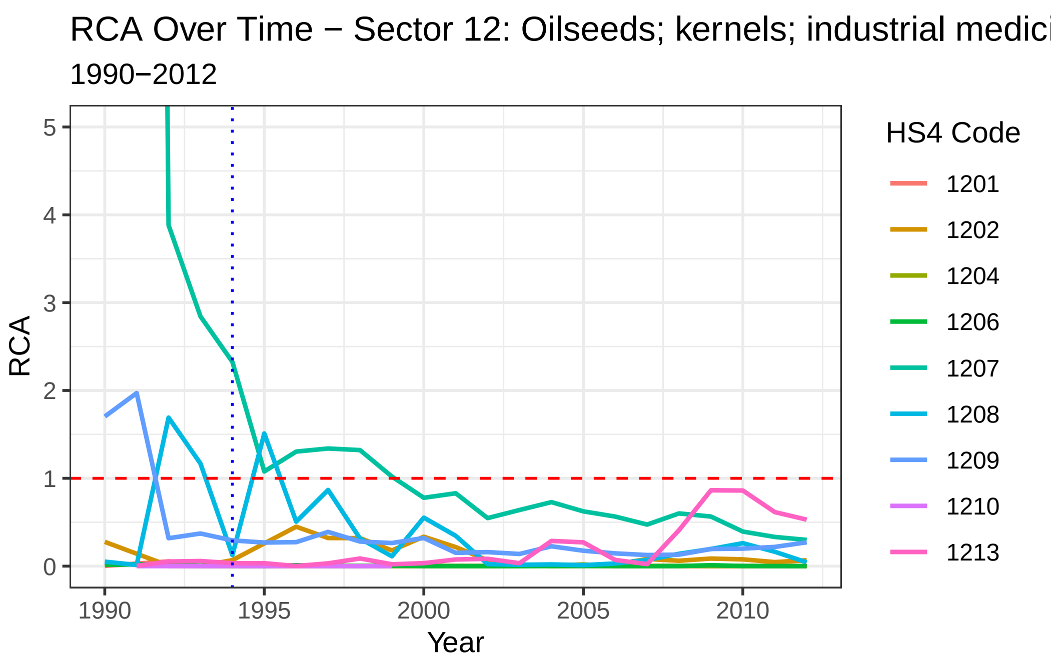
- 1205 Rape or colza seeds; whether or not broken
- 1211 Plants and parts of plants (including seeds and fruits), of a kind used primarily in perfumery, in pharmacy or for insecticidal, fungicidal or similar purposes, fresh, chilled, frozen or dried, whether or not cut, crushed or powdered
- 1212 Locust beans, seaweeds and other algae, sugar beet, sugar cane, fresh, chilled, frozen or dried, whether or not ground; fruit stones, kernels and other vegetable products (including unroasted chicory roots) used primarily for human consumption, n.e.c.

- 1214 Swedes, mangolds, fodder roots, hay, lucerne (alfalfa), clover, sainfoin, forage kale, lupines, vetches and similar forage products, whether or not in the form of pellets

```
rca_sector_12 <- rca_big_df |>
  filter(sector == "12", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_12, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 12: Oilseeds; kernels; industrial medicinal plants; fo
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

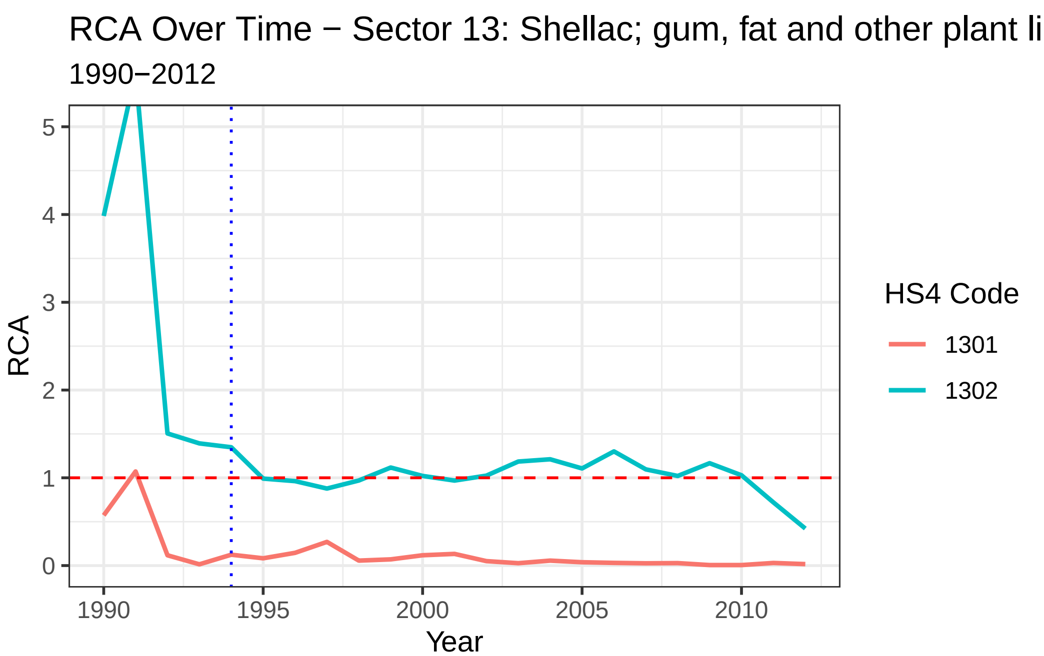


13 Shellac; gum, fat and other plant liquids, juice

```
rca_sector_13 <- rca_big_df |>
  filter(sector == "13", nchar(cmdCode) == 4)
```

Plot sector 13

```r
ggplot(rca_sector_13, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 13: Shellac; gum, fat and other plant liquids, juice"
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



14 Planting materials for planting; other plant products

```r
rca_sector_14 <- rca_big_df |>
  filter(sector == "14", nchar(cmdCode) == 4)
```

Plot sector 14

```r
ggplot(rca_sector_14, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
```

```
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 14: Planting materials for planting; other plant produ
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```

15 Animal and vegetable oils and fats; refined edible oils and fats

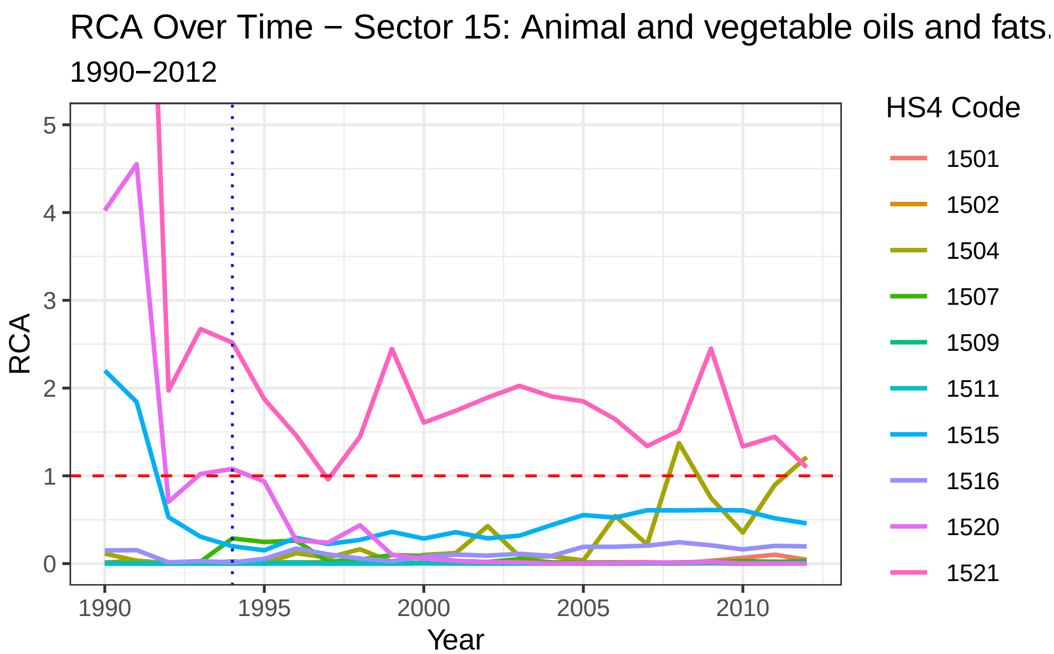
We Omit the following:

- 1503
- 1505
- 1506
- 1508
- 1510
- 1511 to 1514
- 1517 to 1519

```r
rca_sector_15 <- rca_big_df |>
  filter(sector == "15", nchar(cmdCode) == 4)
```

Plot sector 15

```r
ggplot(rca_sector_15, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 15: Animal and vegetable oils and fats...",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



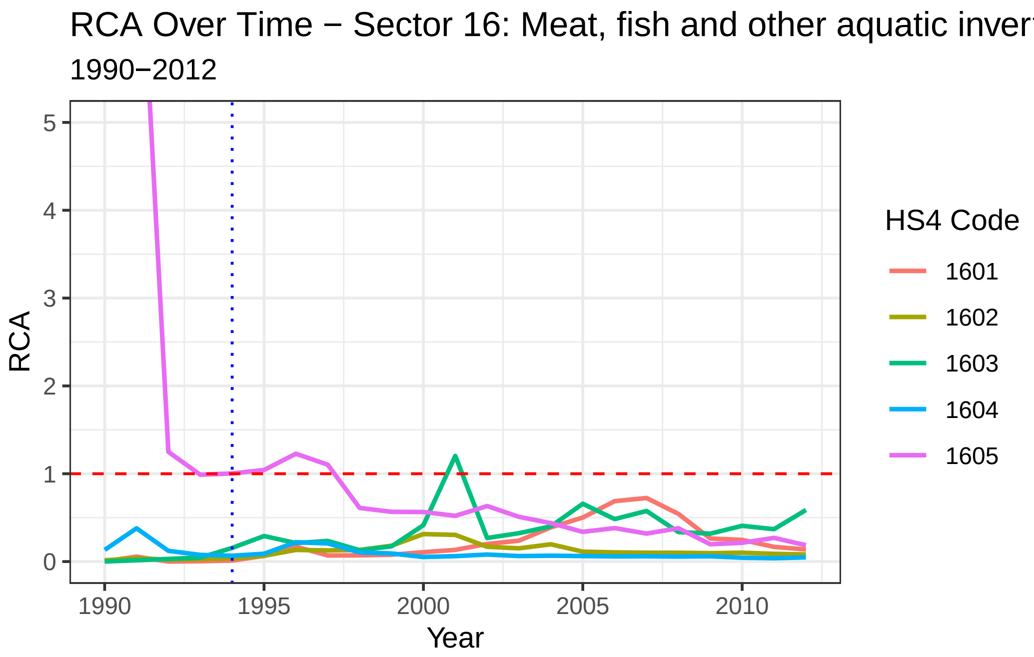
16 Meat, fish and other aquatic invertebrate products

RCA Sector 16

```r
rca_sector_16 <- rca_big_df |>
  filter(sector == "16", nchar(cmdCode) == 4)
```

Plot sector 10

```r
ggplot(rca_sector_16, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 16: Meat, fish and other aquatic invertebrate products
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



17 Sugar and confectionery

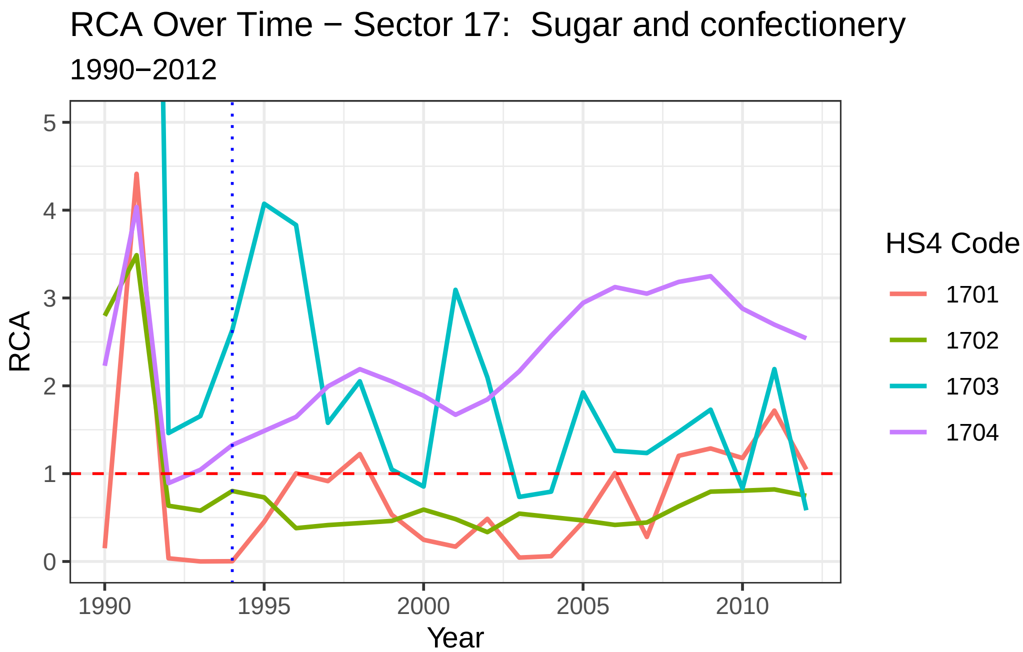
RCA Sector 17

```r
rca_sector_17 <- rca_big_df |>
  filter(sector == "17", nchar(cmdCode) == 4)
```

Plot sector 10

```r
ggplot(rca_sector_17, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
```

```
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 17:  Sugar and confectionery",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



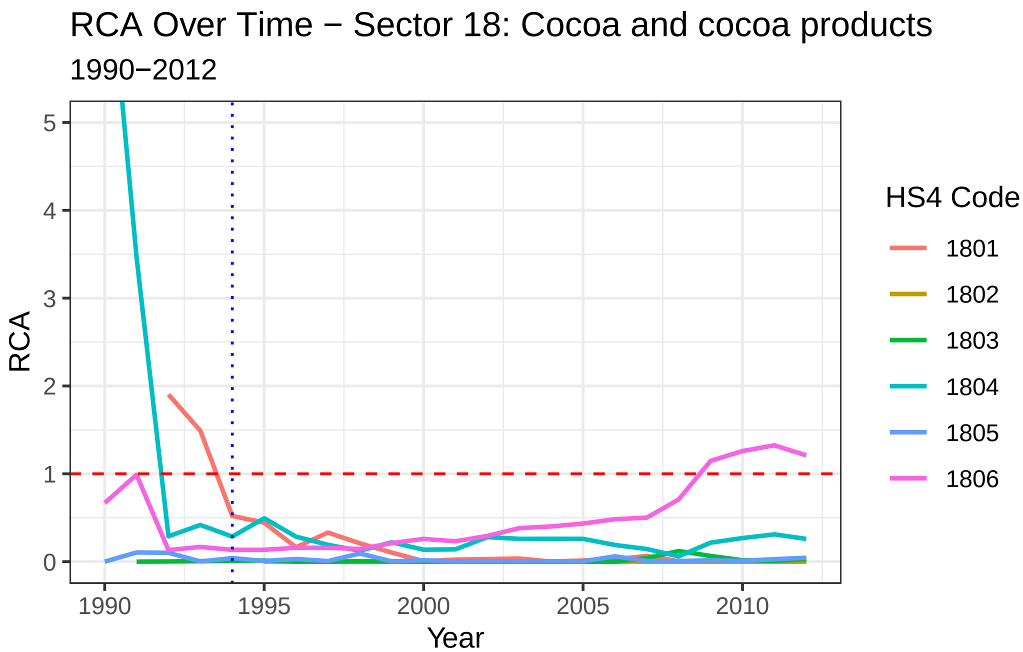
18 Cocoa and cocoa products

RCA Sector 18

```
rca_sector_18 <- rca_big_df |>
  filter(sector == "18", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_18, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 18: Cocoa and cocoa products",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



19 Cereal flour, starch, etc. or dairy products; cakes

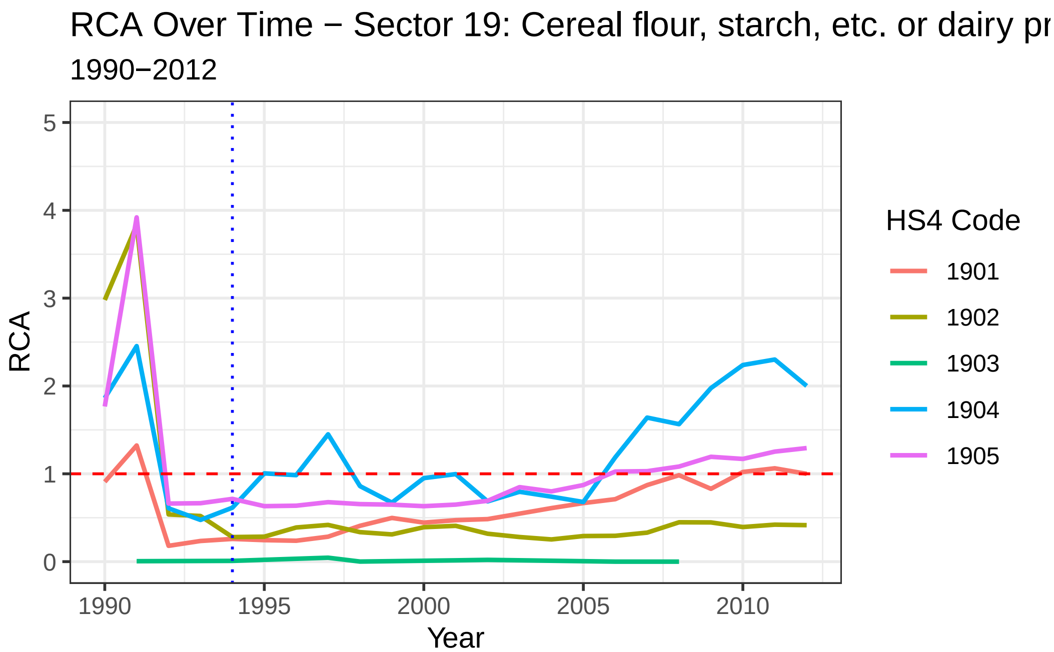
RCA Sector 19

No HS Code is omitted

```
rca_sector_19 <- rca_big_df |>
  filter(sector == "19", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_19, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 19: Cereal flour, starch, etc. or dairy products; cak
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



20 Products of vegetables, fruits or other parts of plants

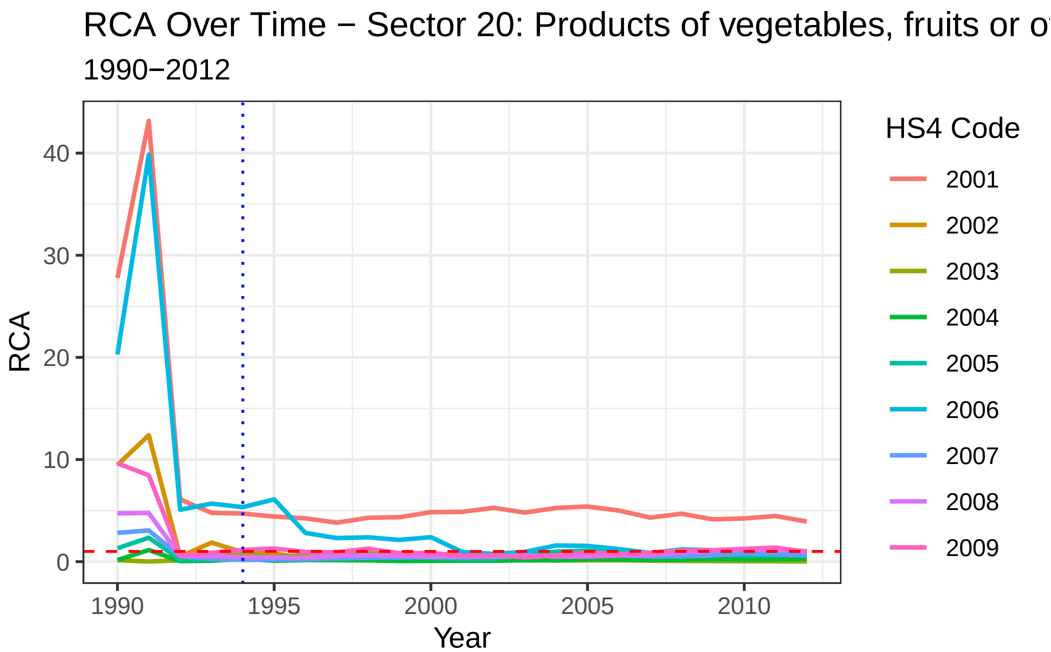
RCA Sector 20

No HS Code is omitted

```
rca_sector_20 <- rca_big_df |>
  filter(sector == "20", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_20, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 43)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 20: Products of vegetables, fruits or other parts of p
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



21 Miscellaneous food

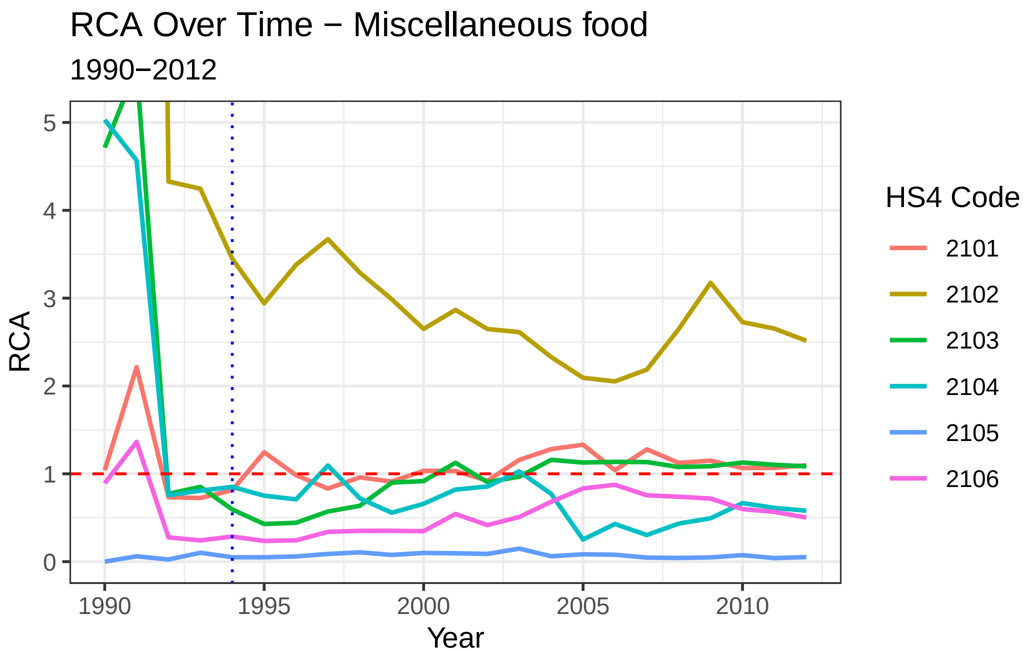
RCA Sector 21

No HS Code is omitted

```
rca_sector_21 <- rca_big_df |>
  filter(sector == "21", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_21, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Miscellaneous food",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



22 Beverages, wine and vinegar

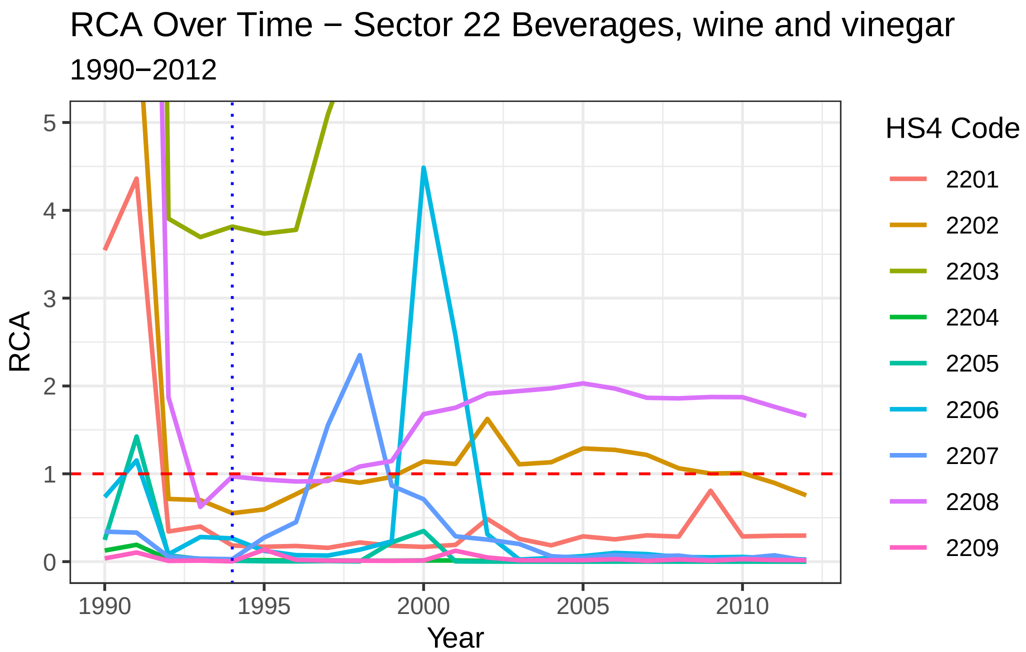
RCA Sector 22

No HS Code is omitted

```
rca_sector_22 <- rca_big_df |>
  filter(sector == "22", nchar(cmdCode) == 4)
```

Plot sector 22

```
ggplot(rca_sector_22, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 22 Beverages, wine and vinegar",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



23 Food industry residues and waste; formulated feed

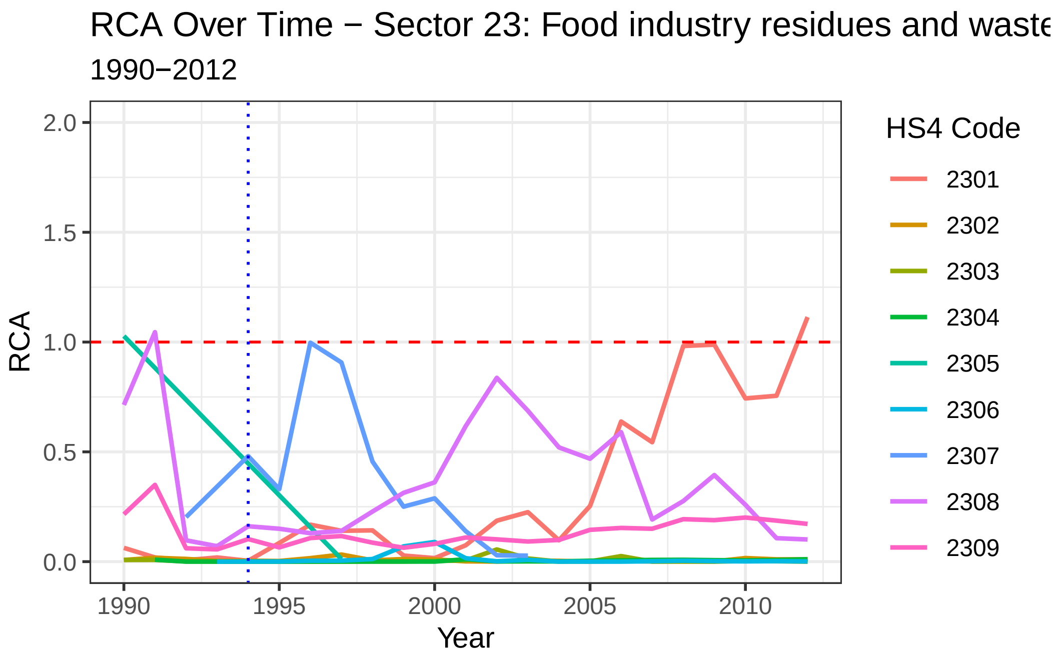
RCA Sector 23

No HS Code is omitted

```
rca_sector_23 <- rca_big_df |>
  filter(sector == "23", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_23, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 2)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 23: Food industry residues and waste; formulated feed'
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



24 Tobacco, tobacco and tobacco substitute products

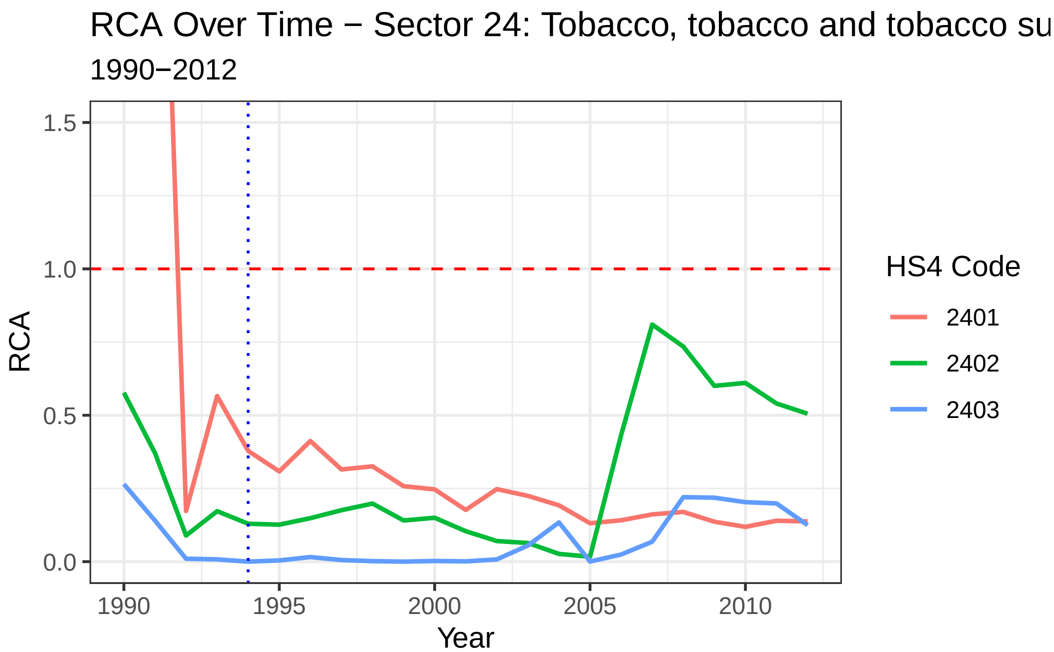
RCA Sector 24

No HS Code is omitted.

```
rca_sector_24 <- rca_big_df |>
  filter(sector == "24", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_24, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 1.5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 24: Tobacco, tobacco and tobacco substitute products"
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



50 Silk

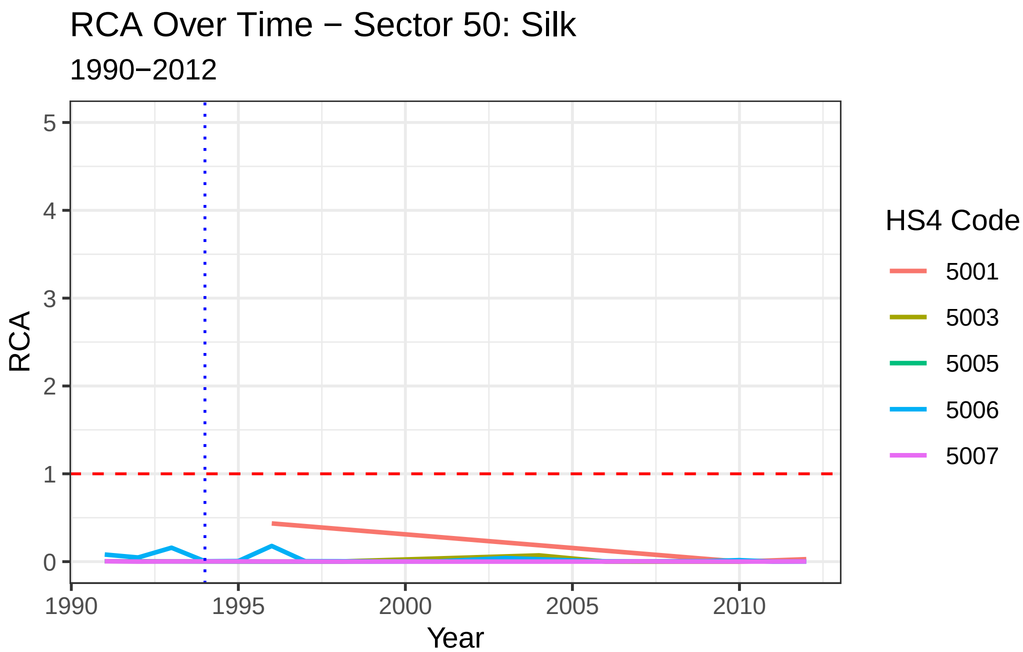
RCA Sector 50

No HS Code is omitted

```
rca_sector_50 <- rca_big_df |>
  filter(sector == "50", nchar(cmdCode) == 4)
```

Plot sector 50

```
ggplot(rca_sector_50, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 50: Silk",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



51 Wool and other animal hair

RCA Sector 51

We omit the following HS Codes

- 5103 Waste of wool or of fine or coarse animal hair, including yarn waste but excluding garnetted stock

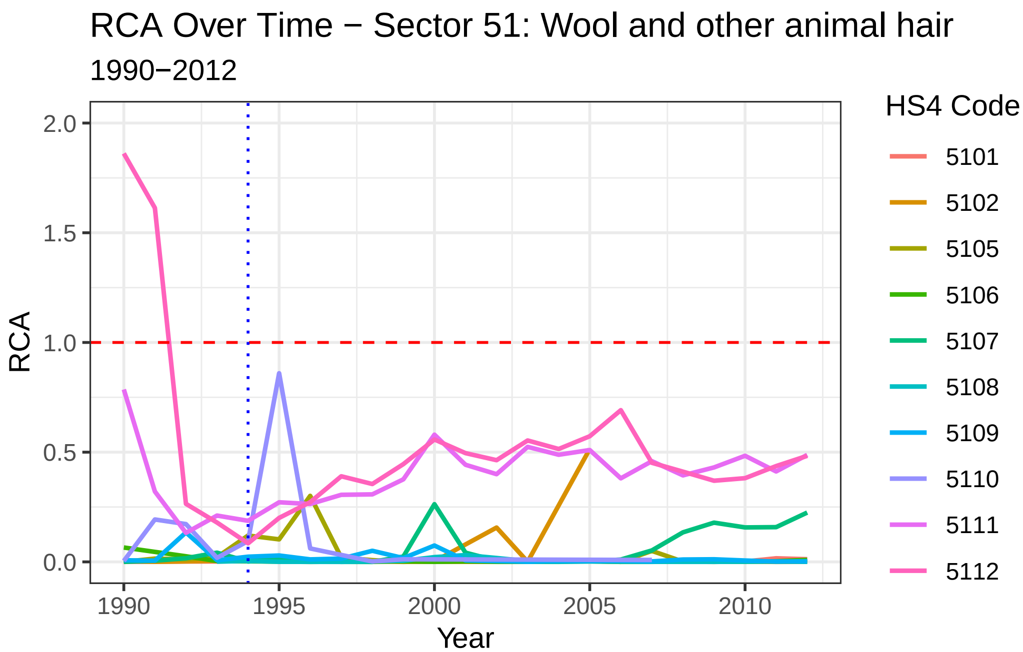
- 5104 Wool, or fine or coarse animal hair; garnetted stock
- 5113 Woven fabrics of coarse animal hair or of horsehair

```
rca_sector_51 <- rca_big_df |>
  filter(sector == "51", nchar(cmdCode) == 4)
```

Plot sector 10

```
ggplot(rca_sector_51, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 2)) + # cap at 10, adjust as needed
```

```r
  labs(title = "RCA Over Time - Sector 51: Wool and other animal hair",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



52 Cotton

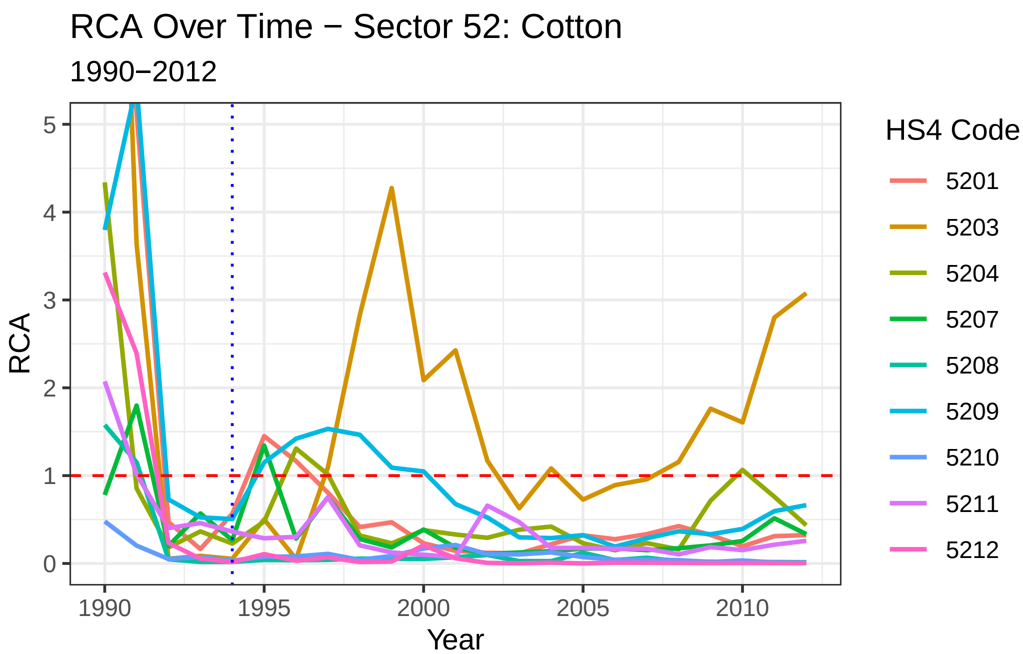
Omit

- 5205 Cotton yarn (other than sewing thread), containing 85% or more by weight of cotton, not put up for retail sale
- 5206 Cotton yarn (other than sewing thread), containing less than 85% by weight of cotton, not put up for retail sale

```r
rca_sector_52 <- rca_big_df |>
  filter(sector == "52", nchar(cmdCode) == 4)
```

Plot sector 10

```r
ggplot(rca_sector_52, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 5)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 52: Cotton",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



53 Other plant fibers

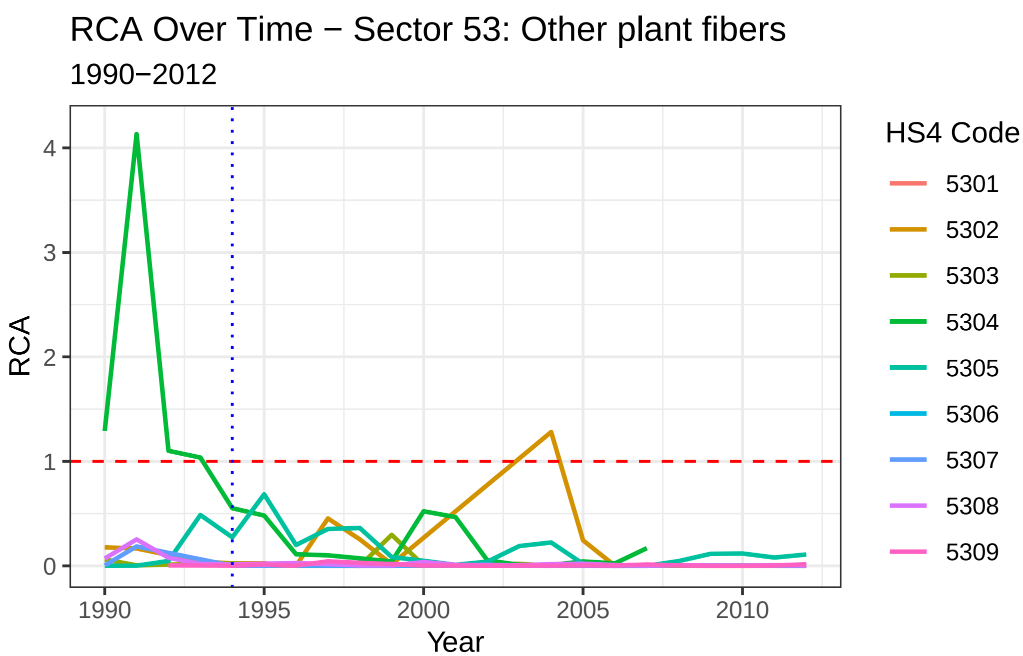
We omit the following:

- 5310 Woven fabrics of jute, other textile bast fibres of heading no. 5303
- 5311 Woven fabrics of other vegetable textile fibres; woven fabrics of paper yarn

```r
rca_sector_53 <- rca_big_df |>
  filter(sector == "53", nchar(cmdCode) == 4)
```

Plot sector 53

```r
ggplot(rca_sector_53, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 1994, linetype = "dotted", color = "blue") +
  coord_cartesian(ylim = c(0, 4.2)) + # cap at 10, adjust as needed
  labs(title = "RCA Over Time - Sector 53: Other plant fibers",
       subtitle = "1990-2012", x = "Year", y = "RCA", color = "HS4 Code") +
  theme_bw()
```



Load Variables for Regression

```r
# load exchange rate dataset and tidy up
exch_rate <- read_csv("regression/variables/exch_rates.csv")

# (https://databank.worldbank.org/reports.aspx?source=2&series=PA.NUS.FCRF&country=MEX)


# load dataset for fdi inflow and tidy up

fdi_inflow <- read_csv("regression/variables/fdi_inflow_to_mx.csv") # worldbank filtered out
```

```
#fdi_inflow <- fdi_inflow |>
  #  select(Time, `Foreign direct investment, net inflows (BoP, current US$) [BX.KLT.DINV.CD
  #rename(period = "Time", fdi_inflow_usd = `Foreign direct investment, net inflows (BoP, cu
 # na.omit(fdi_inflow)


# load datasets for agriculture fdi

agriculture_fdi_pt_1 <- read_csv("regression/variables/agriculture_fdi_pt_1.csv") ## https:/

fdi_agri_mexico_2 <- read_csv("regression/variables/fdi_agri_mexico_2.csv")

agriculture_fdi_pt_1 <- agriculture_fdi_pt_1 |>
  select(period, agri_fdi)


agriculture_fdi_pt_2 <- fdi_agri_mexico_2 |>
  filter(anio >= 2001 & anio <= 2012) |>
  mutate(fn_millones_de_dolares = as.numeric(fn_millones_de_dolares)) |>
  group_by(anio) |>
  summarise(total_agri_fdi = sum(fn_millones_de_dolares, na.rm = TRUE), .groups = "drop") |>
  rename(period = anio)

agriculture_fdi_total <- agriculture_fdi_pt_1 |>
  mutate(agri_fdi = agri_fdi / 1e6) |>
  rename(total_agri_fdi = agri_fdi) |>
  bind_rows(agriculture_fdi_pt_2) |>
  arrange(period)

agriculture_fdi_total
```

```
# A tibble: 23 x 2
   period total_agri_fdi
    <dbl>          <dbl>
 1   1990           61.1
 2   1991           44.9
 3   1992           39.3
 4   1993           34.4
 5   1994           10.3
 6   1995           10
 7   1996           31.7
 8   1997            9.5
```

```
 9   1998           27.6
10   1999           85.5
# i 13 more rows
```

Reg/EDA Df

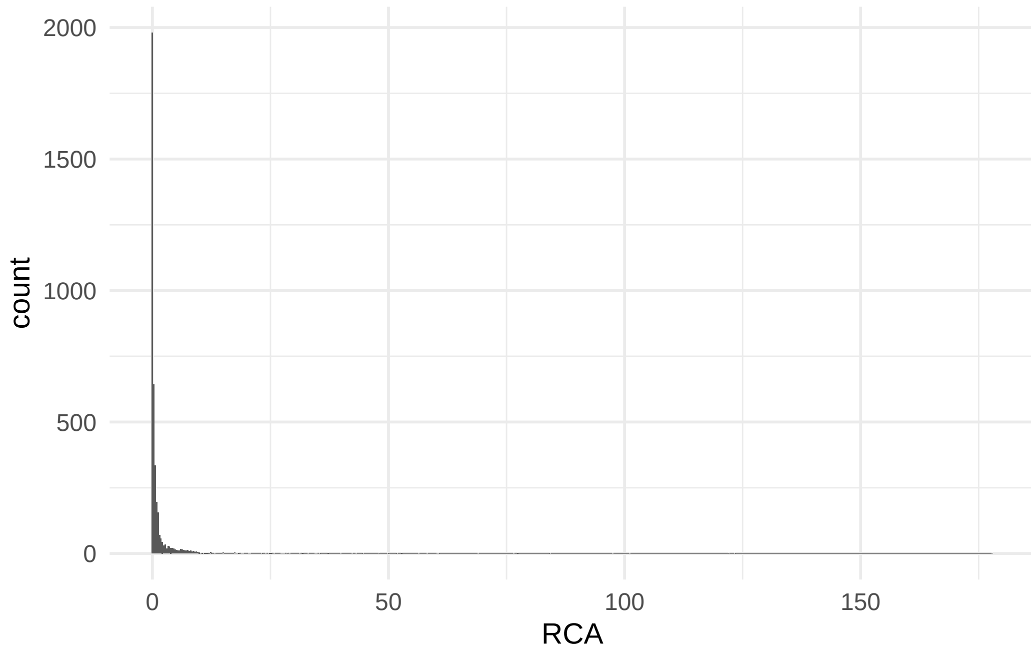
```
## Create main Dataframe
rca_panel <- rca_big_df |>
  left_join(fdi_inflow, by = "period") |>
  left_join(exch_rate, by = "period") |>
  left_join(agriculture_fdi_total, by = "period") |>
  filter(nchar(cmdCode) == 4) |>
  mutate(log_rca = log(RCA))


## store all cmdCodes as a vector

hs4_codes <- rca_panel |>
  pull(cmdCode)
```

EDA

Now that we have our main dataframe `rca_panel`, we can conduct an EDA to see whether or not we need to do any transformations before we model.

1) Distribution

```
rca_panel |>
  select(RCA) |>
  ggplot(aes(x = RCA)) +
  geom_histogram(binwidth = 0.3) +
  theme_minimal()
```



We observe that our RCA is heavily right skewed. Hence, we *might* have to do a log(RCA) transformation due to the nature of the distribution.

2) Time

Show the time frame (1992-2000)

```
rca_panel|>
  distinct(period) |>
  arrange(period)
```

```
# A tibble: 23 x 1
   period
    <dbl>
 1   1990
 2   1991
 3   1992
 4   1993
 5   1994
 6   1995
 7   1996
 8   1997
 9   1998
```

```
10   1999
# i 13 more rows
```

3) Num of products

```
rca_panel |>
  distinct(cmdCode) |> #gives us distinct values for cmdCode
  nrow() #gives us the number of rows
```

```
[1] 194
```

The number of HSCodes that we are analyzing is 194.

4) Balanced

Does every product appear in every year?

```
rca_panel |>
  count(cmdCode) |>
  summary()
```

```
   cmdCode                 n
 Length:194         Min.   : 2.00
 Class :character   1st Qu.:19.25
 Mode  :character   Median :23.00
                    Mean   :20.25
                    3rd Qu.:23.00
                    Max.   :23.00
```

5) Correlations

We must check for correlations between our explanatory variables

```
rca_panel |>
  distinct(period, post_nafta, exch_rate, fdi_inflow_usd, total_agri_fdi) |>
  select(post_nafta, exch_rate, total_agri_fdi )|>
  cor()
```

```
                 post_nafta  exch_rate total_agri_fdi
post_nafta      1.00000000 0.77167205    -0.02806659
exch_rate       0.77167205 1.00000000     0.09090548
total_agri_fdi -0.02806659 0.09090548     1.00000000
```

6) Transform the variables

```r
rca_panel <- rca_panel|>
  mutate(
    log_rca = log(RCA),  # if RCA is right-skewed
    log_exch = log(exch_rate),
    asinh_fdi = asinh(total_agri_fdi) # sinh = (e^x - e^-x)/2
  )
```

7) Num of Prods RCA > 1

```r
rca_greq1 <- rca_panel |>
  filter(RCA >= 1)

## what are the products with RCA >= 1? regardless of time

rca_greq1 |>
  select(cmdCode) |>
  unique()
```

```
# A tibble: 95 x 1
   cmdCode
   <chr>
 1 0102
 2 0101
 3 0202
 4 0205
 5 0207
 6 0209
 7 0201
 8 0502
 9 0503
10 0508
# i 85 more rows
```

8) Products which did not start off with comparative advantage before nafta, and ended up with one after?

```
#products that did not have a CA before NAFTA
no_ca_before <- rca_panel |>
  filter(post_nafta == 0) |> ## before nafta
  group_by(cmdCode) |> #group by commodity Code
  summarise(had_ca = any(has_CA)) |> # if any values in has_CA is true, i.e does have a CA
  filter(had_ca == FALSE) |>
  pull(cmdCode)

no_ca_before
```

```
  [1] "0103" "0104" "0105" "0201" "0203" "0206" "0208" "0209" "0210" "0401"
 [11] "0402" "0403" "0404" "0405" "0406" "0407" "0408" "0410" "0501" "0504"
 [21] "0505" "0506" "0507" "0701" "0801" "0808" "0809" "0902" "0903" "0906"
 [31] "0907" "0908" "0909" "0910" "1001" "1003" "1004" "1005" "1006" "1008"
 [41] "1101" "1103" "1104" "1105" "1106" "1107" "1109" "1201" "1202" "1204"
 [51] "1206" "1210" "1213" "1401" "1501" "1502" "1504" "1507" "1509" "1511"
 [61] "1516" "1601" "1602" "1603" "1604" "1803" "1805" "1806" "1903" "2003"
 [71] "2105" "2204" "2207" "2209" "2301" "2302" "2303" "2304" "2306" "2307"
 [81] "2309" "2402" "2403" "5005" "5006" "5007" "5101" "5102" "5105" "5106"
 [91] "5107" "5108" "5109" "5110" "5111" "5210" "5301" "5302" "5303" "5305"
[101] "5307" "5308" "5309"
```

```
# of those products which ones developed a comparative advantage after 1994?
gained_ca_after <- rca_panel  |>
  filter(post_nafta == 1, cmdCode %in% no_ca_before) |>
  group_by(cmdCode) |>
  summarise(gained_ca = any(has_CA)) |>
  filter(gained_ca == TRUE) |>
  pull(cmdCode)

gained_ca_after
```

```
 [1] "0201" "0209" "1103" "1106" "1504" "1603" "1806" "2207" "2301" "5302"
```

9) Descriptive overview

```
rca_panel |>
  group_by(post_nafta) |>
  summarise(
    mean_rca = mean(RCA, na.rm = TRUE),
    median_rca = median(RCA, na.rm = TRUE),
    sd_rca = sd(RCA, na.rm = TRUE),
    min_rca = min(RCA, na.rm = TRUE),
    max_rca = max(RCA, na.rm = TRUE),
    pct_with_RCA = mean(has_CA)*100
  )
```

```
# A tibble: 2 x 7
  post_nafta mean_rca median_rca sd_rca      min_rca max_rca pct_with_RCA
       <dbl>    <dbl>      <dbl>  <dbl>        <dbl>   <dbl>        <dbl>
1          0    4.61       0.203  15.3  0.000000264    178.          32.9
2          1    0.836      0.136   2.19 0.0000000326    37.2          18.5
```

10) Prevalence in RCA by sector

```
rca_panel |>
  group_by(sector, post_nafta) |>
  summarise(
    n_products= n_distinct(cmdCode), #counts num of distinct values
    pct_with_CA = mean(has_CA) * 100,
    mean_rca = mean(RCA, na.rm = TRUE)
  )
```

```
# A tibble: 52 x 5
# Groups:   sector [26]
   sector post_nafta n_products pct_with_CA mean_rca
   <chr>       <dbl>      <int>       <dbl>    <dbl>
 1 01              0          5       29.4    7.53
 2 01              1          5       23.8    0.935
 3 02              0          9       20.6    0.743
 4 02              1         10        6.67   0.326
 5 04              0          9        0      0.0155
 6 04              1          9        0      0.0454
```

```
 7 05               0           9        28.1    3.33
 8 05               1           9         7.26   0.458
 9 07               0          10        85     23.7
10 07               1          10        73.2    3.89
# i 42 more rows
```

11) Year by Year

```
rca_panel |>
  group_by(period) |>
  summarise(
    n_with_CA = sum(has_CA),
    total_products = n_distinct(cmdCode),
    pct_with_CA = mean(has_CA)*100
  )
```

```
# A tibble: 23 x 4
   period n_with_CA total_products pct_with_CA
    <dbl>     <int>          <int>       <dbl>
 1   1990        72            166        43.4
 2   1991        79            168        47.0
 3   1992        31            158        19.6
 4   1993        32            159        20.1
 5   1994        31            167        18.6
 6   1995        34            167        20.4
 7   1996        29            174        16.7
 8   1997        29            173        16.8
 9   1998        29            173        16.8
10   1999        28            176        15.9
# i 13 more rows
```

12) Volatility of RCA

```
 rca_panel |>
  arrange(cmdCode, period) |>
  group_by(cmdCode) |>
  summarise(
    switches = sum(has_CA != lag(has_CA), na.rm = TRUE)
  ) |>
```

```r
  filter(switches > 1) |>
  arrange(desc(switches))
```

```
# A tibble: 49 x 2
   cmdCode switches
   <chr>      <int>
 1 1701           9
 2 0802           8
 3 1703           7
 4 2009           7
 5 0901           6
 6 1904           6
 7 2101           6
 8 5203           6
 9 1302           5
10 2006           5
# i 39 more rows
```

13 Regressor variation

```r
rca_panel |>
  group_by(nafta_phase) |>
  summarise(
    mean_exch_rate = mean(exch_rate, na.rm = TRUE),
    mean_fdi = mean(fdi_inflow_usd, na.rm = TRUE),
    mean_agri_fdi = mean(total_agri_fdi, na.rm = TRUE),
    n_obs = n()
  )
```

```
# A tibble: 3 x 5
  nafta_phase mean_exch_rate     mean_fdi mean_agri_fdi n_obs
  <chr>                <dbl>        <dbl>         <dbl> <int>
1 post                  7.66 12497714312.          38.7  1195
2 pre                   3.01  4038717358.          45.1   651
3 <NA>                 11.4  24524747142.          44.1  2083
```

14) RCA Trajectories

```
get_rca_trajectory <- function(tempcmdcde) {
  rca_panel |>
    filter(cmdCode == tempcmdcde) |>
    select(period, cmdCode, RCA, has_CA, exch_rate, fdi_inflow_usd) |>
    arrange(period) |>
    mutate(rca_change = RCA - lag(RCA))  # how much RCA shifted each year
}

map(unique(rca_panel$cmdCode), get_rca_trajectory) |>
  list_rbind()
```

```
# A tibble: 3,929 x 7
   period cmdCode    RCA has_CA exch_rate fdi_inflow_usd rca_change
    <dbl> <chr>    <dbl> <lgl>      <dbl>          <dbl>      <dbl>
 1   1990 0101    0.804  FALSE       2.81     2634000000    NA
 2   1991 0101    1.22   TRUE        3.02     4762000000     0.416
 3   1992 0101    0.268  FALSE       3.09     4393000000    -0.952
 4   1993 0101    0.204  FALSE       3.12     4389000000    -0.0635
 5   1994 0101    0.0470 FALSE       3.38    10972500000    -0.157
 6   1995 0101    0.236  FALSE       6.42     9526290000     0.189
 7   1996 0101    0.125  FALSE       7.60     9185600000    -0.111
 8   1997 0101    0.0370 FALSE       7.92    12829800000    -0.0875
 9   1998 0101    0.0747 FALSE       9.14    12756764558     0.0376
10   1999 0101    0.0488 FALSE       9.56    13941043232    -0.0258
# i 3,919 more rows
```

15) How long did it take for products to gain RCA after NAFTA?

```
rca_panel |>
  filter(post_nafta == 1, has_CA == TRUE) |>
  group_by(cmdCode) |>
  summarise(first_ca_year = min(period)) |>
  mutate(years_after_nafta = first_ca_year - 1994) |>
  arrange(years_after_nafta) |>
  arrange(desc(years_after_nafta))
```

```
# A tibble: 63 x 3
```

```
   cmdCode first_ca_year years_after_nafta
   <chr>           <dbl>             <dbl>
 1 0201             2012                18
 2 2301             2012                18
 3 1901             2010                16
 4 1806             2009                15
 5 2008             2009                15
 6 1504             2008                14
 7 0705             2007                13
 8 1103             2006                12
 9 1905             2006                12
10 2005             2005                11
# i 53 more rows
```

16) Consistency in RCAS

```r
rca_panel |>
  filter(post_nafta == 1) |>
  group_by(cmdCode) |>
  summarise(
    years_with_CA = sum(has_CA),
    total_years = n(),
    consistency = years_with_CA/total_years
  ) |>
  mutate(ca_type = case_when(
    consistency == 1 ~ "Persistent CA",
    consistency >= 0.5 ~ "Majority CA",
    consistency > 0 & consistency < 0.5 ~ "Sporadic CA",
    consistency == 0 ~ "No CA"
  )) |>
  count(ca_type)
```

```
# A tibble: 4 x 2
  ca_type           n
  <chr>         <int>
1 Majority CA      16
2 No CA           131
3 Persistent CA    16
4 Sporadic CA      31
```

17) Avg RCA By sector

```
rca_panel |>
  group_by(sector, post_nafta) |>
  summarise(
    mean_rca = mean(RCA),
    median_rca  = median(RCA),
    n_products = n_distinct(cmdCode)
  ) |>
  pivot_wider(
    names_from = post_nafta,
    values_from = c(mean_rca, median_rca, n_products),
    names_prefix = "post_"
  )
```

```
# A tibble: 26 x 7
# Groups:   sector [26]
   sector mean_rca_post_0 mean_rca_post_1 median_rca_post_0 median_rca_post_1
   <chr>            <dbl>           <dbl>             <dbl>             <dbl>
 1 01               7.53            0.935             0.0203            0.0116
 2 02               0.743           0.326             0.0522            0.0551
 3 04               0.0155          0.0454            0.00275           0.0148
 4 05               3.33            0.458             0.511             0.112
 5 07              23.7             3.89              7.13              3.28
 6 08              11.0             1.92              1.86              0.850
 7 09               3.49            0.269             0.156             0.0415
 8 10               0.212           0.0525            0.00620           0.00576
 9 11               1.98            0.474             0.0201            0.107
10 12               3.26            0.212             0.0516            0.0462
# i 16 more rows
# i 2 more variables: n_products_post_0 <int>, n_products_post_1 <int>
```

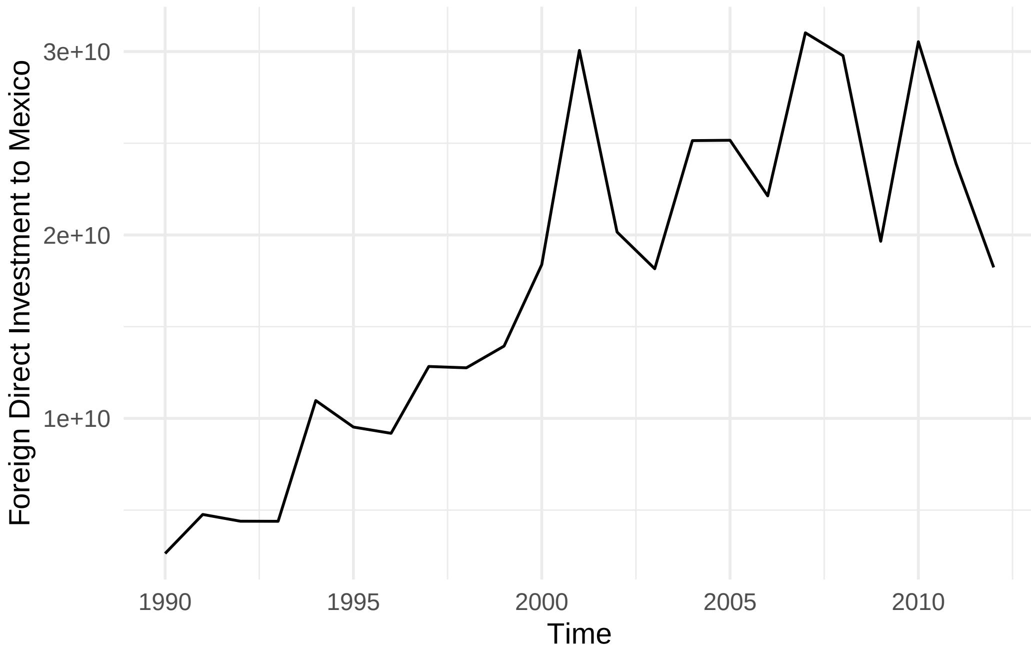
18) Largest RCA Gains?

```
rca_panel |>
  group_by(sector, post_nafta) |>
  summarise(mean_rca = mean(RCA)) |>
  pivot_wider(names_from = post_nafta, values_from = mean_rca, names_prefix = "period_") |>
  mutate(rca_gain = period_1 - period_0) |>
  arrange(desc(rca_gain))
```

```
# A tibble: 26 x 4
# Groups:   sector [26]
   sector period_0 period_1 rca_gain
   <chr>     <dbl>    <dbl>    <dbl>
 1 04       0.0155   0.0454   0.0299
 2 23       0.161    0.147   -0.0135
 3 50       0.0425   0.0190  -0.0236
 4 51       0.175    0.125   -0.0507
 5 10       0.212    0.0525  -0.159
 6 53       0.407    0.0590  -0.348
 7 02       0.743    0.326   -0.417
 8 24       0.646    0.195   -0.451
 9 19       1.35     0.702   -0.647
10 16       0.966    0.284   -0.683
# i 16 more rows
```

19) FDI over time

```r
ggplot(data = rca_panel, aes(x = period, y = fdi_inflow_usd)) +
  geom_line() +
  theme_minimal() +
  xlab("Time") +
  ylab("Foreign Direct Investment to Mexico")
```




```r
ggplot(data = rca_panel, aes(x = period, y = total_agri_fdi)) +
  geom_line() +
  theme_minimal() +
  xlab("Time") +
  ylab("Agriculture Foreign Direct Investment to Mexico")
```

MODELS

Model 1 Fixed Effects Regresission

```r
model <- feols(RCA ~ post_nafta + log(exch_rate) + log(fdi_inflow_usd) | cmdCode,
               data = rca_panel,
               cluster = ~cmdCode)

summary(model)
```

```
OLS estimation, Dep. Var.: RCA
Observations: 3,929
Fixed-effects: cmdCode: 194
Standard-errors: Clustered (cmdCode)
                     Estimate Std. Error  t value   Pr(>|t|)
post_nafta          -3.049748   0.599862 -5.08409 8.7037e-07 ***
log(exch_rate)       0.241514   0.101063  2.38973 1.7823e-02 *
log(fdi_inflow_usd) -0.566835   0.150788 -3.75915 2.2585e-04 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 5.36165     Adj. R2: 0.321928
                Within R2: 0.060454
```

Model 4 USE THIS

```r
model4 <- feols(log(RCA) ~ post_nafta + log(exch_rate) + log(total_agri_fdi)|cmdCode,
               data = rca_panel,
               cluster = ~cmdCode)

summary(model4)
```

```
OLS estimation, Dep. Var.: log(RCA)
Observations: 3,581
Fixed-effects: cmdCode: 194
Standard-errors: Clustered (cmdCode)
                     Estimate Std. Error   t value   Pr(>|t|)
post_nafta          -0.622017   0.154363 -4.029563 8.0338e-05 ***
log(exch_rate)       0.038146   0.147407  0.258781 7.9608e-01
```

```
log(total_agri_fdi)  0.042889   0.025788  1.663147 9.7906e-02 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 1.50984     Adj. R2: 0.759364
                Within R2: 0.023345
```

Model 4 Specification

$$\log(\text{RCA}_{it}) = \beta_1 \text{PostNAFTA}_t + \beta_2 \log(\text{ExchRate}_t) + \beta_3 \log(\text{AgriFFDI}_t) + \alpha_i + \varepsilon_{it}$$

where i indexes HS4 product codes, t indexes years, and:

- $\log(\text{RCA}_{it})$ is the log of the Balassa (1965) Revealed Comparative Advantage index for product i in year t
- PostNAFTA_t is a binary indicator equal to 1 for $t \geq 1994$ and 0 otherwise
- $\log(\text{ExchRate}_t)$ is the log of the Mexican peso–US dollar exchange rate (pesos per USD)
- $\log(\text{AgriFFDI}_t)$ is the log of total foreign direct investment inflows into Mexico's agricultural sector
- α_i are product-level fixed effects absorbing time-invariant heterogeneity across HS4 codes
- ε_{it} is the idiosyncratic error term, with standard errors clustered at the α_i (cmdCode) level

The model is estimated via Ordinary Least Squares with product-level fixed effects using the `feols` function from the `fixest` package in R [@bergé2018]. The within-transformation removes α_i, so β_1 identifies the average change in log(RCA) associated with NAFTA implementation, holding exchange rate and agricultural FDI constant, within product codes over time.

Interpretation

The interpretation of model 4 will be as follows.

Log(Exchange Rate)

A 1 percent increase in the exchange rate is assoiated with a 0.43 percent increase in RCA (at the 1 percent level)

nafta_phaseearly

Log(RCA) was 0.41 (units) lower than in the pre-NAFTA baseline, holding exchange rate and agriculture_FDI constant (significant at the 5 percent level)

** asinh(agri_fdi)**

A rise in agriculture FDI is associated with a decrease in RCA (counterintuitive).

A 1 percent increase in FDI in Mexico's Agriculture Sector is linked with a -0.142 percent decrease in RCA, holding all else constant.

- FDI may have been used to increase domestic production rather than building export competitiveness in the global market.
- FDI may flow into sectors that lose comparative advantage.
- Aggregate agriculture FDI may have been flowing disproportionately into subsectors (i.e fertilizer, etc..) so FDI is not actually reaching the products driving the RCA variation.

TABLES NEEDS FIXING

```r
# Main Model Table (Claude AI)

model4_tidy <- broom::tidy(model4, conf.int = TRUE) |>
  mutate(
    term = case_when(
      term == "nafta_phaseearly" ~ "NAFTA Phase: Early",
      term == "nafta_phasepost"  ~ "NAFTA Phase: Post",
      term == "log(exch_rate)"   ~ "Log Exchange Rate",
      term == "asinh(agri_fdi)"  ~ "IHS Agricultural FDI",
      TRUE ~ term
    ),
    stars = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.1  ~ "*",
      TRUE           ~ ""
    ),
    estimate_stars = paste0(round(estimate, 3), stars),
    std.error = paste0("(", round(std.error, 3), ")")
  ) |>
  select(term, estimate_stars, std.error, p.value)

# Build gt table
model4_tidy |>
  gt() |>
  tab_header(
    title    = "Fixed Effects Estimation of NAFTA's Impact on RCA",
    subtitle = "Dependent variable: Log(RCA)"
```

```
  ) |>
  cols_label(
    term = "Variable",
    estimate_stars = "Coefficient",
    std.error = "Std. Error",
    p.value = "p-value"
  ) |>
  fmt_number(
    columns = p.value,
    decimals = 3
  ) |>
  tab_spanner(
    label = "Log(RCA)",
    columns = c(estimate_stars, std.error, p.value)
  ) |>
  tab_footnote(
    footnote = "*** p < 0.01, ** p < 0.05, * p < 0.1"  #shows statistical significance
  ) |>
  tab_footnote(
    footnote = "Clustered standard errors at the cmdCode level. Reference category: Pre-NAFTA
  ) |>
  tab_source_note(
    source_note = "Source: UN Comtrade, Secretaría de Economía, Banco de México."
  ) |>
  tab_options(
    heading.title.font.size = 14,
    heading.subtitle.font.size = 12,
    column_labels.font.weight  = "bold",
    table.width = pct(100)
  )
```

Transition table

```
#Generated by Claude

transition_table <- rca_panel |>
  filter(post_nafta == 0) |>
  group_by(cmdCode) |>
  summarise(had_ca = any(has_CA)) |>
  inner_join(
```

[

Fixed Effects Estimation of NAFTA's Impact on RCA
Dependent variable: Log(RCA)] Fixed Effects Estimation of NAFTA's Impact on RCA

		Log(RCA)	
	Variable	Coefficient	Std. Error
Dependent variable: Log(RCA)	post_nafta	-0.622***	(0.154)
	Log Exchange Rate	0.038	(0.147)
	log(total_agri_fdi)	0.043*	(0.026)

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Clustered standard errors at the cmdCode level. Reference category: Pre-NAFTA.
Source: UN Comtrade, Secretaría de Economía, Banco de México.

```
    rca_panel |>
      filter(post_nafta == 1) |>
      group_by(cmdCode) |>
      summarise(gained_ca = any(has_CA)),
    by = "cmdCode"
  ) |>
  mutate(transition = case_when(
    !had_ca &  gained_ca ~ "Gained RCA post-NAFTA",
    had_ca  & !gained_ca ~ "Lost RCA post-NAFTA",
    had_ca  &  gained_ca ~ "Maintained RCA",
    !had_ca & !gained_ca ~ "Never had RCA"
  )) |>
  count(transition) |>
  rename(
    `Transition Status` = transition,
    `Number of Products` = n
  )


transition_table |>
  gt() |>
  tab_header(
    title    = "Comparative Advantage Transitions Around NAFTA (1994)",
    subtitle = "Mexico's Agricultural Sector - HS Codes 01-53"
  ) |>
  tab_spanner(
```

[

Comparative Advantage Transitions Around NAFTA (1994)

Mexico's Agricultural Sector — HS Codes 01–53] Comparative Advantage Transitions Around NAFTA (1994)

Mexico's Agricultural Sector — HS Codes 01–53

Pre vs. Post-NAFTA CA Status
Transition Status
Gained RCA post-NAFTA
Lost RCA post-NAFTA
Maintained RCA
Never had RCA

Source: UN Comtrade. RCA calculated using Balassa (1965) index. CA defined as RCA 1.

```
    label   = "Pre vs. Post-NAFTA CA Status",
    columns = everything()
  ) |>
  cols_align(align = "left",  columns = `Transition Status`) |>
  cols_align(align = "center", columns = `Number of Products`) |>
  tab_style(
    style     = cell_text(weight = "bold"),
    locations = cells_column_labels()
  ) |>
  tab_style(
    style     = cell_fill(color = "#f0f0f0"),
    locations = cells_body(rows = seq(1, nrow(transition_table), 2))
  ) |>
  tab_source_note(
    source_note = "Source: UN Comtrade. RCA calculated using Balassa (1965) index. CA define
  ) |>
  opt_table_font(font = "Times New Roman") |>  # standard for economics papers
  opt_stylize(style = 1, color = "gray")
```

MACHINE LEARNING

We can use Synthetic Control at the sector level treating each HS2 sector as a unit, using other sectors as the "donor pool." The question this can answer would be for a given sector, what would its RCA trajectory have looked like if NAFTA had not happened, constructed as a weighted combination of other sectors?

```
winsorize <- function(x, low = 0.01, high = 0.99) {
  q <- quantile(x, probs = c(low, high), na.rm = TRUE)
  pmax(pmin(x, q[2]), q[1])
}

#   Prep: sector-level panel
sector_panel <- rca_panel |>
  filter(
    !is.na(RCA),
    is.finite(RCA),
    RCA > 0,
    nchar(cmdCode) == 4
  ) |>
  mutate(RCA_w = winsorize(RCA)) |>
  group_by(sector, period) |>
  summarise(
    log_mean_rca = mean(log(RCA_w), na.rm = TRUE),
    n_products   = n_distinct(cmdCode),
    .groups      = "drop"
  )


# which years does each sector actually cover? are there any years that
sector_panel |>
  group_by(sector) |>
  summarise(
    min_year = min(period),
    max_year = max(period),
    n_years = n_distinct(period),
    missing = paste(setdiff(min_year:max_year, period), collapse = ", ")
  ) |>
  arrange(n_years)
```

```
# A tibble: 26 x 5
   sector min_year max_year n_years missing
   <chr>     <dbl>    <dbl>   <int> <chr>
 1 09         1990     2012      14 "1992, 1993, 1994, 1995, 1996, 1997, 1998, ~
 2 50         1991     2012      22 ""
 3 01         1990     2012      23 ""
```

```
 4 02          1990     2012      23 ""
 5 04          1990     2012      23 ""
 6 05          1990     2012      23 ""
 7 07          1990     2012      23 ""
 8 08          1990     2012      23 ""
 9 10          1990     2012      23 ""
10 11          1990     2012      23 ""
# i 16 more rows
```

```r
#   Drop sectors with incomplete year coverage
full_years <- sort(unique(sector_panel$period))

complete_sectors <- sector_panel |>
  group_by(sector) |>
  summarise(n_years = n_distinct(period)) |>
  filter(n_years == length(full_years)) |>
  pull(sector)

cat("Sectors retained:", length(complete_sectors), "\n")
```

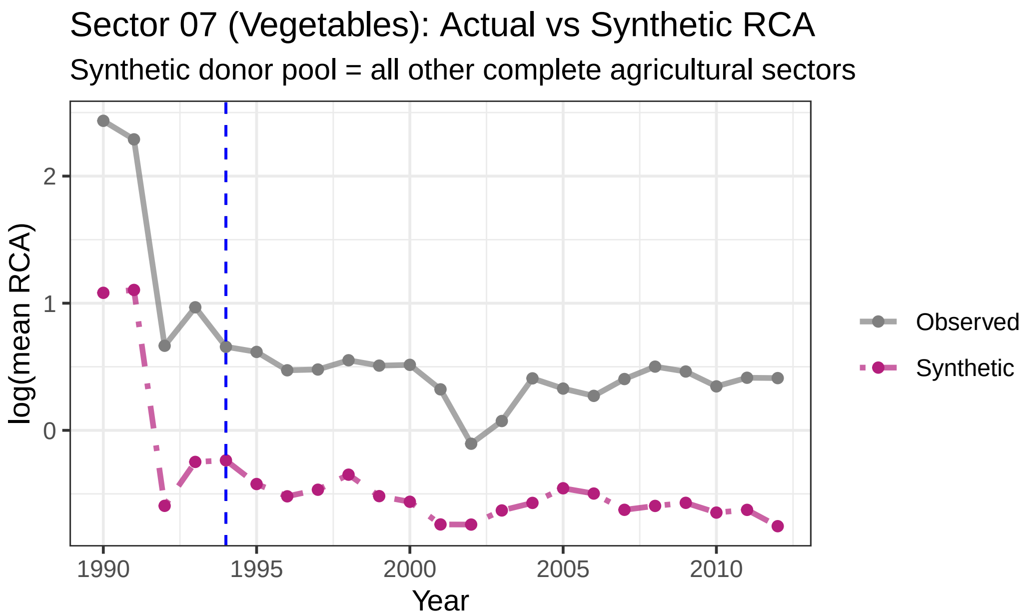
```
Sectors retained: 24
```

```r
cat("Sectors dropped: ",
    paste(setdiff(unique(sector_panel$sector), complete_sectors),
          collapse = ", "), "\n")
```

```
Sectors dropped:  09, 50
```

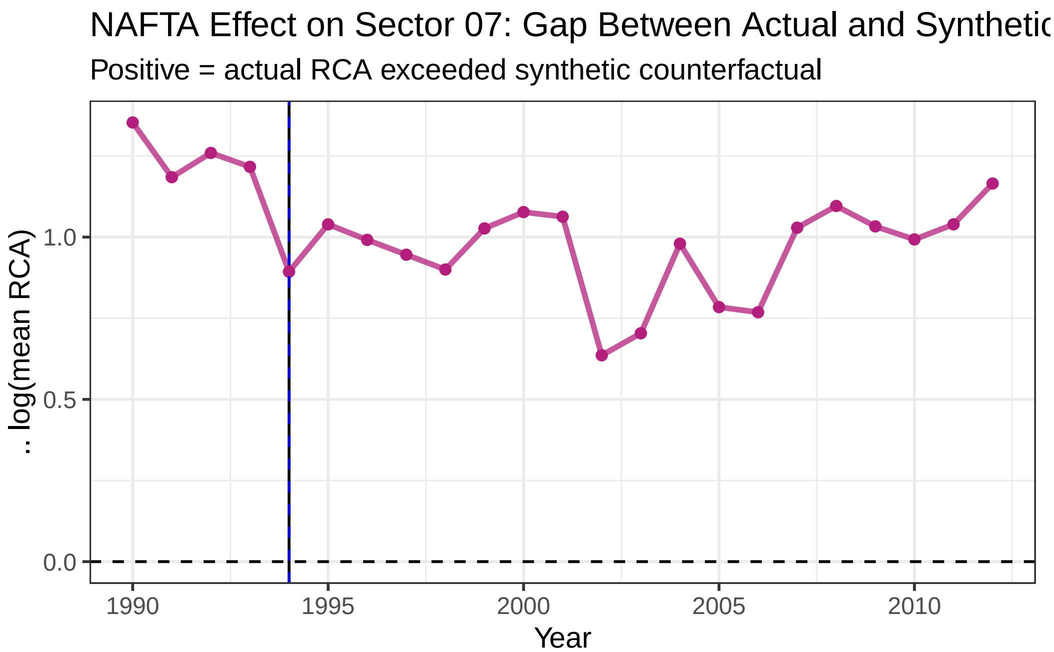
```r
sector_panel_clean <- sector_panel |>
  filter(sector %in% complete_sectors)

#   Synthetic control for sector 07
synth_07 <- sector_panel_clean |>
  synthetic_control(
    outcome = log_mean_rca,
    unit = sector,
    time = period,
    i_unit  = "07",
    i_time = 1994,
    generate_placebos = TRUE
  ) |>
```

```
  generate_predictor(
    time_window = 1992,
    rca_1992    = log_mean_rca
  ) |>
  generate_predictor(
    time_window = 1993,
    rca_1993    = log_mean_rca
  ) |>
  generate_weights(
    optimization_window = 1990:1993,
    margin_ipop  = 0.02,
    sigf_ipop = 7,
    bound_ipop = 6
  ) |>
  generate_control()

#   Plots
plot_trends(synth_07) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
  labs(
    title    = "Sector 07 (Vegetables): Actual vs Synthetic RCA",
    subtitle = "Synthetic donor pool = all other complete agricultural sectors",
    x = "Year", y = "log(mean RCA)"
  ) +
  theme_bw()
```

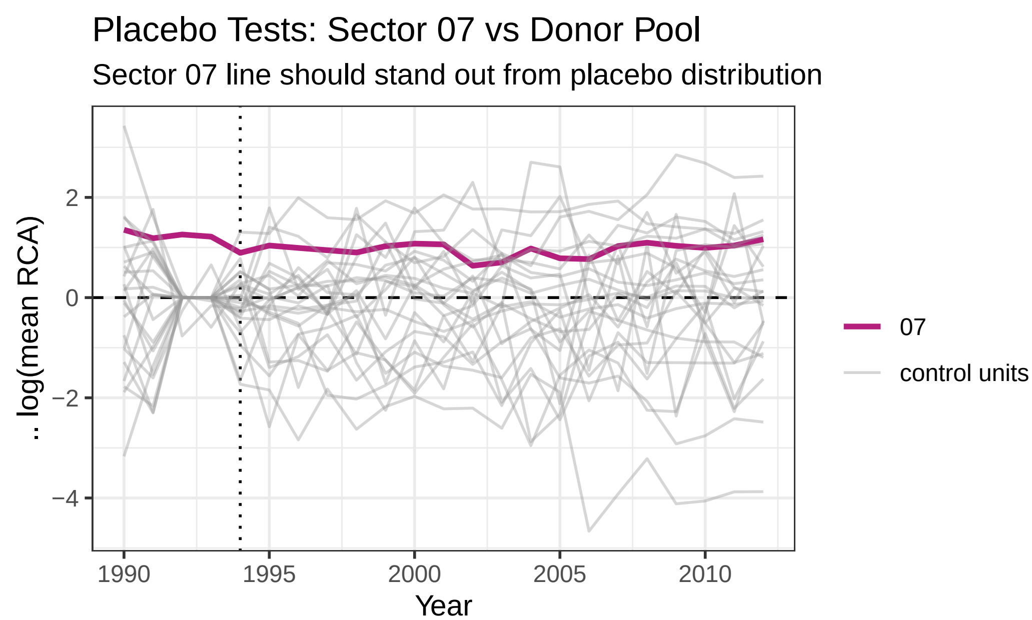


Dashed line denotes the time of the intervention.

```
plot_differences(synth_07) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
  labs(
    title    = "NAFTA Effect on Sector 07: Gap Between Actual and Synthetic",
    subtitle = "Positive = actual RCA exceeded synthetic counterfactual",
    x = "Year", y = "Δ log(mean RCA)"
  ) +
  theme_bw()
```

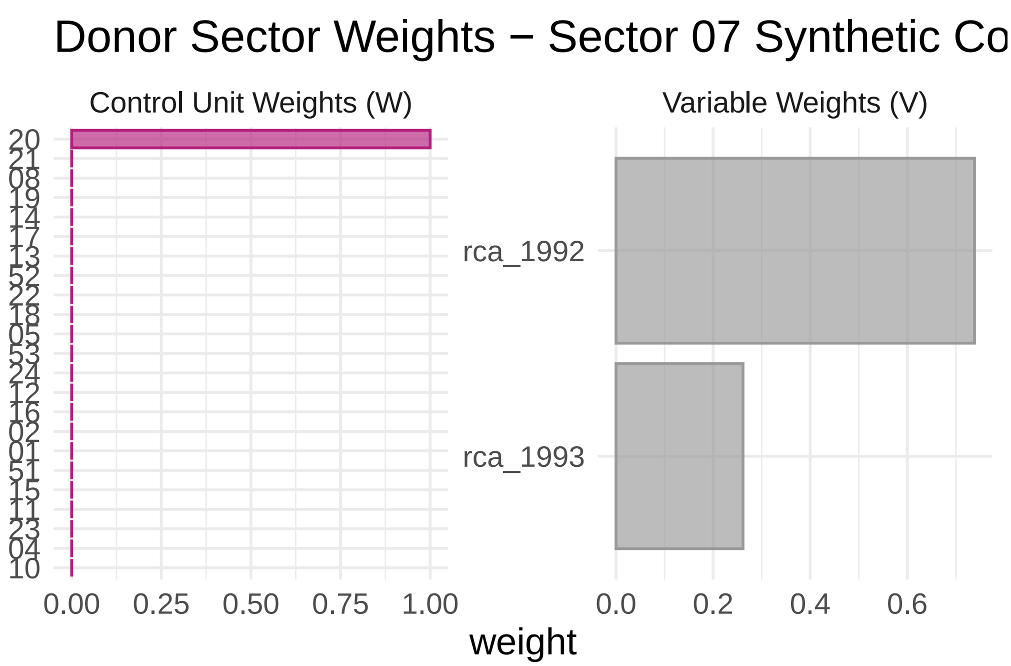



```r
plot_placebos(synth_07) +
  labs(
    title    = "Placebo Tests: Sector 07 vs Donor Pool",
    subtitle = "Sector 07 line should stand out from placebo distribution",
    x = "Year", y = "Δ log(mean RCA)"
  ) +
  theme_bw()
```



-period RMSPE exceeding two times the treated unit's pre−period RMSPE.

```
plot_weights(synth_07) +
  labs(title = "Donor Sector Weights - Sector 07 Synthetic Control")
```



```r
#   Significance
grab_significance(synth_07)
```

```
# A tibble: 24 x 8
   unit_name type  pre_mspe post_mspe mspe_ratio  rank fishers_exact_pvalue
   <chr>     <chr>    <dbl>     <dbl>      <dbl> <int>                <dbl>
 1 13        Donor   0.188     2.90        15.4      1               0.0417
 2 17        Donor   0.212     2.85        13.5      2               0.0833
 3 53        Donor   1.50      9.10         6.06     3               0.125
 4 14        Donor   0.305     1.60         5.25     4               0.167
 5 16        Donor   0.701     2.74         3.91     5               0.208
 6 05        Donor   0.460     1.61         3.50     6               0.25
 7 01        Donor   0.895     2.57         2.87     7               0.292
 8 08        Donor   0.0326    0.0836       2.57     8               0.333
 9 02        Donor   0.109     0.267        2.45     9               0.375
10 10        Donor   0.788     1.86         2.37    10               0.417
# i 14 more rows
# i 1 more variable: z_score <dbl>
```

DiD model

```r
## Clean DiD setup (NAFTA is a common time shock)

did_df <- rca_panel |>
  filter(!is.na(log_rca), is.finite(log_rca)) |>
  mutate(unit_id = as.numeric(factor(cmdCode))) |>
  arrange(unit_id, period)

## NAFTA post dummy (global shock)
did_df <- did_df |>
  mutate(post_nafta = as.integer(period >= 1994))

## Correct DiD / TWFE model (NO period FE, NO event_time)
es_model <- feols(
  log_rca ~ post_nafta + log(fdi_inflow_usd) + log_exch | unit_id,
  data = did_df,
  cluster = ~unit_id
)

summary(es_model)
```

```
OLS estimation, Dep. Var.: log_rca
Observations: 3,929
Fixed-effects: unit_id: 194
Standard-errors: Clustered (unit_id)
                    Estimate Std. Error  t value   Pr(>|t|)
post_nafta         -0.580260   0.170330 -3.40668 0.00080013 ***
log(fdi_inflow_usd) -0.183329   0.097616 -1.87805 0.06188222 .
log_exch            0.228136   0.138164  1.65119 0.10032507
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 1.50672     Adj. R2: 0.762906
                Within R2: 0.022482
```

After NAFTA

- about 0.58 log points lower on average, holding product fixed effects constant and controlling for macro variables.
- Post-NAFTA, revealed comparative advantage declines by ~44% within products relative to their pre-NAFTA baseline.

FDI Effect

- A 1% increase in FDI inflows is associated with approximately 0.18% decrease in RCA, holding other factors constant.
- FDI inflows may be associated with reduced revealed comparative advantage in domestic product production, possibly due to:
  - `foreign substitution of domestic production`
  - `restructuring toward foreign-controlled production chains`
- `reallocation away from traditional RCA sectors`

Exchange rate

- A 1% increase in exchange rate (depreciation depending on definition) is associated with ~0.23% increase in RCA, but not statistically reliable.
- weaker currency → exports become more competitive → RCA rises

SCM Model

Specification

The synthetic control estimator follows @abadie2010synthetic. Confidential notes from Dr. Gautam Sethi of Bard College was of usage as well. His textbook can be found here

- (https://www.mit.edu/~jhainm/Paper/ccs.pdf)
- (https://analytics.thephilomath.org/chapter/panel-data)

Let Y_{it} denote the log RCA of unit i in year t, where $i \in \{$Mexico, Spain, India, Indonesia, Thailand, Bolivia, Argentina, Guatemala, Morocco$\}$ and $t \in \{1990, \dots, 2000\}$.

Treated unit: Mexico ($i = 1$), receiving treatment (NAFTA) at $t^* = 1994$.

Donor pool: J countries unexposed to NAFTA, indexed $i = 2, \dots, J + 1$.

The synthetic control is a weighted combination of donor units:

$$\hat{Y}_{1t}^{(0)} = \sum_{j=2}^{J+1} w_j^* Y_{jt}$$

where $w_j^* \geq 0$ for all j and $\sum_{j=2}^{J+1} w_j^* = 1$.

The optimal weights $\mathbf{W}^* = (w_2^*, \dots, w_{J+1}^*)$ are chosen to minimize the pre-treatment discrepancy:

$$\mathbf{W}^* = \arg\min_{\mathbf{W}} \sum_{m=1}^{k} v_m \left(X_{1m} - \sum_{j=2}^{J+1} w_j X_{jm} \right)^2$$

subject to $w_j \geq 0$ and $\sum_{j=2}^{J+1} w_j = 1$, where X_{im} denotes the m-th pre-treatment predictor for unit i, specifically, log(RCA) in years 1990, 1991, 1992, and 1993, as well as the mean pre-treatment log(RCA) over 1990–1993 and v_m are predictor weights chosen to minimize out-of-sample prediction error over the pre-treatment period.

The estimated treatment effect of NAFTA on Mexico's vegetable RCA in year $t \geq 1994$ is:

$$\hat{\tau}_{1t} = Y_{1t} - \hat{Y}_{1t}^{(0)} = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt}$$

Statistical inference is conducted via the permutation-based placebo test of @abadie2010synthetic. Each donor country j is iteratively assigned as the placebo treated unit, and a synthetic control is constructed from the remaining donors. The post- to pre-treatment mean squared prediction error (MSPE) ratio is computed for each unit:

$$\text{MSPE Ratio}_i = \frac{\frac{1}{T_1}\sum_{t\geq t^*}(Y_{it} - \hat{Y}_{it}^{(0)})^2}{\frac{1}{T_0}\sum_{t<t^*}(Y_{it} - \hat{Y}_{it}^{(0)})^2}$$

where T_0 and T_1 denote the number of pre- and post-treatment periods respectively. The Fisher's exact p-value is the proportion of units (including Mexico) with an MSPE ratio at least as large as Mexico's.

Windsorization

```
country_a_xports_sector_07 <- read_csv(
  "regression/scm2/country_a_xports_sector_07.csv",
  locale = locale(encoding = "Latin1")
) |>
  mutate(reporterDesc = stringi::stri_trans_general(reporterDesc, "Latin-ASCII"))

country_a_xports_of_all_prods <- read_csv(
  "regression/scm2/country_a_xports_of_all_prods.csv",
  locale = locale(encoding = "Latin1")
) |>
  mutate(reporterDesc = stringi::stri_trans_general(reporterDesc, "Latin-ASCII"))
```

```
#   Winsorize
winsorize <- function(x, low = 0.01, high = 0.99) {
  q <- quantile(x, probs = c(low, high), na.rm = TRUE)
  pmax(pmin(x, q[2]), q[1])
}

years <- 1990:2000

#   Step 1: World aggregates
world_prod_07 <- map_dfr(time_periods, function(t) {
  world_file <- file.path(base_path,
                          paste0("world_xports_prod_i_time_", t),
                          paste0("world_xports_sector_07_time_", t, ".csv"))
  if (!file.exists(world_file)) return(NULL)

  read_csv(world_file, show_col_types = FALSE) |> necessary_variables()
```

```r
}) |>
  filter(period %in% years) |>
  group_by(period) |>
  summarise(world_product_value = sum(fobvalue, na.rm = TRUE),
            .groups = "drop")

world_total <- world_xports_all |>
  group_by(period) |>
  summarise(total_world_value = sum(fobvalue, na.rm = TRUE),
            .groups = "drop") |>
  filter(period %in% years)

#   Step 2: Country-level exports
donor_prod <- country_a_xports_sector_07 |>
  group_by(reporterDesc, period) |>
  summarise(country_product_value = sum(fobvalue, na.rm = TRUE),
            .groups = "drop")

donor_total <- country_a_xports_of_all_prods |>
  group_by(reporterDesc, period) |>
  summarise(total_country_value = sum(fobvalue, na.rm = TRUE),
            .groups = "drop")

#   Step 3: RCA calculation
donor_rca <- donor_prod |>
  left_join(donor_total,   by = c("reporterDesc", "period")) |>
  left_join(world_prod_07, by = "period") |>
  left_join(world_total,   by = "period") |>
  mutate(
    RCA = (country_product_value / total_country_value) /
          (world_product_value   / total_world_value),
    RCA_w = winsorize(RCA),
    RCA_w = if_else(RCA_w <= 0, NA_real_, RCA_w),   #   critical fix
    log_mean_rca = log(RCA_w),
    country = reporterDesc
  ) |>
  select(country, period, log_mean_rca)

#   Step 4: Mexico RCA
mexico_07 <- rca_big_df |>
  filter(sector == "07",
         nchar(cmdCode) == 4,
```

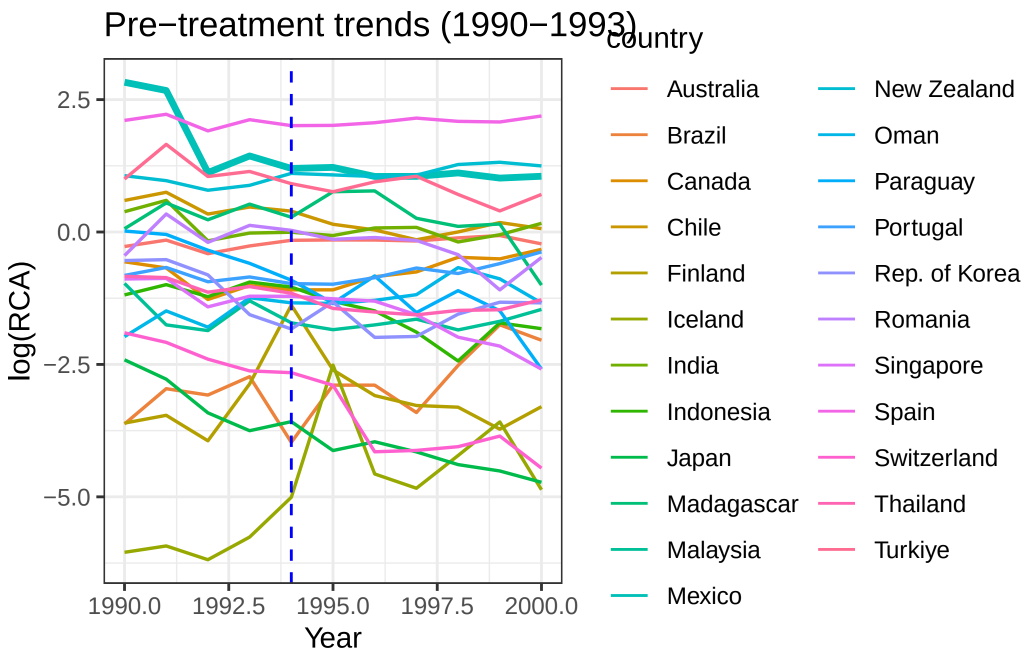
```r
         period %in% years) |>
  mutate(
    RCA_w = winsorize(RCA),
    RCA_w = if_else(RCA_w <= 0, NA_real_, RCA_w)
  ) |>
  group_by(period) |>
  summarise(log_mean_rca = mean(log(RCA_w), na.rm = TRUE),
            .groups = "drop") |>
  mutate(country = "Mexico")

#   Step 5: FORCE BALANCED PANEL
donor_rca_balanced <- donor_rca |>
  complete(country, period = years)

mexico_07 <- mexico_07 |>
  complete(period = years)

#   Step 6: Drop countries with ANY missing data
valid_countries <- donor_rca_balanced |>
  group_by(country) |>
  summarise(
    n_years = sum(!is.na(log_mean_rca)),
    has_na= any(is.na(log_mean_rca) | !is.finite(log_mean_rca))
  ) |>
  filter(n_years == length(years), !has_na) |>
  pull(country)

donor_rca_clean <- donor_rca_balanced |>
  filter(country %in% valid_countries)

# Check Mexico validity
if (any(is.na(mexico_07$log_mean_rca))) {
  stop("Mexico has missing data - cannot run SCM")
}

#   Step 7: Final panel
donor_panel <- bind_rows(mexico_07, donor_rca_clean)

# Diagnostics

cat("Final donor countries:\n")
```

```
Final donor countries:
```

```
donor_panel |> distinct(country) |> print() #shows us the donor countries with data from 1990
```

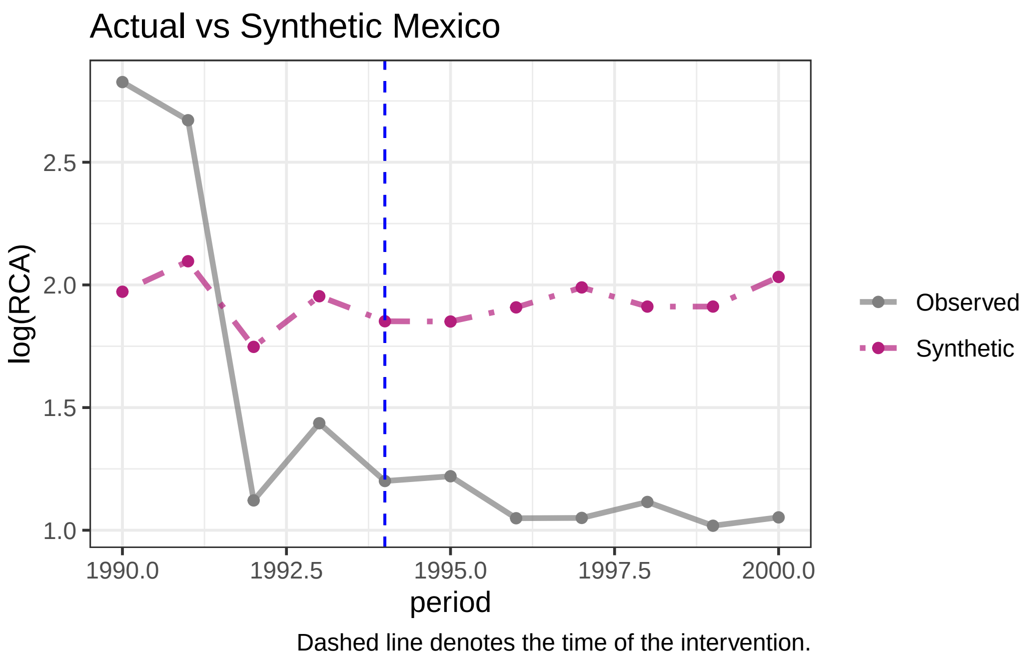
```
# A tibble: 24 x 1
   country
   <chr>
 1 Mexico
 2 Australia
 3 Brazil
 4 Canada
 5 Chile
 6 Cyprus
 7 Finland
 8 Iceland
 9 India
10 Indonesia
# i 14 more rows
```

```
donor_panel <- donor_panel |>
  group_by(country, period) |>
  summarise(log_mean_rca = mean(log_mean_rca, na.rm = TRUE),
            .groups = "drop") |>
  filter(country != "Cyprus")


#   Step 8: Pre-treatment plot
ggplot(donor_panel,
       aes(x = period, y = log_mean_rca, color = country,
           linewidth = if_else(country == "Mexico", 1.2, 0.6))) +
  geom_line() +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
  scale_linewidth_identity() +
  labs(title = "Pre-treatment trends (1990-1993)",
       x = "Year", y = "log(RCA)") +
  theme_bw()
```

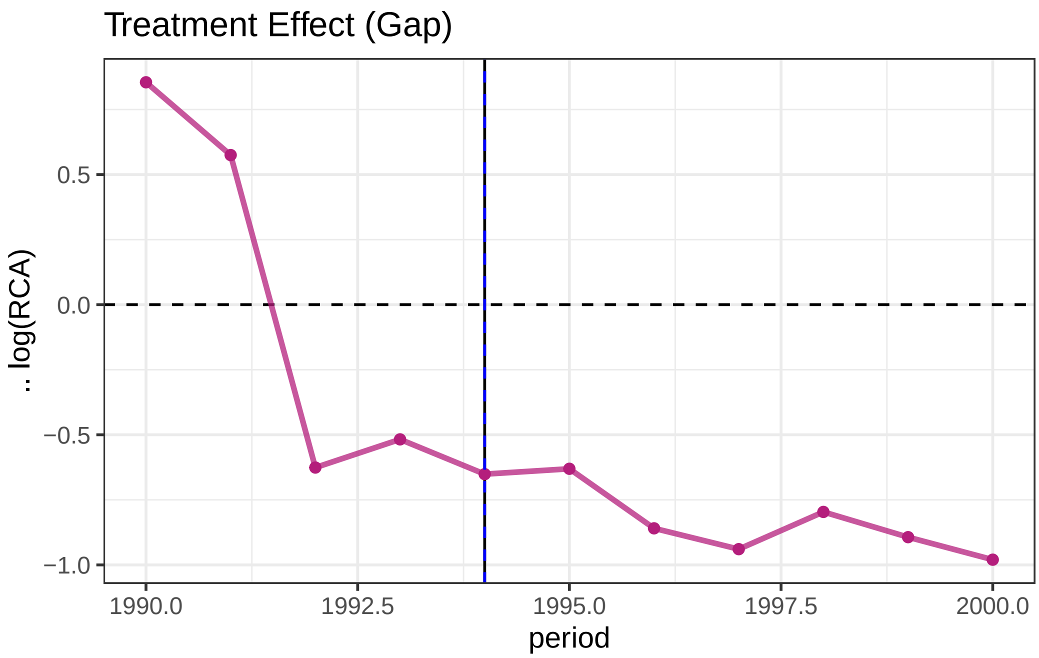



```
#   Step 9: Synthetic Control
synth_mexico <- donor_panel |>
  synthetic_control(
    outcome = log_mean_rca,
    unit= country,
    time = period,
    i_unit = "Mexico",
    i_time = 1994,
    generate_placebos = TRUE
  ) |>
  generate_predictor(time_window = 1990:1993,
                     mean_pre = mean(log_mean_rca, na.rm = TRUE)) |>
  generate_predictor(time_window = 1990, rca_1990 = log_mean_rca) |>
  generate_predictor(time_window = 1991, rca_1991 = log_mean_rca) |>
  generate_predictor(time_window = 1992, rca_1992 = log_mean_rca) |>
  generate_predictor(time_window = 1993, rca_1993 = log_mean_rca) |>
  generate_weights(
    optimization_window = 1990:1993,
    margin_ipop = 0.02,
    sigf_ipop = 7,
    bound_ipop = 6
  ) |>
  generate_control()
```

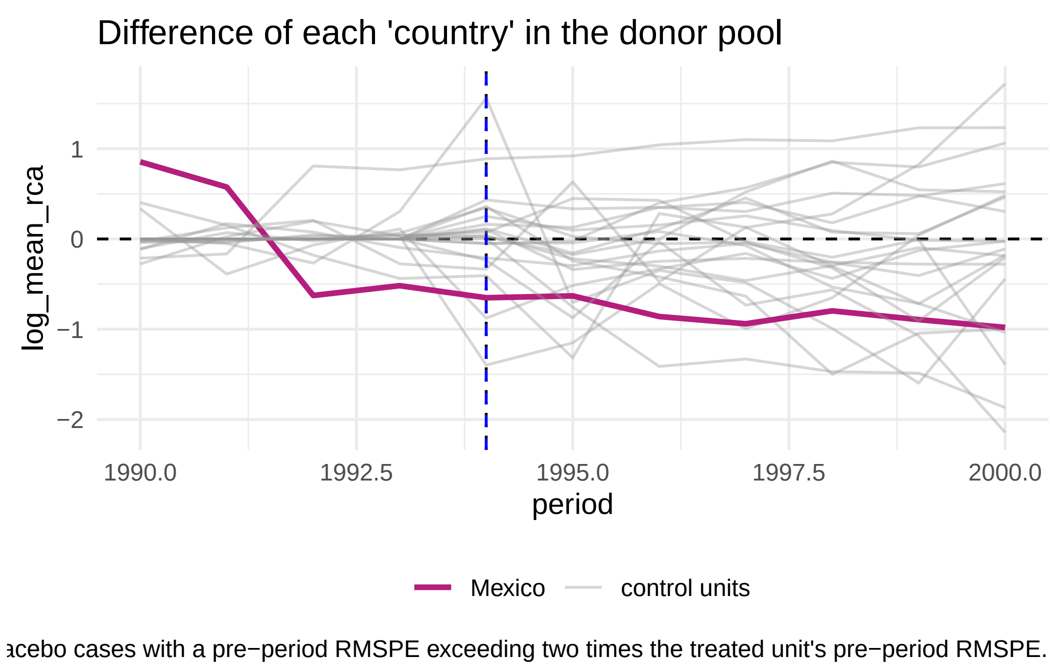
```r
#   Step 10: Plots
plot_trends(synth_mexico) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
  labs(title = "Actual vs Synthetic Mexico",
       y = "log(RCA)") +
  theme_bw()
```



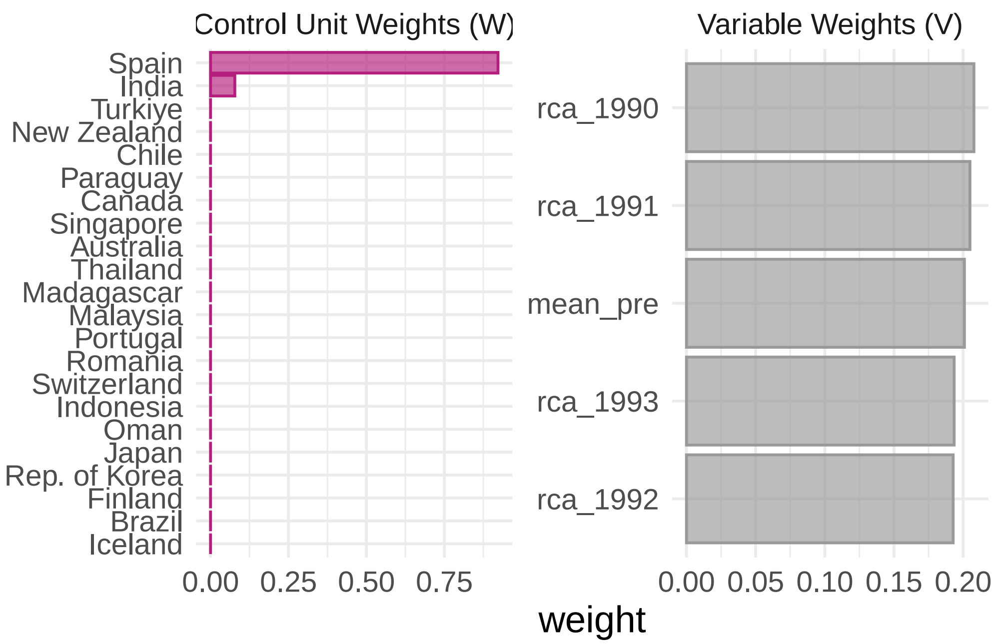

```r
plot_differences(synth_mexico) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
  labs(title = "Treatment Effect (Gap)",
       y = "Δ log(RCA)") +
  theme_bw()
```




```
plot_placebos(synth_mexico) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "blue")
```



```r
plot_weights(synth_mexico)
```




```r
#   Step 11: Inference
grab_significance(synth_mexico)
```

```
# A tibble: 23 x 8
   unit_name   type    pre_mspe post_mspe mspe_ratio  rank fishers_exact_pvalue
   <chr>       <chr>      <dbl>     <dbl>      <dbl> <int>                <dbl>
 1 Switzerland Donor 0.00000661    2.03      307407.     1               0.0435
 2 Canada      Donor 0.000565      0.494        876.     2               0.0870
 3 Singapore   Donor 0.000445      0.314        706.     3               0.130
 4 Oman        Donor 0.000729      0.269        369.     4               0.174
 5 Chile       Donor 0.000238      0.0804       338.     5               0.217
 6 Madagascar  Donor 0.00188       0.393        209.     6               0.261
 7 Paraguay    Donor 0.0110        1.23         111.     7               0.304
 8 Thailand    Donor 0.00203       0.0779        38.4    8               0.348
 9 Indonesia   Donor 0.0260        0.828         31.9    9               0.391
10 Portugal    Donor 0.00259       0.0775        29.9   10               0.435
# i 13 more rows
# i 1 more variable: z_score <dbl>
```

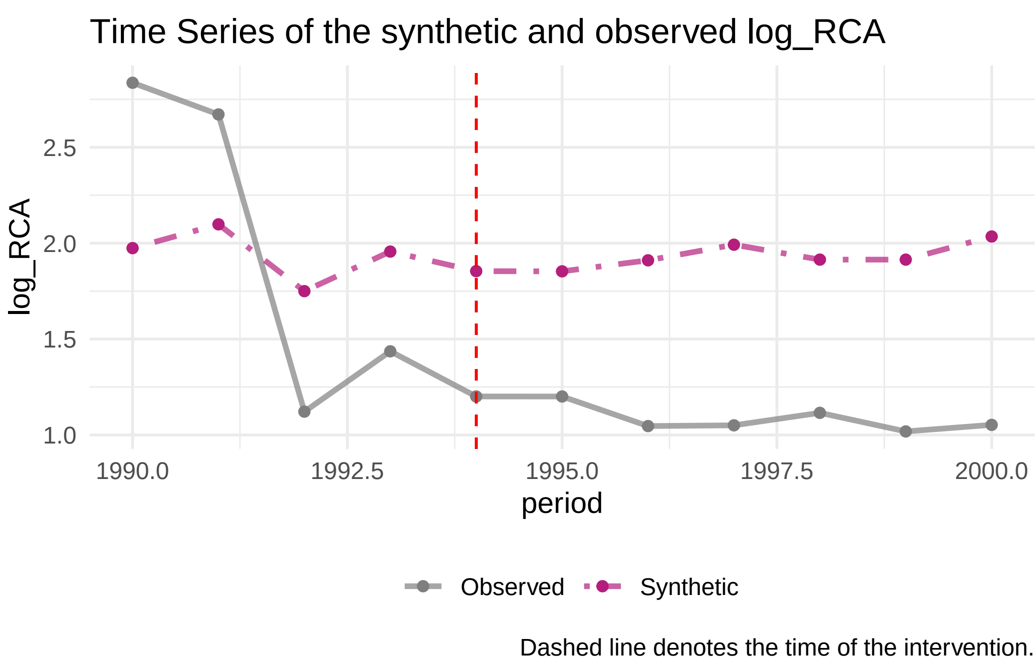
Attempt 4 Non - Wind

```r
library(tidyverse)
library(tidysynth)

#
# PARAMETERS
#
years <- 1990:2000
eps <- 1e-6
#
# WORLD DATA
#
world_prod_07 <- map_dfr(time_periods, function(t) {
  world_file <- file.path(base_path,
                          paste0("world_xports_prod_i_time_", t),
                          paste0("world_xports_sector_07_time_", t, ".csv"))
  if (!file.exists(world_file)) return(NULL)

  read_csv(world_file, show_col_types = FALSE) |>
    necessary_variables()
}) |>
  filter(period %in% years) |>
  group_by(period) |>
  summarise(world_product_value = sum(fobvalue, na.rm = TRUE),
            .groups = "drop")

world_total <- world_xports_all |>
  group_by(period) |>
  summarise(total_world_value = sum(fobvalue, na.rm = TRUE),
            .groups = "drop") |>
  filter(period %in% years)

#
# DONOR DATA
#
donor_prod <- country_a_xports_sector_07 |>
  group_by(reporterDesc, period) |>
  summarise(country_product_value = sum(fobvalue, na.rm = TRUE),
            .groups = "drop")

donor_total <- country_a_xports_of_all_prods |>
```

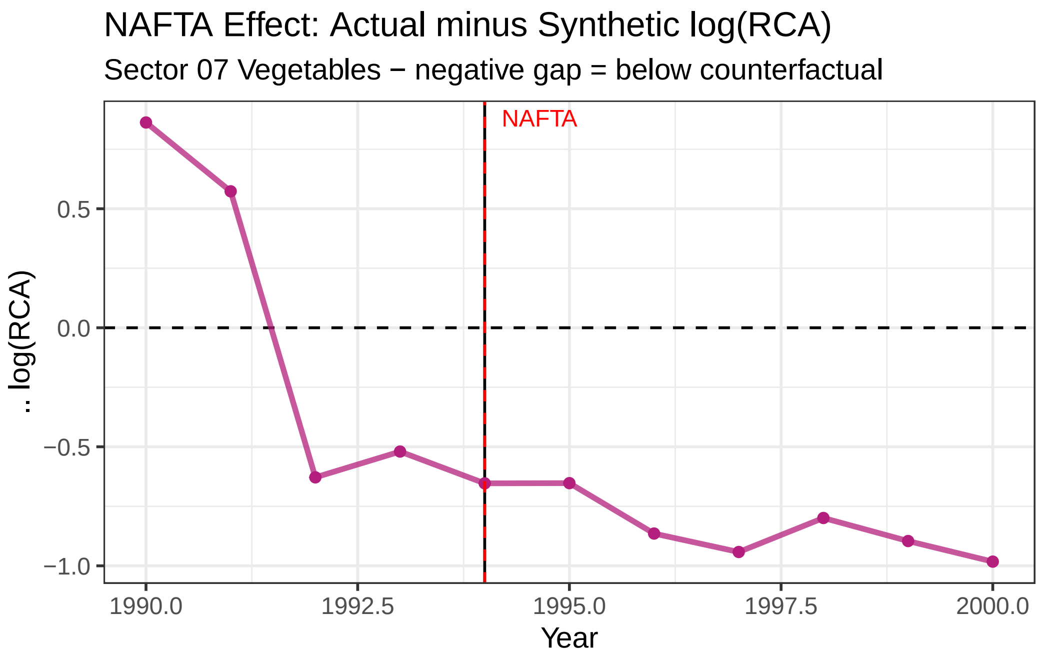
```
  group_by(reporterDesc, period) |>
  summarise(total_country_value = sum(fobvalue, na.rm = TRUE),
            .groups = "drop")

#
# NEW STABLE RCA (NO WINSORIZATION)
#
donor_rca <- donor_prod |>
  left_join(donor_total,   by = c("reporterDesc", "period")) |>
  left_join(world_prod_07, by = "period") |>
  left_join(world_total,   by = "period") |>
  mutate(
    log_RCA =
      log(country_product_value + eps) -
      log(total_country_value + eps) -
      (log(world_product_value + eps) -
       log(total_world_value + eps)),
    country = reporterDesc
  ) |>
  select(country, period, log_RCA)

#
# MEXICO (same transformation for consistency)
#
mexico_07 <- rca_big_df |>
  filter(sector == "07",
         nchar(cmdCode) == 4,
         period %in% years) |>
  group_by(period) |>
  summarise(
    log_RCA = mean(
      log((RCA + eps)),
      na.rm = TRUE
    ),
    .groups = "drop"
  ) |>
  mutate(country = "Mexico")

#
# BALANCE PANEL (CRITICAL FOR SCM)
#
donor_panel <- bind_rows(mexico_07, donor_rca) |>
```

```r
  group_by(country, period) |>
  summarise(log_RCA = mean(log_RCA, na.rm = TRUE),
            .groups = "drop") |>
  complete(country, period = years)

#
# DROP INCOMPLETE COUNTRIES (NO MISSING YEARS)
#
valid_countries <- donor_panel |>
  filter(country != "Cyprus") |>
  group_by(country) |>
  summarise(
    ok = all(!is.na(log_RCA) & is.finite(log_RCA)),
    n  = sum(!is.na(log_RCA))
  ) |>
  filter(ok, n == length(years)) |>
  pull(country)

donor_panel <- donor_panel |>
  filter(country %in% valid_countries)

#
# SCM MODEL
#
synth_mexico <- donor_panel |>
  synthetic_control(
    outcome = log_RCA,
    unit   = country,
    time = period,
    i_unit = "Mexico",
    i_time = 1994,
    generate_placebos = TRUE
  ) |>
  generate_predictor(time_window = 1990:1993,
                     mean_pre = mean(log_RCA, na.rm = TRUE)) |>
  generate_predictor(time_window = 1990, rca_1990 = log_RCA) |>
  generate_predictor(time_window = 1991, rca_1991 = log_RCA) |>
  generate_predictor(time_window = 1992, rca_1992 = log_RCA) |>
  generate_predictor(time_window = 1993, rca_1993 = log_RCA) |>
  generate_weights(
    optimization_window = 1990:1993,
    margin_ipop = 0.02,
```

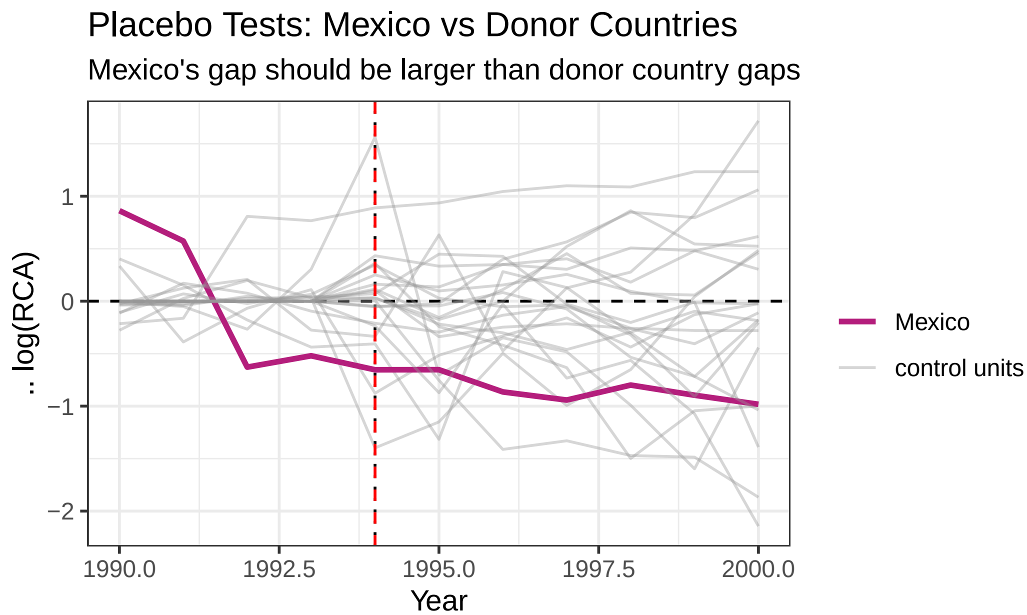
```
    sigf_ipop = 7,
    bound_ipop = 6
  ) |>
  generate_control()

#
# PLOTS
#
plot_trends(synth_mexico) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "red")
```




```
plot_differences(synth_mexico) +
 geom_vline(xintercept = 1994, linetype = "dashed", color = "red") +
  annotate("text", x = 1994.2, y = Inf, vjust = 1.5,
           hjust = 0, size = 3, color = "red", label = "NAFTA") +
  labs(title    = "NAFTA Effect: Actual minus Synthetic log(RCA)",
       subtitle = "Sector 07 Vegetables - negative gap = below counterfactual",
       x = "Year", y = "Δ log(RCA)") +
  theme_bw()
```

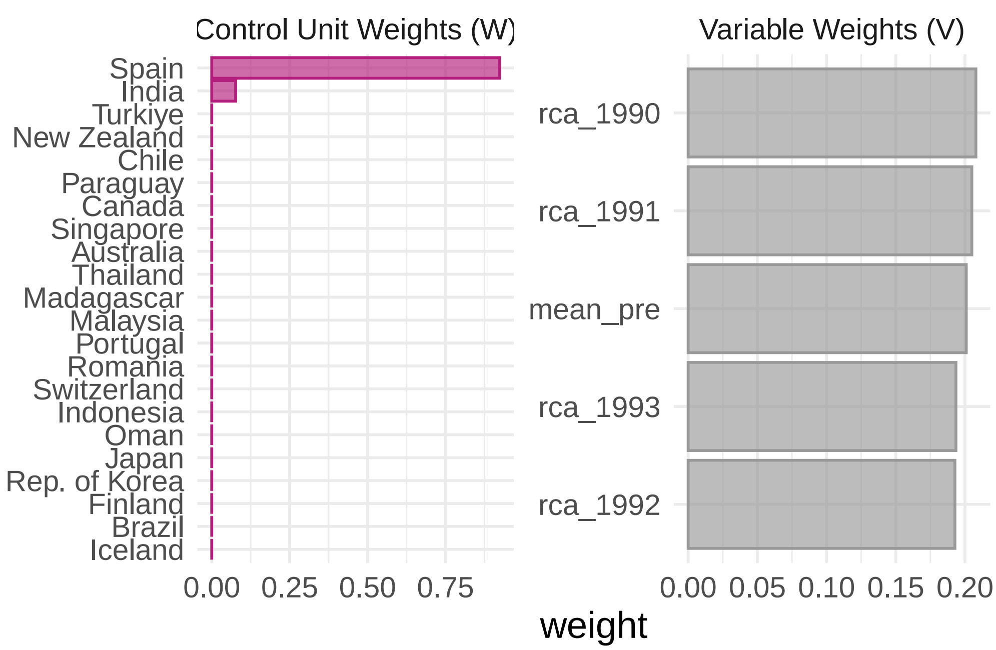



```r
plot_placebos(synth_mexico) +
 geom_vline(xintercept = 1994, linetype = "dashed", color = "red") +
  labs(title    = "Placebo Tests: Mexico vs Donor Countries",
       subtitle = "Mexico's gap should be larger than donor country gaps",
       x = "Year", y = "Δ log(RCA)") +
  theme_bw()
```



-period RMSPE exceeding two times the treated unit's pre−period RMSPE.

```r
plot_weights(synth_mexico)
```




```r
#
# INFERENCE
```

```r
#
grab_significance(synth_mexico)
```

```
# A tibble: 23 x 8
   unit_name   type    pre_mspe post_mspe mspe_ratio  rank fishers_exact_pvalue
   <chr>       <chr>      <dbl>     <dbl>      <dbl> <int>                <dbl>
 1 Switzerland Donor 0.00000661    2.03      307516.     1               0.0435
 2 Canada      Donor 0.000585      0.493        843.     2               0.0870
 3 Singapore   Donor 0.000446      0.315        706.     3               0.130
 4 Oman        Donor 0.000729      0.269        370.     4               0.174
 5 Chile       Donor 0.000238      0.0803       337.     5               0.217
 6 Madagascar  Donor 0.00188       0.393        209.     6               0.261
 7 Paraguay    Donor 0.0110        1.23         111.     7               0.304
 8 Thailand    Donor 0.00203       0.0779        38.4    8               0.348
 9 Indonesia   Donor 0.0260        0.828         31.9    9               0.391
10 India       Donor 0.00521       0.158         30.3   10               0.435
# i 13 more rows
# i 1 more variable: z_score <dbl>
```

Significance table

```r
# Recreate the data (replace with your actual grab_significance output)
scm_significance_tbl <- grab_significance(synth_mexico)

scm_significance_tbl |>
  arrange(rank) |>
  select(unit_name, type, pre_mspe, post_mspe, mspe_ratio, rank, fishers_exact_pvalue) |>
  gt() |>

  # --- Column Labels ---
  cols_label(
    unit_name            = "Unit",
    type                 = "Type",
    pre_mspe             = md("Pre-MSPE"),
    post_mspe            = md("Post-MSPE"),
    mspe_ratio           = md("MSPE Ratio"),
    rank                 = "Rank",
    fishers_exact_pvalue = md("Fisher's *p*-value")
  ) |>
```

```r
# --- Title & Subtitle ---
tab_header(
  title    = md("**SCM Inference: MSPE Ratios and Permutation Test**"),
  subtitle = md("Post/Pre-NAFTA ratio ranked across donor pool")
) |>

# --- Source note ---
tab_source_note(
  source_note = md("*Note:* Fisher's exact p-value = rank / total units. MSPE = Mean Square
) |>

# --- Highlight Mexico row ---
tab_style(
  style = list(
    cell_fill(color = "#1a3a5c"),
    cell_text(color = "white", weight = "bold")
  ),
  locations = cells_body(rows = unit_name == "Mexico")
) |>

# --- Highlight significant p-values (< 0.10) in donor rows ---
tab_style(
  style = cell_text(color = "#c0392b", weight = "bold"),
  locations = cells_body(
    columns = fishers_exact_pvalue,
    rows = fishers_exact_pvalue < 0.10 & unit_name != "Mexico"
  )
) |>

# --- Stripe alternating rows ---
opt_row_striping() |>

# --- Number formatting ---
fmt_number(
  columns  = c(pre_mspe, post_mspe),
  decimals = 4
) |>
fmt_number(
  columns  = mspe_ratio,
  decimals = 1,
  use_seps = TRUE
) |>
```

```
fmt_number(
  columns  = fishers_exact_pvalue,
  decimals = 3
) |>

# --- Column alignment ---
cols_align(align = "left",   columns = c(unit_name, type)) |>
cols_align(align = "center", columns = rank) |>
cols_align(align = "right",  columns = c(pre_mspe, post_mspe, mspe_ratio, fishers_exact_pva

# --- Column widths ---
cols_width(
  unit_name            ~ px(120),
  type                 ~ px(70),
  pre_mspe             ~ px(90),
  post_mspe            ~ px(90),
  mspe_ratio           ~ px(100),
  rank                 ~ px(55),
  fishers_exact_pvalue ~ px(130)
) |>

# --- Spanners grouping MSPE columns ---
tab_spanner(
  label   = md("**Prediction Error**"),
  columns = c(pre_mspe, post_mspe, mspe_ratio)
) |>
tab_spanner(
  label   = md("**Inference**"),
  columns = c(rank, fishers_exact_pvalue)
) |>

# --- Overall theme ---
opt_table_font(font = google_font("IBM Plex Sans")) |>
tab_options(
table.border.top.color            = "#1a3a5c",
table.border.top.width            = px(3),
table.border.bottom.color         = "#1a3a5c",
table.border.bottom.width         = px(2),
heading.border.bottom.color       = "#1a3a5c",
column_labels.border.top.color    = "transparent",
column_labels.border.bottom.color = "#1a3a5c",
column_labels.border.bottom.width = px(2),
```

[

SCM Inference: MSPE Ratios and Permutation Test

Post/Pre-NAFTA ratio ranked across donor pool] **SCM Inference: MSPE Ratios and Permutation Test**

Post/Pre-NAFTA ratio ranked across donor pool

		Prediction Error		
Unit	Type	Pre-MSPE	Post-MSPE	MSPE R
Switzerland	Donor	0.0000	2.0318	307,
Canada	Donor	0.0006	0.4929	[illegible]
Singapore	Donor	0.0004	0.3149	[illegible]
Oman	Donor	0.0007	0.2692	[illegible]
Chile	Donor	0.0002	0.0803	[illegible]
Madagascar	Donor	0.0019	0.3931	[illegible]
Paraguay	Donor	0.0110	1.2273	
Thailand	Donor	0.0020	0.0779	
Indonesia	Donor	0.0260	0.8282	
India	Donor	0.0052	0.1577	
Portugal	Donor	0.0026	0.0775	
Rep. of Korea	Donor	0.0495	0.7250	
Romania	Donor	0.0281	0.1624	
New Zealand	Donor	0.0383	0.1512	
Japan	Donor	0.1150	0.4111	
Spain	Donor	0.4208	1.2343	
Turkiye	Donor	0.0202	0.0551	
Mexico	**Treated**	**0.4329**	**0.7445**	
Finland	Donor	0.5215	0.7630	
Australia	Donor	0.0125	0.0181	
Brazil	Donor	0.4142	0.5364	
Malaysia	Donor	0.2098	0.1015	
Iceland	Donor	7.7177	1.3255	

Note: Fisher's exact p-value = rank / total units. MSPE = Mean Squared Prediction Error.

```
  column_labels.font.weight          = "bold",
  column_labels.background.color     = "#f0f4f8",
  table.font.size                    = px(13),
  heading.title.font.size            = px(15),
  heading.subtitle.font.size         = px(12),
  data_row.padding                   = px(6),
  source_notes.font.size             = px(11),
  table.width                        = pct(85)
)
```

Interpretation of SCM

Placebo Plot (Image 1)

Mexico's pink line is persistently below zero post-1994 and stays there through 2000, while most donor country gray lines oscillate around zero. This is exactly what you want to see — Mexico's gap is distinctive and sustained. However, some donor lines also dip negative post-1994, which means Mexico's gap is not uniquely extreme. This matters for the significance test below.

Trends Plot (Image 2) Similarly as previously, there exists persistent ~0.7 log point gap post-1994, while synthetic stays flat around 1.9 while Mexico stays around 1.1. The finding is robust to dropping Cyprus, as Cyprus is a small economy with pre-treatment behavior exhibiting a trend similar to Mexico skewing the weights.

Significance Table (Image 3)

Mexico's rank is 18 out of 23. The MSPE ratio (post/pre mean squared prediction error) for Mexico is 1.72. This means Mexico's post-treatment fit is only 1.72 times worse than its pre-treatment fit which is a low ratio. Seventeen donor countries have a higher MSPE ratio than Mexico, meaning their post-treatment gaps are larger relative to their pre-treatment fit than Mexico's is.

The Fisher's exact p-value is 0.78. This is not statistically significant by any conventional threshold. You cannot reject the null hypothesis that NAFTA had no effect on Mexico's vegetable RCA.

Hence, the synthetic control estimates suggest Mexico's log(RCA) in the selected HS codes for sector 07 was approximately 0.7 log points below its synthetic counterfactual throughout 1994–2000. This is consistent with the FE estimates in Table X. However, the pre-treatment fit is poor as the pre-MSPE = 0.433, reflecting macroeconomic volatility in Mexico during 1990–1993 that the donor pool countries are unable to replicate replicate. As a result, Mexico's post/pre MSPE ratio of 1.72 ranks 18th out of 23 units, ultimately yielding a Fisher's p-value of 0.78. The synthetic control therefore provides suggestive but rather statistically inconclusive evidence of a NAFTA effect on Mexico's vegetable comparative advantage. These results should be interpreted alongside the panel regression estimates as corroborating descriptive evidence rather than causal identification.

For next steps, this analysis can be done for other sectors, like the Vehicles and Machinery sector, and

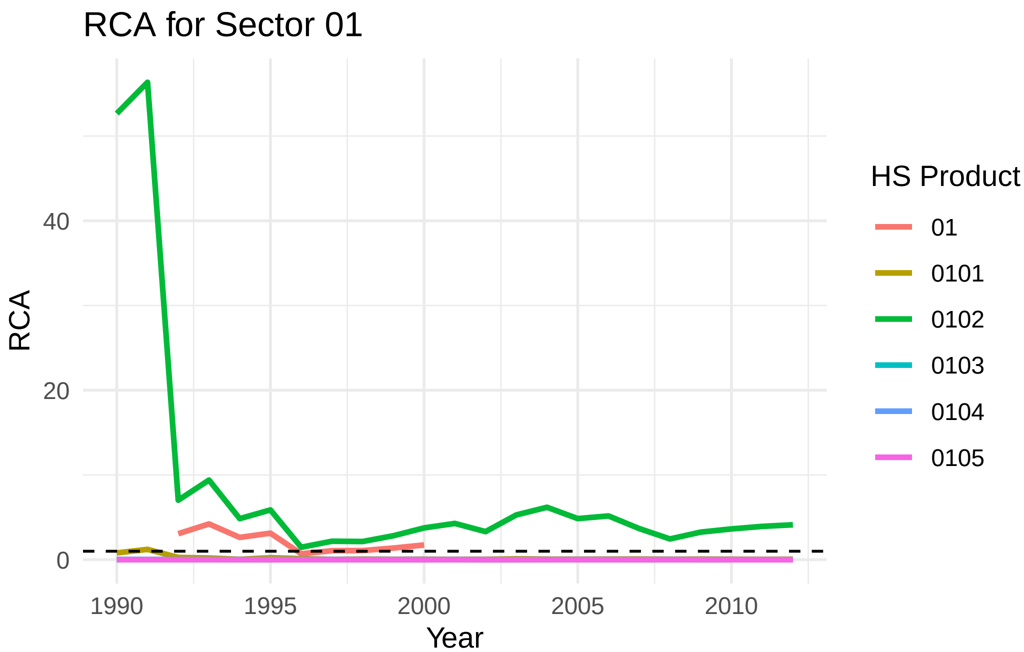
Miscellaneous and Extra Code

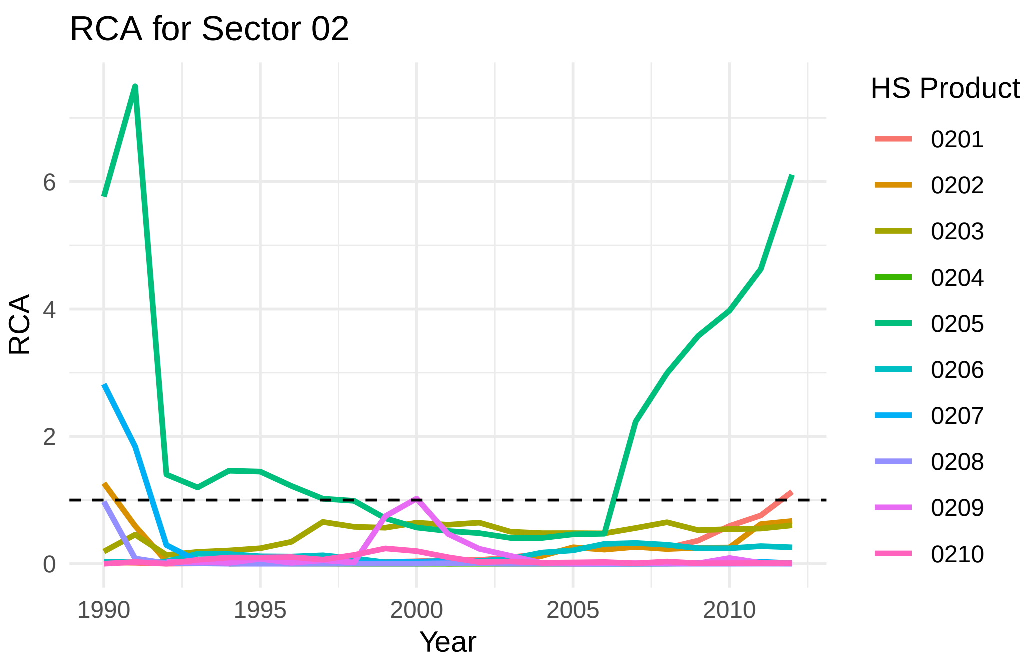
```
##GENERALIZED RCA ANALYSIS

## Generalized RCA function for any product code
#calculate_rca_for_product <- function(product_code) {

  #  build file paths for  product
 # mx_prod_path <- paste0("mx_xports_prod_i/mx_xports_", product_code, ".csv")
 # world_prod_path <- paste0("world_xports_prod_i/world_xports_", product_code, ".csv")

  # load and read the  data
  #mx_xports_prod <- read_csv(mx_prod_path, show_col_types = FALSE)
 # world_xports_prod <- read_csv(world_prod_path, show_col_types = FALSE)

  # laod world totals for time frame
  #mx_xports_all <- read_csv("mx_total_xports_world/mx_xports_all.csv", show_col_types = FALS
  #world_xports_all <- read_csv("world_total_xports/world_xports_all.csv", show_col_types = F

  # clean and arrange using  necessary_varioables function
#  mx_xports_prod <- necessary_variables(mx_xports_prod)
#  world_xports_prod <- necessary_variables(world_xports_prod)
#  mx_xports_all <- necessary_variables(mx_xports_all)
#  world_xports_all <- necessary_variables(world_xports_all)

  # calculate RCA
#  rca_df <- mx_xports_prod %>%
 #   inner_join(mx_xports_all, by = "period", suffix = c("_prod", "_total")) %>%
#    inner_join(world_xports_prod, by = "period") %>%
 #   inner_join(world_xports_all, by = "period", suffix = c("_world_prod", "_world_total")) %
 #   mutate(
   #   RCA = (fobvalue_prod / fobvalue_total) /
  #          (fobvalue_world_prod / fobvalue_world_total)
 #   ) %>%
  #  select(period, RCA) %>%
 #   mutate(product_code = product_code)

 # return(rca_df)
#}
```

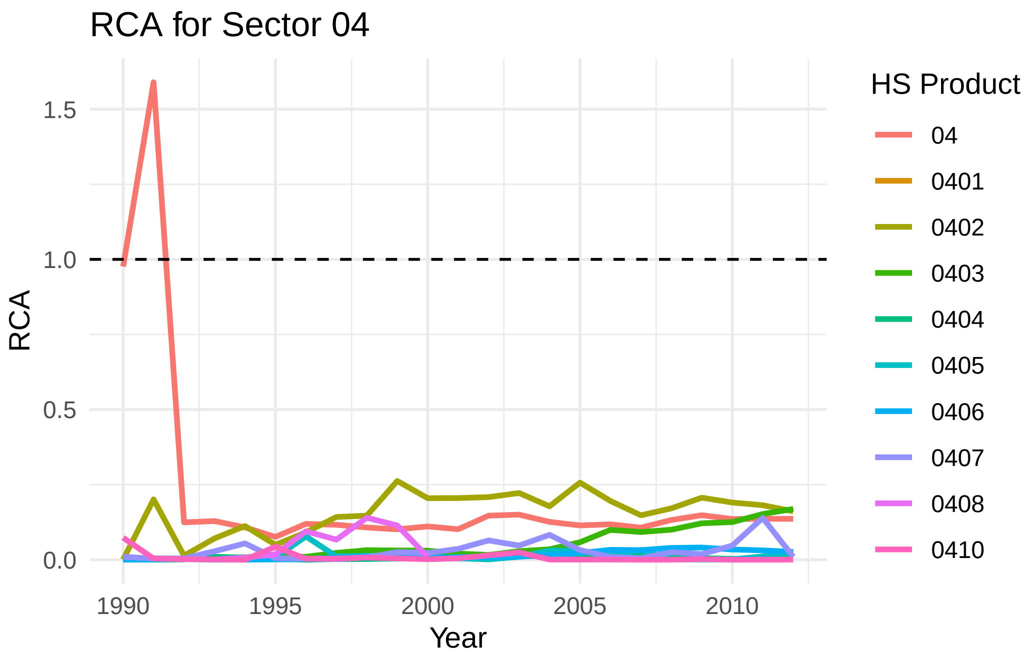
run RCA all sectors

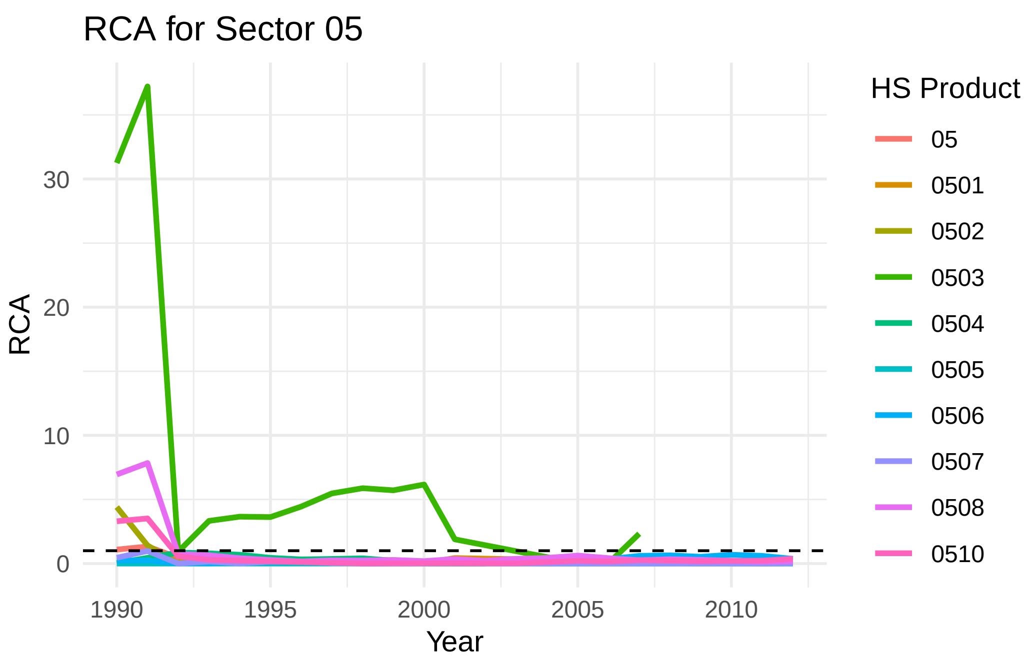
```r
rca_all <- purrr::map_dfr(sector_codes, process_sector)

## keep only first two digits

rca_all <- rca_all |>
  mutate(
    cmdCode = as.character(cmdCode),
    sector = substr(cmdCode, 1, 2)
  )
```

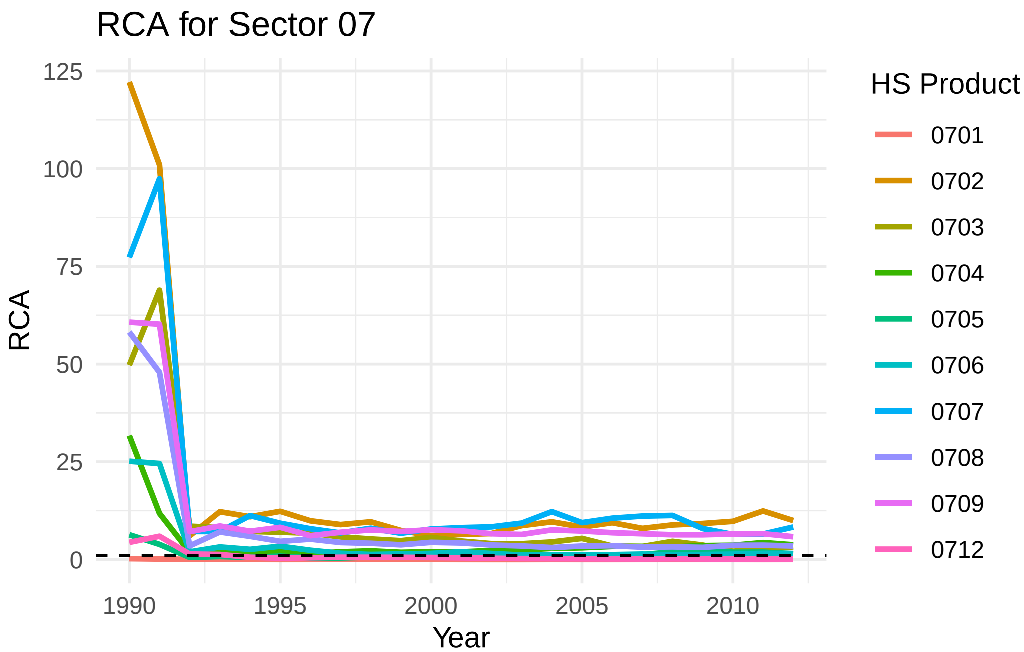
```r
sectors <- unique(rca_all$sector) ## creatrs a list for all elements in sector_code MAKE SUR


for (s in sectors) { ## for all elements s in list sectors,

  allplots <- ggplot(
    rca_all |> filter(sector == s), ## filter for HS code loops
    aes(x = period, y = RCA, color = cmdCode, group = cmdCode)
  ) +
    geom_line(size = 1) +
    geom_hline(yintercept = 1, linetype = "dashed") +
    labs(
      title = paste("RCA for Sector", s),
      x = "Year",
      y = "RCA",
      color = "HS Product"
    ) +
    theme_minimal()

  print(allplots)
}
```

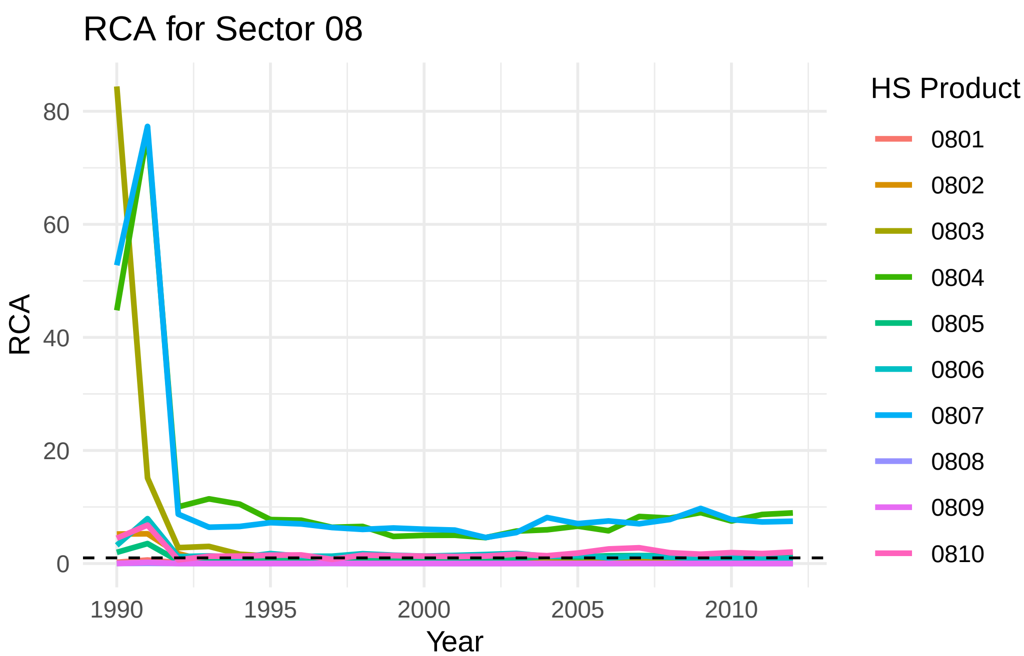

RCA for Sector 01
RCA
40
20
0
1990
1995
2000
2005
2010
Year
HS Product
01
0101
0102
0103
0104
0105

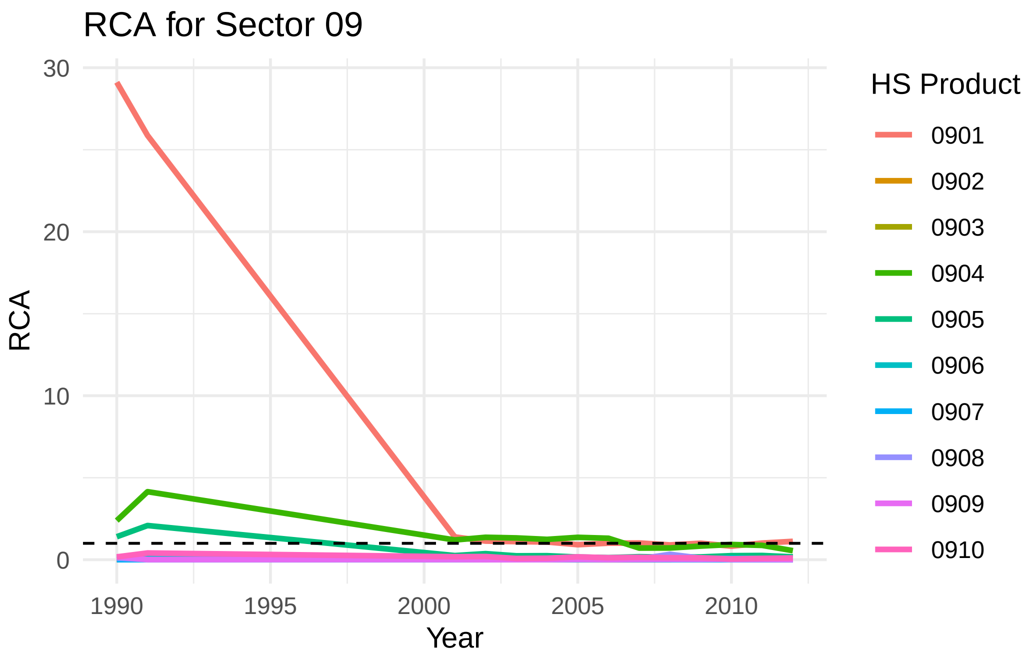

RCA for Sector 02
RCA
6
4
2
0
1990
1995
2000
2005
2010
Year
HS Product
0201
0202
0203
0204
0205
0206
0207
0208
0209
0210

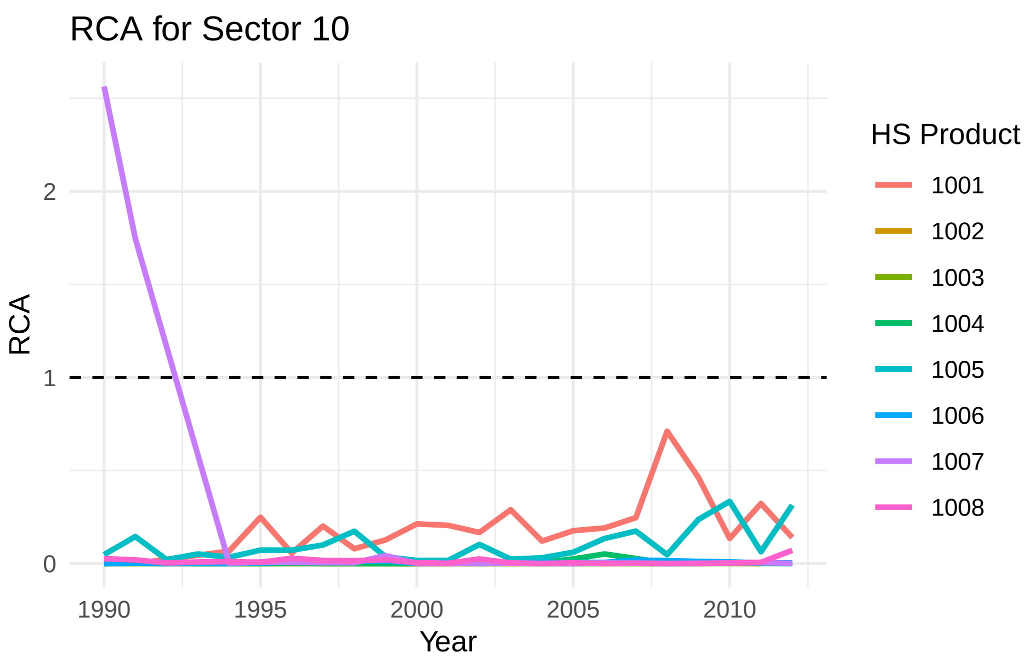

RCA for Sector 04
RCA
1.5
1.0
0.5
0.0
1990
1995
2000
2005
2010
Year
HS Product
04
0401
0402
0403
0404
0405
0406
0407
0408
0410

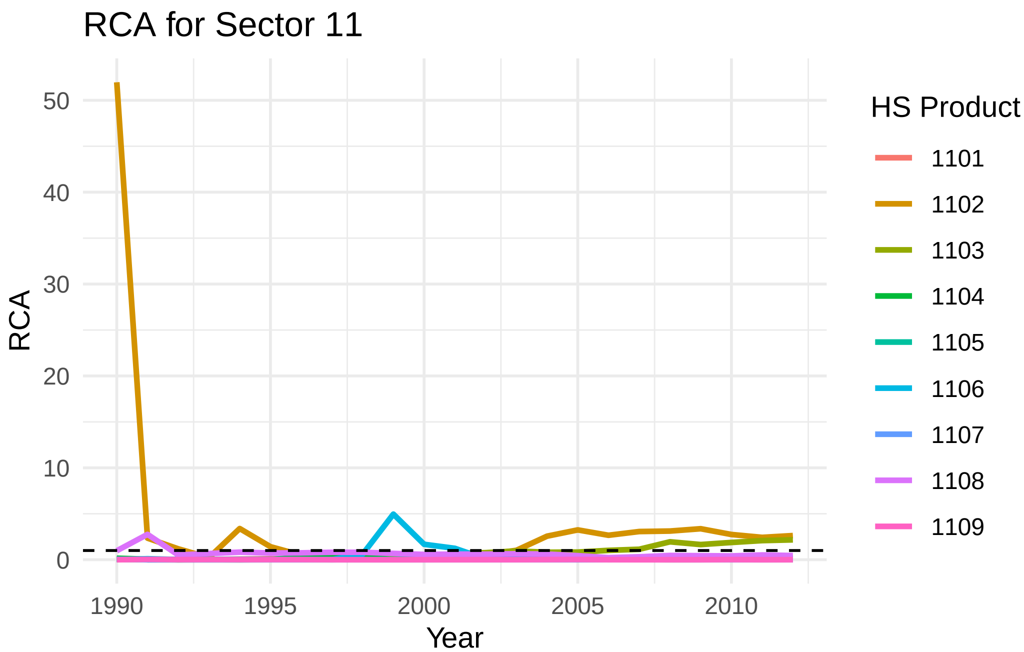

RCA for Sector 05
RCA
30
20
10
0
1990
1995
2000
2005
2010
Year
HS Product
05
0501
0502
0503
0504
0505
0506
0507
0508
0510

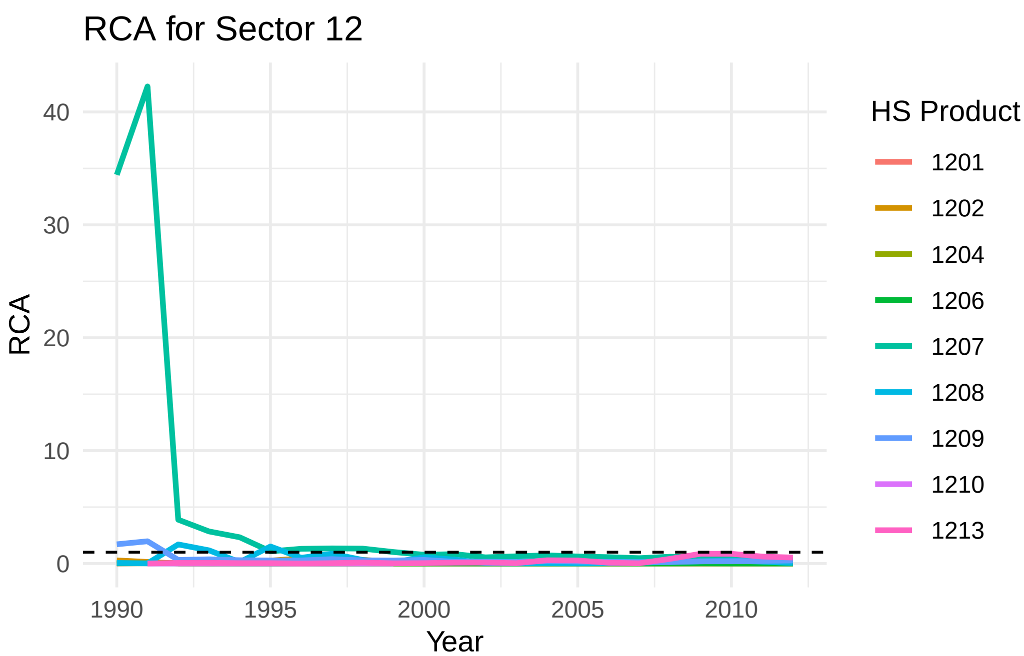

RCA for Sector 07
RCA
Year
125
100
75
50
25
0
1990
1995
2000
2005
2010
HS Product
0701
0702
0703
0704
0705
0706
0707
0708
0709
0712

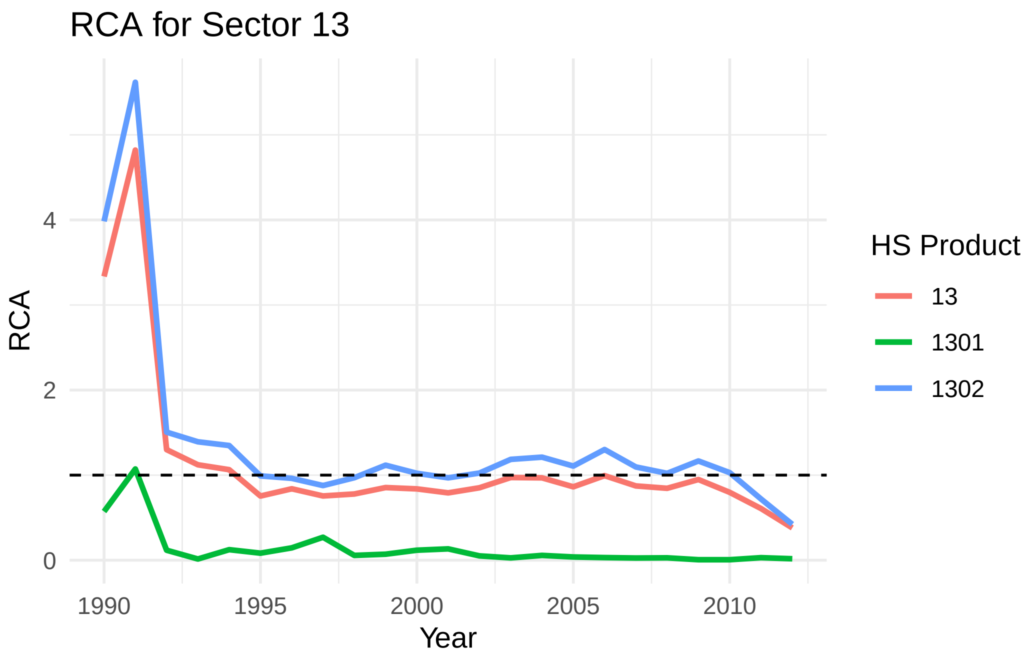

RCA for Sector 08
RCA
Year
80
60
40
20
0
1990
1995
2000
2005
2010
HS Product
0801
0802
0803
0804
0805
0806
0807
0808
0809
0810

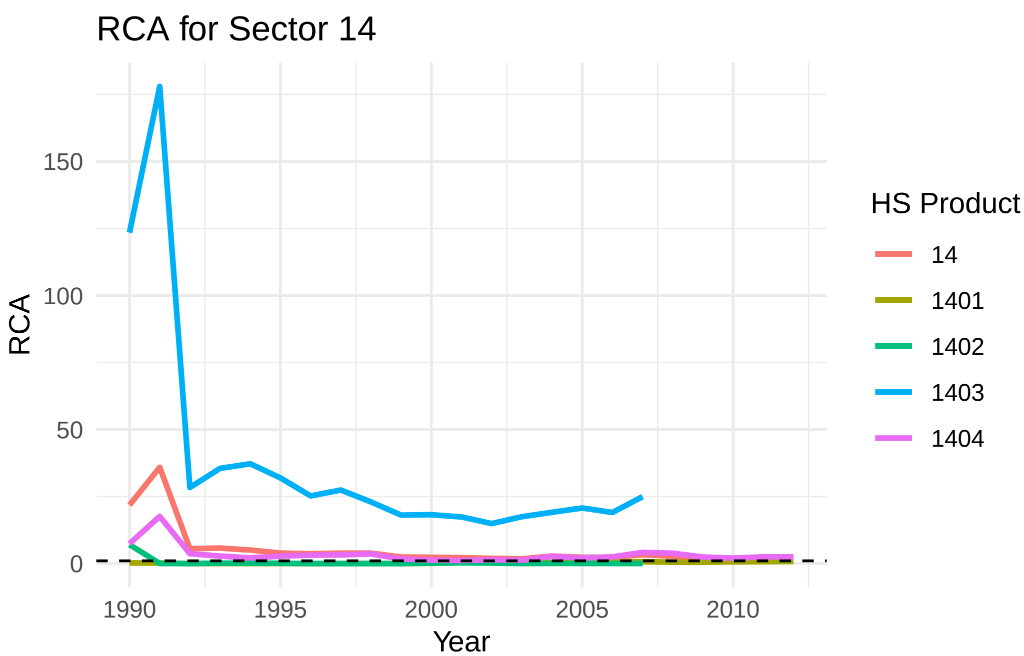

RCA for Sector 09
30
20
10
0
RCA
1990
1995
2000
2005
2010
Year
HS Product
0901
0902
0903
0904
0905
0906
0907
0908
0909
0910

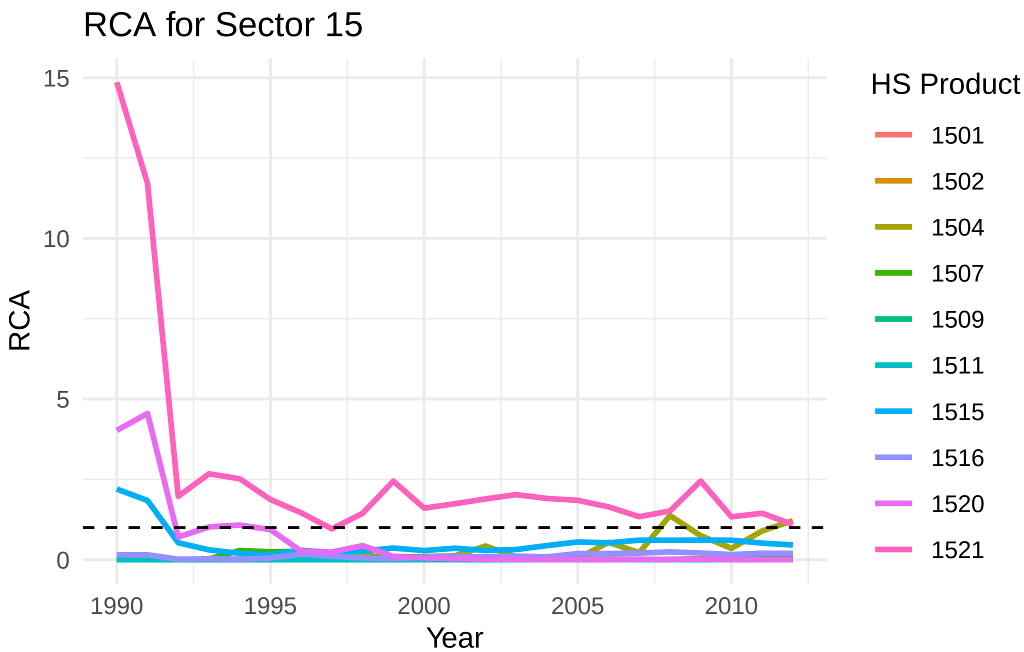

RCA for Sector 10
2
1
0
RCA
1990
1995
2000
2005
2010
Year
HS Product
1001
1002
1003
1004
1005
1006
1007
1008

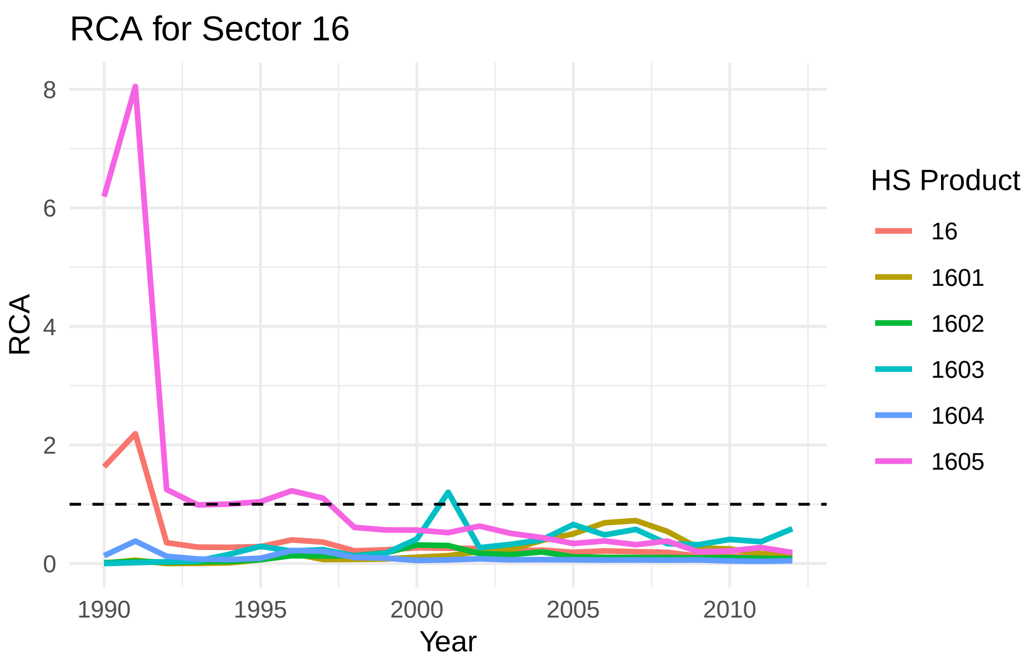

RCA for Sector 11
RCA
Year
HS Product
1101
1102
1103
1104
1105
1106
1107
1108
1109

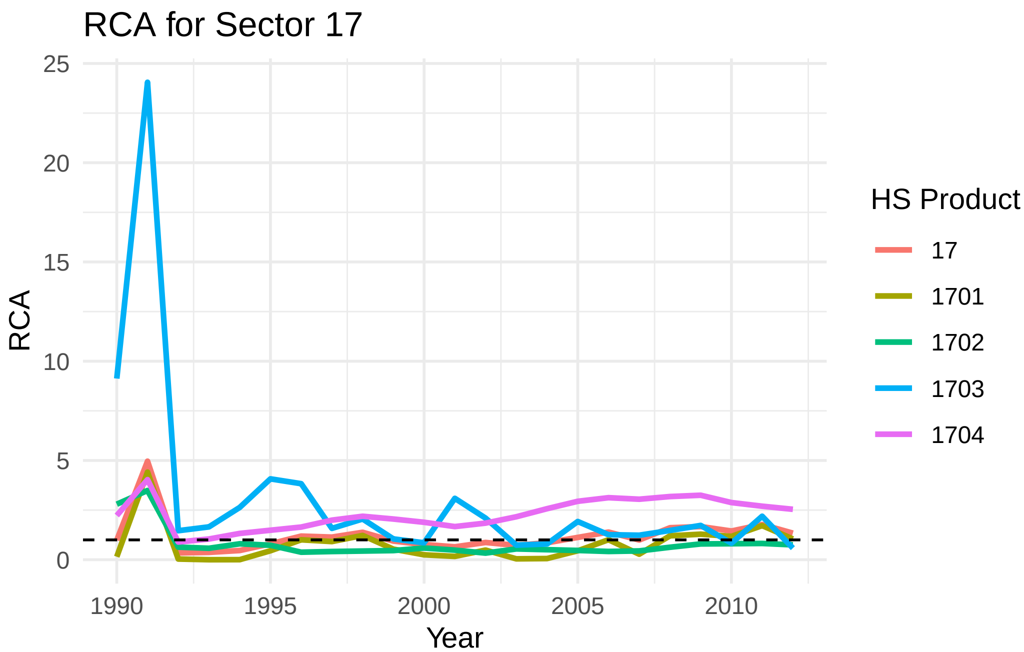

RCA for Sector 12
RCA
Year
HS Product
1201
1202
1204
1206
1207
1208
1209
1210
1213

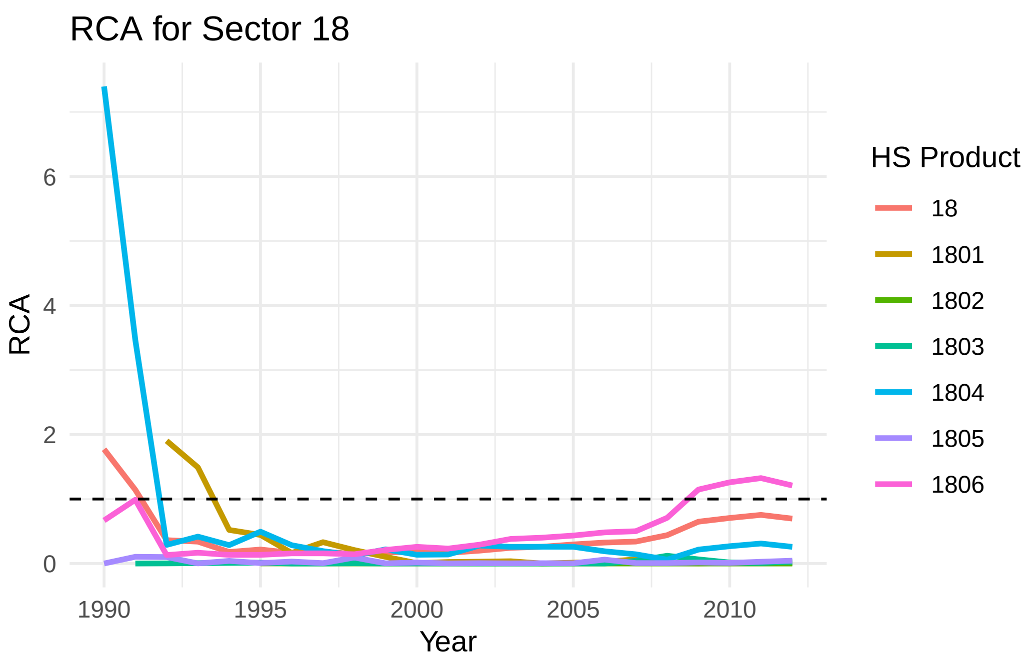

RCA for Sector 13
RCA
0
2
4
1990
1995
2000
2005
2010
Year
HS Product
13
1301
1302

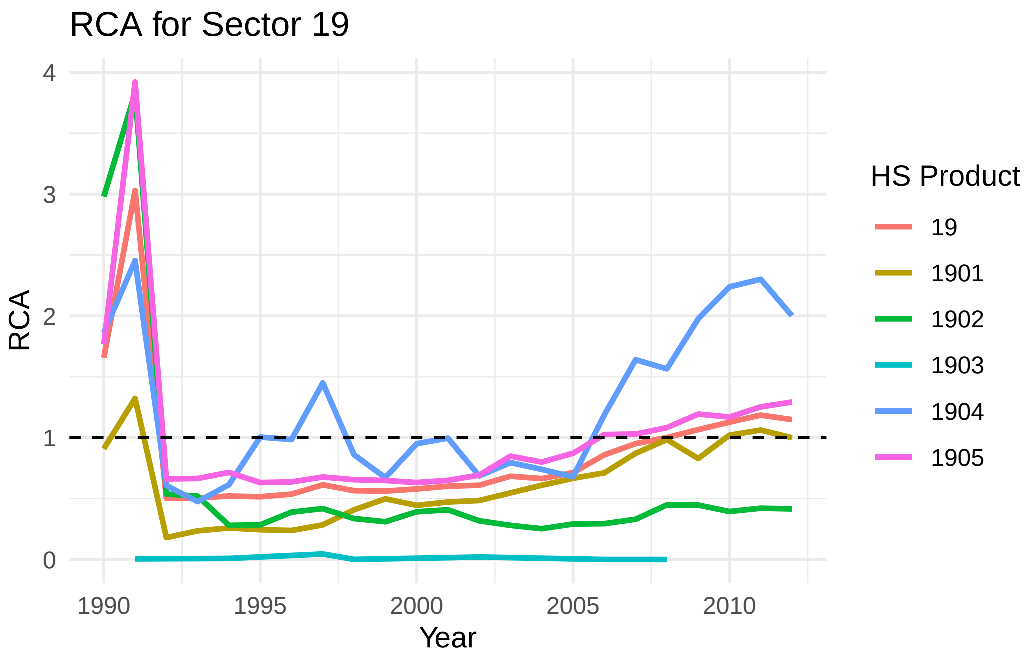

RCA for Sector 14
RCA
0
50
100
150
1990
1995
2000
2005
2010
Year
HS Product
14
1401
1402
1403
1404

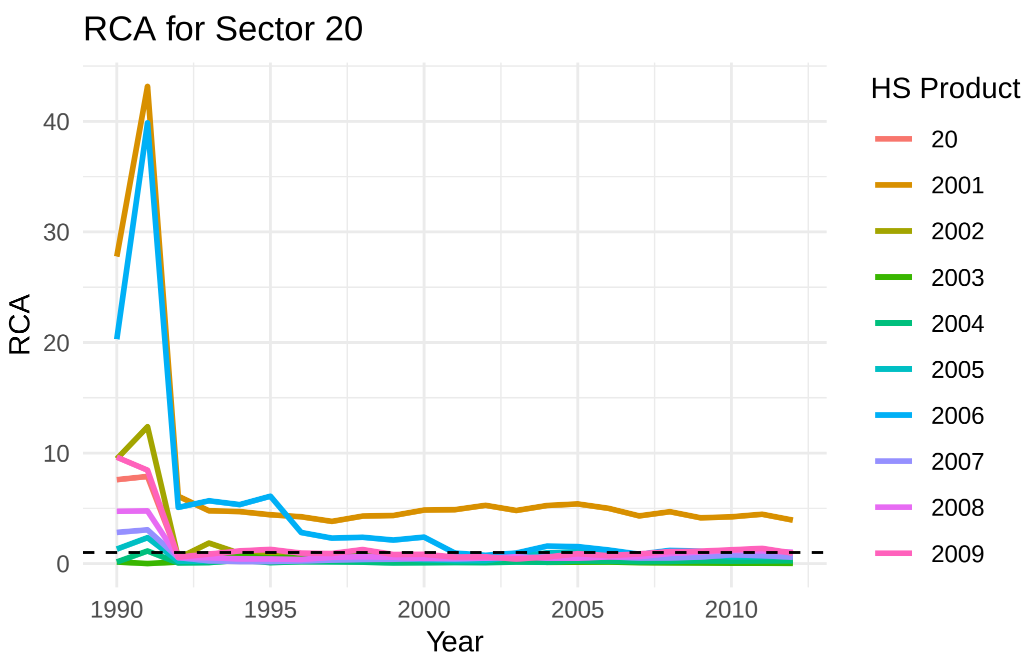

RCA for Sector 15
RCA
Year
HS Product
1501
1502
1504
1507
1509
1511
1515
1516
1520
1521

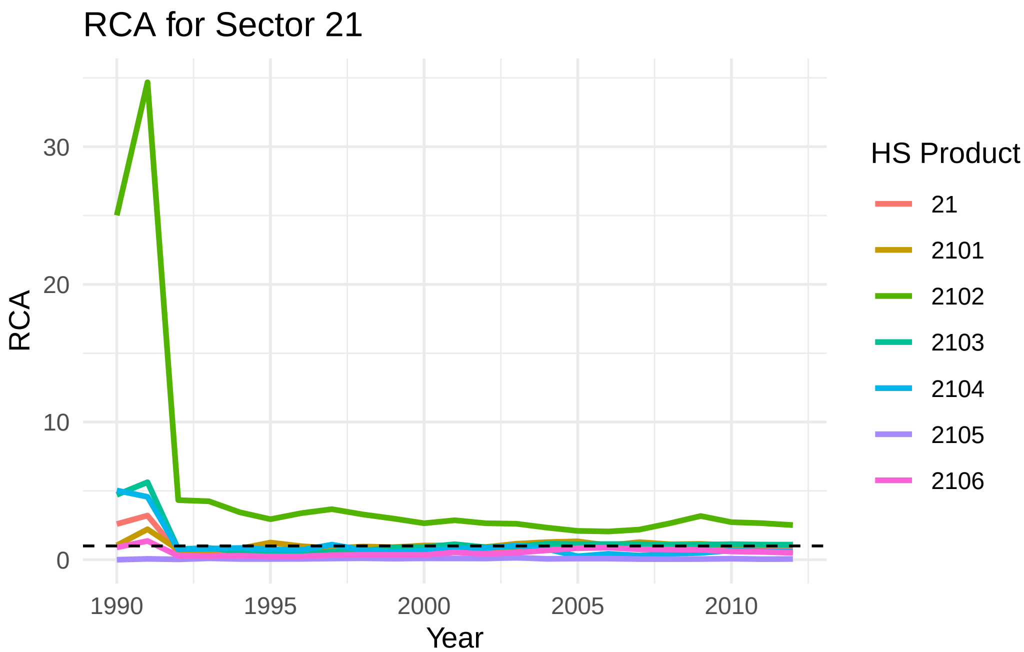

RCA for Sector 16
RCA
Year
HS Product
16
1601
1602
1603
1604
1605

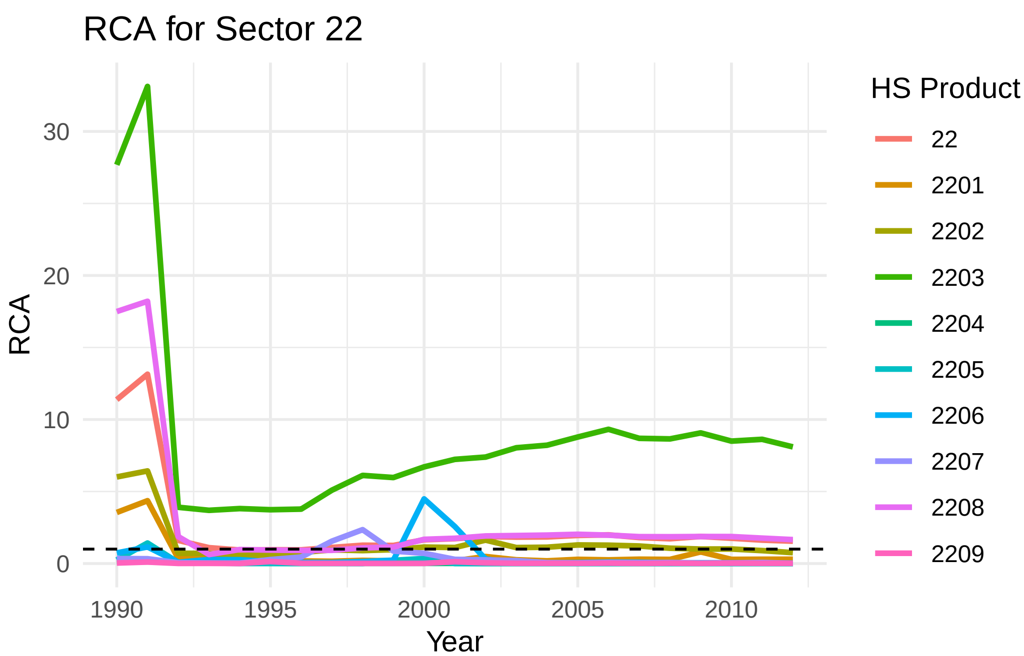

RCA for Sector 17
RCA
Year
1990
1995
2000
2005
2010
0
5
10
15
20
25
HS Product
17
1701
1702
1703
1704

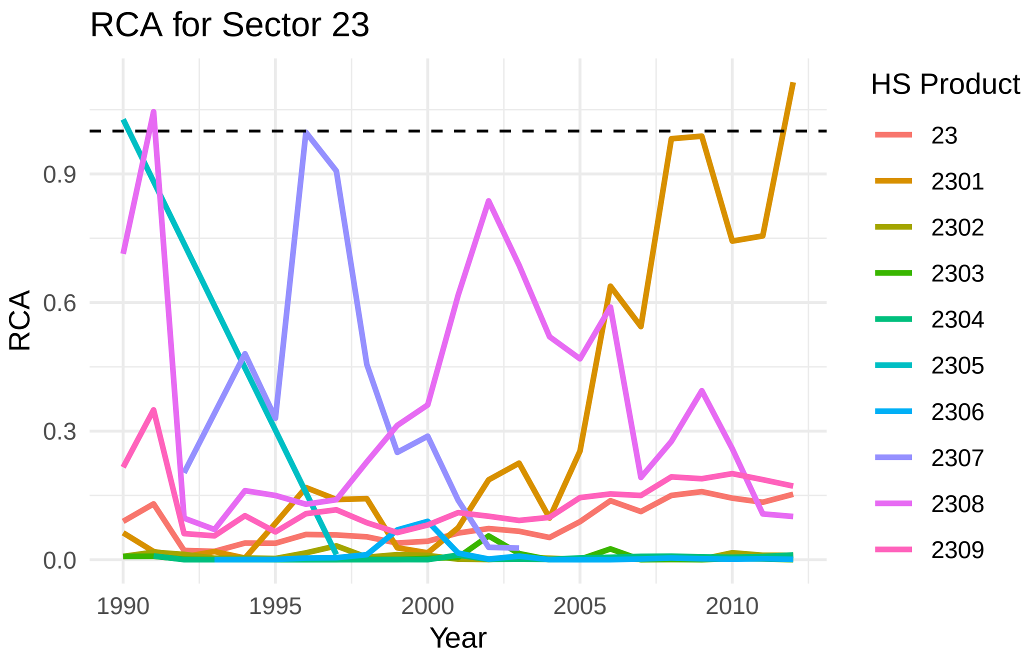

RCA for Sector 18
RCA
Year
1990
1995
2000
2005
2010
0
2
4
6
HS Product
18
1801
1802
1803
1804
1805
1806

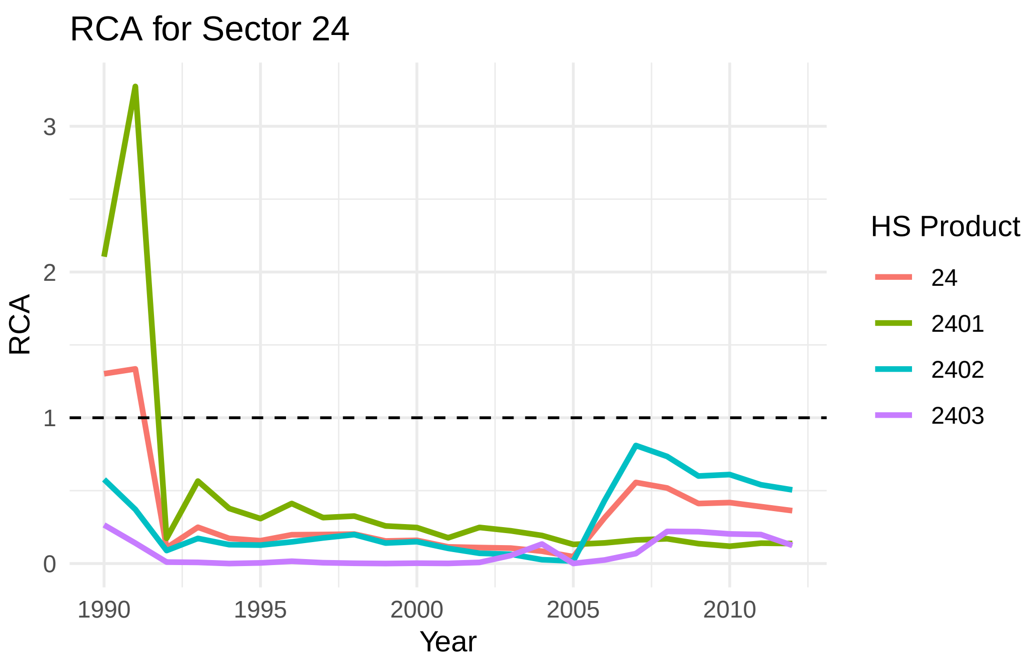

RCA for Sector 19
RCA
Year
4
3
2
1
0
1990
1995
2000
2005
2010
HS Product
19
1901
1902
1903
1904
1905

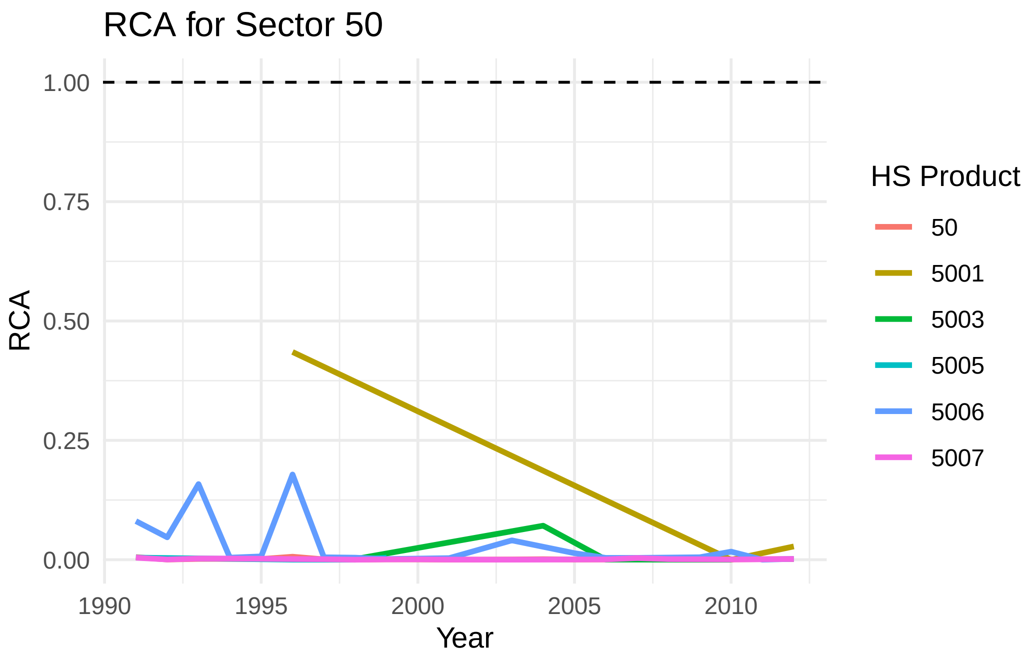

RCA for Sector 20
RCA
Year
40
30
20
10
0
1990
1995
2000
2005
2010
HS Product
20
2001
2002
2003
2004
2005
2006
2007
2008
2009

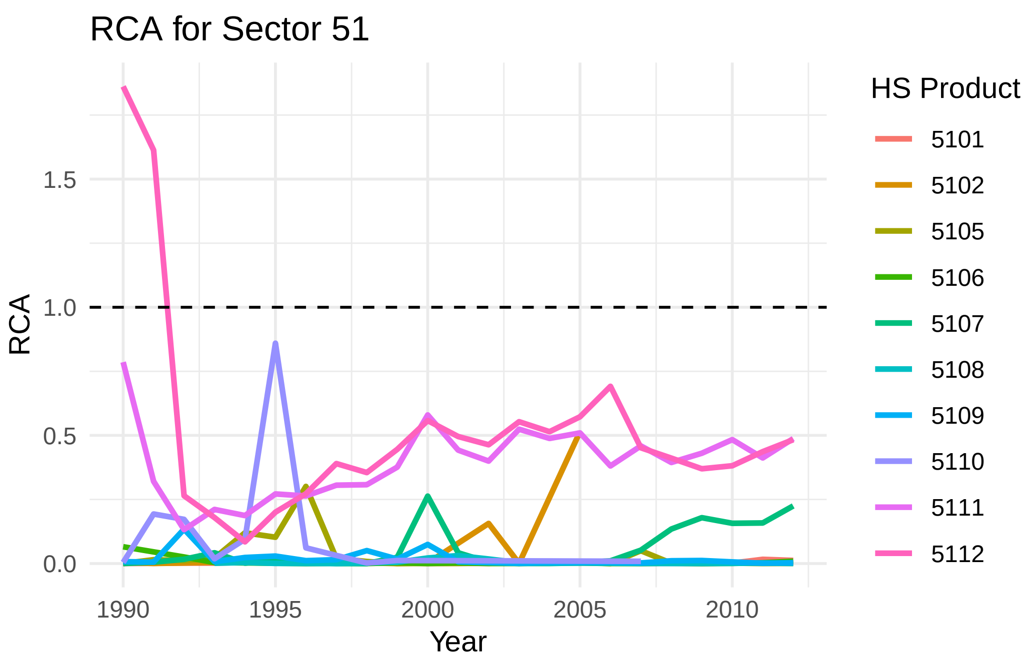

RCA for Sector 21
RCA
30
20
10
0
1990
1995
2000
2005
2010
Year
HS Product
21
2101
2102
2103
2104
2105
2106

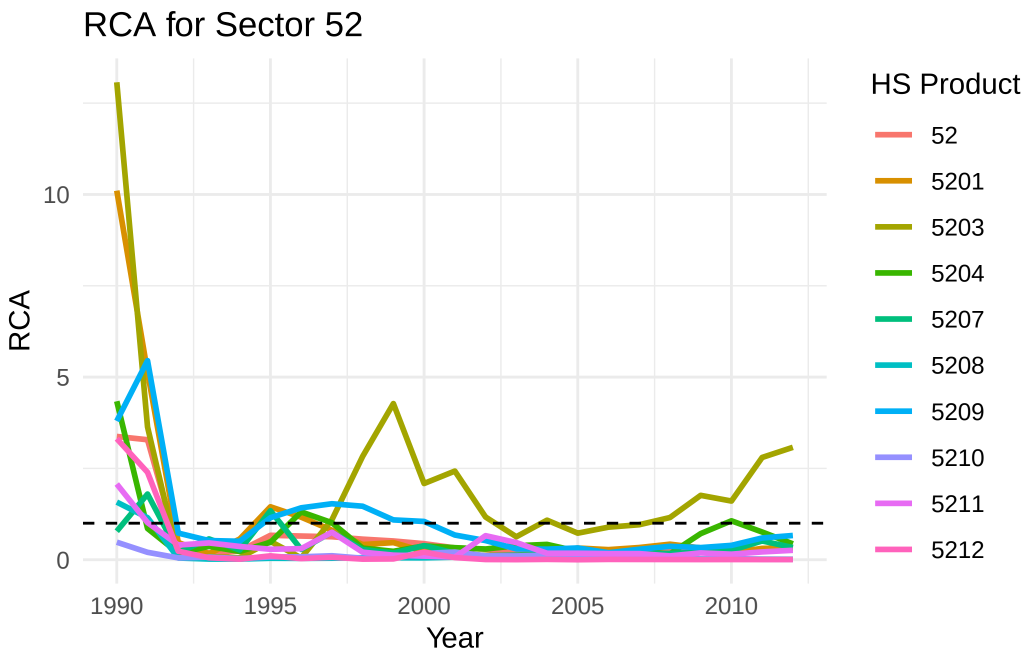

RCA for Sector 22
RCA
30
20
10
0
1990
1995
2000
2005
2010
Year
HS Product
22
2201
2202
2203
2204
2205
2206
2207
2208
2209


RCA for Sector 23
RCA
0.0
0.3
0.6
0.9
1990
1995
2000
2005
2010
Year
HS Product
23
2301
2302
2303
2304
2305
2306
2307
2308
2309

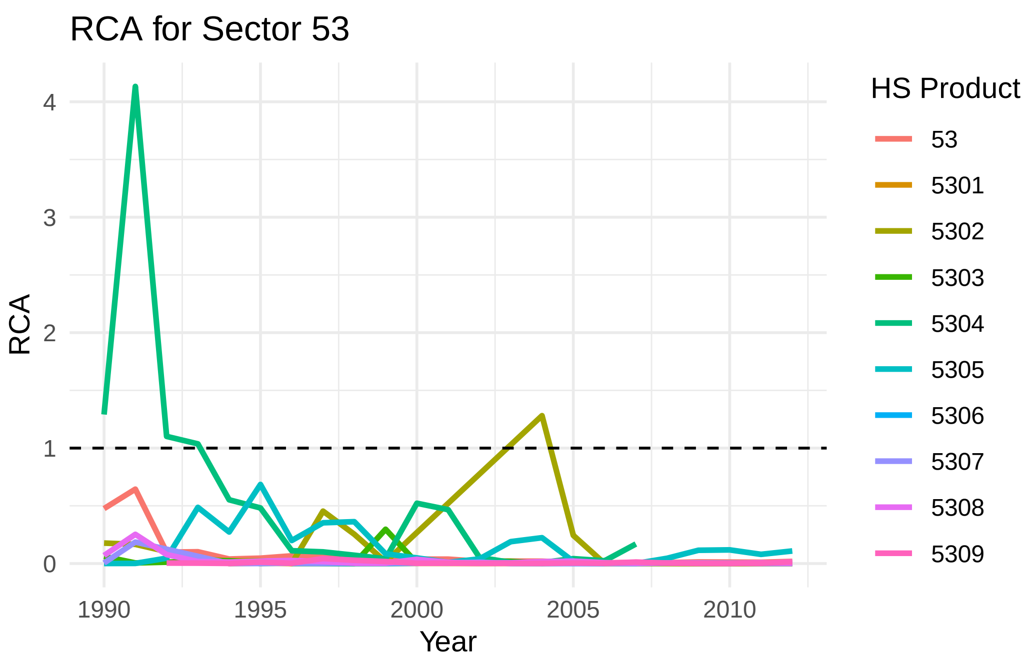

RCA for Sector 24
RCA
0
1
2
3
1990
1995
2000
2005
2010
Year
HS Product
24
2401
2402
2403


RCA for Sector 50
1.00
0.75
0.50
0.25
0.00
RCA
1990
1995
2000
2005
2010
Year
HS Product
50
5001
5003
5005
5006
5007


RCA for Sector 51
1.5
1.0
0.5
0.0
RCA
1990
1995
2000
2005
2010
Year
HS Product
5101
5102
5105
5106
5107
5108
5109
5110
5111
5112



RCA for Sector 52
RCA
10
5
0
1990
1995
2000
2005
2010
Year
HS Product
52
5201
5203
5204
5207
5208
5209
5210
5211
5212



RCA for Sector 53
RCA
4
3
2
1
0
1990
1995
2000
2005
2010
Year
HS Product
53
5301
5302
5303
5304
5305
5306
5307
5308
5309

SCM Failed Attempts

Attempt 1

```
#
# library(tidysynth)
#
# #   Winsorize function
# winsorize <- function(x, low = 0.01, high = 0.99) {
#   q <- quantile(x, probs = c(low, high), na.rm = TRUE)
#   pmax(pmin(x, q[2]), q[1])
# }
#
# #   Step 1: Load and clean country names
# country_a_xports_prod_i <- read_csv("regression/scm/country_a_xports_prod_i.csv") |>
#   mutate(reporterDesc = case_when(
#     str_detect(reporterDesc, "rkiye") ~ "Turkey",
#     str_detect(reporterDesc, "Bolivia") ~ "Bolivia",
#     TRUE ~ reporterDesc
#   ))
#
# country_a_xports_allprods <- read_csv("regression/scm/country_a_xports_allprods.csv") |>
#   mutate(reporterDesc = case_when(
#     str_detect(reporterDesc, "rkiye") ~ "Turkey",
#     str_detect(reporterDesc, "Bolivia") ~ "Bolivia",
#     TRUE ~ reporterDesc
#   ))
#
# #   Step 2: World sector 07 exports from existing pipeline files
# world_prod_07 <- map_dfr(time_periods, function(t) {
#   world_file <- file.path(base_path,
#                           paste0("world_xports_prod_i_time_", t),
#                           paste0("world_xports_sector_07_time_", t, ".csv"))
#   if (!file.exists(world_file)) {
#     message("Missing: ", world_file)
#     return(NULL)
#   }
#   read_csv(world_file, show_col_types = FALSE) |> necessary_variables()
# }) |>
#   filter(period >= 1992, period <= 2000) |>
#   group_by(period) |>
```

```
#   summarise(world_product_value = sum(fobvalue, na.rm = TRUE),
#             .groups = "drop")
#
# # World total exports
# world_total <- world_xports_all |>
#   group_by(period) |>
#   summarise(total_world_value = sum(fobvalue, na.rm = TRUE),
#             .groups = "drop") |>
#   filter(period >= 1992, period <= 2000)
#
# #   Step 3: Compute donor country RCAs
#
# donor_prod <- country_a_xports_prod_i |>
#   group_by(reporterDesc, period) |>
#   summarise(country_product_value = sum(fobvalue, na.rm = TRUE),
#             .groups = "drop")
#
# donor_total <- country_a_xports_allprods |>
#   group_by(reporterDesc, period) |>
#   summarise(total_country_value = sum(fobvalue, na.rm = TRUE),
#             .groups = "drop")
#
# donor_rca <- donor_prod |>
#   left_join(donor_total,   by = c("reporterDesc", "period")) |>
#   left_join(world_prod_07, by = "period") |>
#   left_join(world_total,   by = "period") |>
#   mutate(
#     RCA = (country_product_value / total_country_value) /
#                  (world_product_value   / total_world_value),
#     RCA_w = winsorize(RCA),
#     log_mean_rca = log(RCA_w),
#     country = reporterDesc
#   ) |>
#   select(country, period, log_mean_rca)
#
# #   Step 4: Mexico sector 07 RCA
#
# mexico_07 <- rca_big_df |>
#   filter(sector == "07",
#          nchar(cmdCode) == 4,
#          period >= 1992,
#          period <= 2000) |>
```

```
#   mutate(RCA_w = winsorize(RCA)) |>
#   group_by(period) |>
#   summarise(log_mean_rca = mean(log(RCA_w), na.rm = TRUE),
#             .groups = "drop") |>
#   mutate(country = "Mexico")
#
# #   Step 5: Stack and clean donor panel
#
# drop_countries <- c("Kenya", "Costa Rica", "Honduras", "Turkey", "Ecuador")
#
# donor_panel <- bind_rows(mexico_07, donor_rca) |>
#   filter(
#     !country %in% drop_countries,
#     period >= 1992,
#     period <= 2000,
#     !is.na(log_mean_rca),
#     is.finite(log_mean_rca)
#   )
#
# #   Step 6: Completeness check
#
# cat("Countries in donor pool:\n")
# donor_panel |> distinct(country) |> print()
#
# cat("\nCompleteness matrix:\n")
# donor_panel |>
#   pivot_wider(names_from = country, values_from = log_mean_rca) |>
#   arrange(period) |>
#   print(n = 9)
#
# #   Step 7: Pre-treatment trajectory plot
# ggplot(donor_panel,
#        aes(x = period, y = log_mean_rca, color = country,
#            linewidth = if_else(country == "Mexico", 1.2, 0.6))) +
#   geom_line() +
#   geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
#   scale_linewidth_identity() +
#   labs(title    = "Sector 07 log(RCA): Mexico and Donor Countries",
#        subtitle = "Pre-1994 parallel trends check",
#        x = "Year", y = "log(mean RCA)", color = "Country") +
#   theme_bw()
#
```

```
#
# #   Step 8: Synthetic control
# synth_mexico <- donor_panel |>
#   synthetic_control(
#     outcome           = log_mean_rca,
#     unit              = country,
#     time              = period,
#     i_unit            = "Mexico",
#     i_time            = 1994,
#     generate_placebos = TRUE
#   ) |>
#   generate_predictor(time_window = 1992:1993,
#                      mean_pre = mean(log_mean_rca, na.rm = TRUE)) |>
#   generate_predictor(time_window = 1992, rca_1992 = log_mean_rca) |>
#   generate_predictor(time_window = 1993, rca_1993 = log_mean_rca) |>
#   generate_weights(
#     optimization_window = 1992:1993,
#     margin_ipop         = 0.02,
#     sigf_ipop           = 7,
#     bound_ipop          = 6
#   ) |>
#   generate_control()
#
# #   Step 9: Plots
#
# # Actual vs synthetic
#
# plot_trends(synth_mexico) +
#   geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
#   labs(title    = "Mexico Sector 07 (Vegetables): Actual vs Synthetic RCA",
#        subtitle = "Donor pool: Indonesia, Spain, Bolivia, Ecuador, Morocco, Guatemala, Arge
#        x = "Year", y = "log(mean RCA)") +
#   theme_bw()
#
# # Gap plot
# plot_differences(synth_mexico) +
#   geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
#   annotate("text", x = 1994.1, y = Inf, vjust = 1.5,
#            hjust = 0, size = 3, color = "blue", label = "NAFTA") +
#   labs(title    = "NAFTA Effect: Gap Between Actual and Synthetic log(RCA)",
#        subtitle = "Positive = Mexico outperformed its counterfactual",
#        x = "Year", y = "Δ log(mean RCA)") +
```

```
#   theme_bw()
#
# # Placebo tests
# plot_placebos(synth_mexico) +
#   geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
#   labs(title    = "Placebo Tests: Mexico vs Donor Countries",
#        subtitle = "Mexico's post-1994 gap should exceed placebo distribution",
#        x = "Year", y = "Δ log(mean RCA)") +
#   theme_bw()
#
# # Donor weights
# plot_weights(synth_mexico) +
#   labs(title    = "Synthetic Control Donor Weights",
#        subtitle = "Which countries construct Mexico's counterfactual")
#
# #   Step 10: Significance
# grab_signifiance(synth_mexico)
```

Attempt 2

```
#library(tidysynth)

#   Winsorize function (if not already defined)
#winsorize <- function(x, low = 0.01, high = 0.99) {
 # q <- quantile(x, probs = c(low, high), na.rm = TRUE)
  #pmax(pmin(x, q[2]), q[1])
#}

#   Step 1: World sector 07 exports from existing pipeline files
#world_prod_07 <- map_dfr(time_periods, function(t) {
 # world_file <- file.path(base_path,
  #                        paste0("world_xports_prod_i_time_", t),
   #                       paste0("world_xports_sector_07_time_", t, ".csv"))
  #if (!file.exists(world_file)) {
   # message("Missing: ", world_file)
 #   return(NULL)
 # }
 # read_csv(world_file, show_col_types = FALSE) |> necessary_variables()
#}) |>
 # filter(period >= 1990, period <= 2000) |>
```

```
#  group_by(period) |>
 # summarise(world_product_value = sum(fobvalue, na.rm = TRUE),
  #          .groups = "drop")

# World total exports (aggregate - cmdCode = -2)
#world_total <- world_xports_all |>
#  group_by(period) |>
 # summarise(total_world_value = sum(fobvalue, na.rm = TRUE),
  #          .groups = "drop") |>
 # filter(period >= 1990, period <= 2000)

#   Step 2: Donor country RCAs
#donor_prod <- scm_countries_xports_07 |>
#  group_by(reporterDesc, period) |>
 # summarise(country_product_value = sum(fobvalue, na.rm = TRUE),
      #      .groups = "drop")

#donor_total <- scm_countries_xports_all |>
#  group_by(reporterDesc, period) |>
 # summarise(total_country_value = sum(fobvalue, na.rm = TRUE),
   #         .groups = "drop")

#donor_rca <- donor_prod |>
 # left_join(donor_total,   by = c("reporterDesc", "period")) |>
 # left_join(world_prod_07, by = "period") |>
 # left_join(world_total,   by = "period") |>
  #mutate(
  #  RCA          = (country_product_value / total_country_value) /
  #                 (world_product_value   / total_world_value),
  #  RCA_w        = winsorize(RCA),
   # log_mean_rca = log(RCA_w),
  #  country      = reporterDesc
#  ) |>
  #select(country, period, log_mean_rca)

#   Step 3: Mexico sector 07 RCA
#mexico_07 <- rca_big_df |>
 # filter(sector == "07",
  #       nchar(cmdCode) == 4,
   #      period >= 1990,
   #      period <= 2000) |>
 # mutate(RCA_w = winsorize(RCA)) |>
```

```
#  group_by(period) |>
 # summarise(log_mean_rca = mean(log(RCA_w), na.rm = TRUE),
    #        .groups = "drop") |>
 # mutate(country = "Mexico")

#   Step 4: Stack into donor panel
#donor_panel <- bind_rows(mexico_07, donor_rca)

# Sanity check
#onor_panel |> count(country)
#summary(donor_panel$log_mean_rca)

#   Step 5: Pre-treatment trajectory plot
# Visually check whether donor countries track Mexico pre-1994
#ggplot(donor_panel,
 #      aes(x = period, y = log_mean_rca, color = country,
  #         linewidth = if_else(country == "Mexico", 1.2, 0.6))) +
 # geom_line() +
 # geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
# scale_linewidth_identity() +
#  labs(title    = "Sector 07 log(RCA): Mexico and Donor Countries",
#       subtitle = "Pre-1994 parallel trends check",
 #      x = "Year", y = "log(mean RCA)", color = "Country") +
 # theme_bw()

#   Step 6: Synthetic control
#synth_mexico <- donor_panel |>
 # synthetic_control(
 #   outcome           = log_mean_rca,
 #   unit              = country,
 #   time              = period,
 #   i_unit            = "Mexico",
#    i_time            = 1994,
##    generate_placebos = TRUE
#   ) |>
#   generate_predictor(time_window = 1990, rca_1990 = log_mean_rca) |>
#   generate_predictor(time_window = 1991, rca_1991 = log_mean_rca) |>
#   generate_predictor(time_window = 1992, rca_1992 = log_mean_rca) |>
#   generate_predictor(time_window = 1993, rca_1993 = log_mean_rca) |>
#   generate_weights(
#     optimization_window = 1990:1993,
#     margin_ipop         = 0.02,
```

```
#     sigf_ipop           = 7,
#     bound_ipop          = 6
#   ) |>
#   generate_control()
#
# #   Step 7: Plots
#
# # Actual vs synthetic trajectory
# plot_trends(synth_mexico) +
#   geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
#   labs(title    = "Mexico Sector 07 (Vegetables): Actual vs Synthetic RCA",
#        subtitle = "Donor pool: Brazil, Chile, Colombia, Peru",
#        x = "Year", y = "log(mean RCA)") +
#   theme_bw()
#
# # Gap plot: actual minus synthetic
# plot_differences(synth_mexico) +
#   geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
#   annotate("text", x = 1994.2, y = Inf, vjust = 1.5,
#            hjust = 0, size = 3, color = "blue", label = "NAFTA") +
#   labs(title    = "NAFTA Effect: Gap Between Actual and Synthetic log(RCA)",
#        subtitle = "Positive = Mexico outperformed its counterfactual",
#        x = "Year", y = "Δ log(mean RCA)") +
#   theme_bw()
#
# # Placebo tests
# plot_placebos(synth_mexico) +
#   geom_vline(xintercept = 1994, linetype = "dashed", color = "blue") +
#   labs(title    = "Placebo Tests: Mexico vs Donor Countries",
#        subtitle = "Mexico's post-1994 gap should exceed placebo distribution",
#        x = "Year", y = "Δ log(mean RCA)") +
#   theme_bw()
#
# # Donor weights
# plot_weights(synth_mexico) +
#   labs(title    = "Synthetic Control Weights",
#        subtitle = "Which countries construct Mexico's counterfactual")
#
# #   Step 8: Significance
# grab_signifiance(synth_mexico)
```